GOVERNMENT OF ANDHRA PRADESH

Andhra Pradesh Building & Other Construction Workers Welfare Board (APBOCWWB)

DIGITAL ILAC PLATFORM

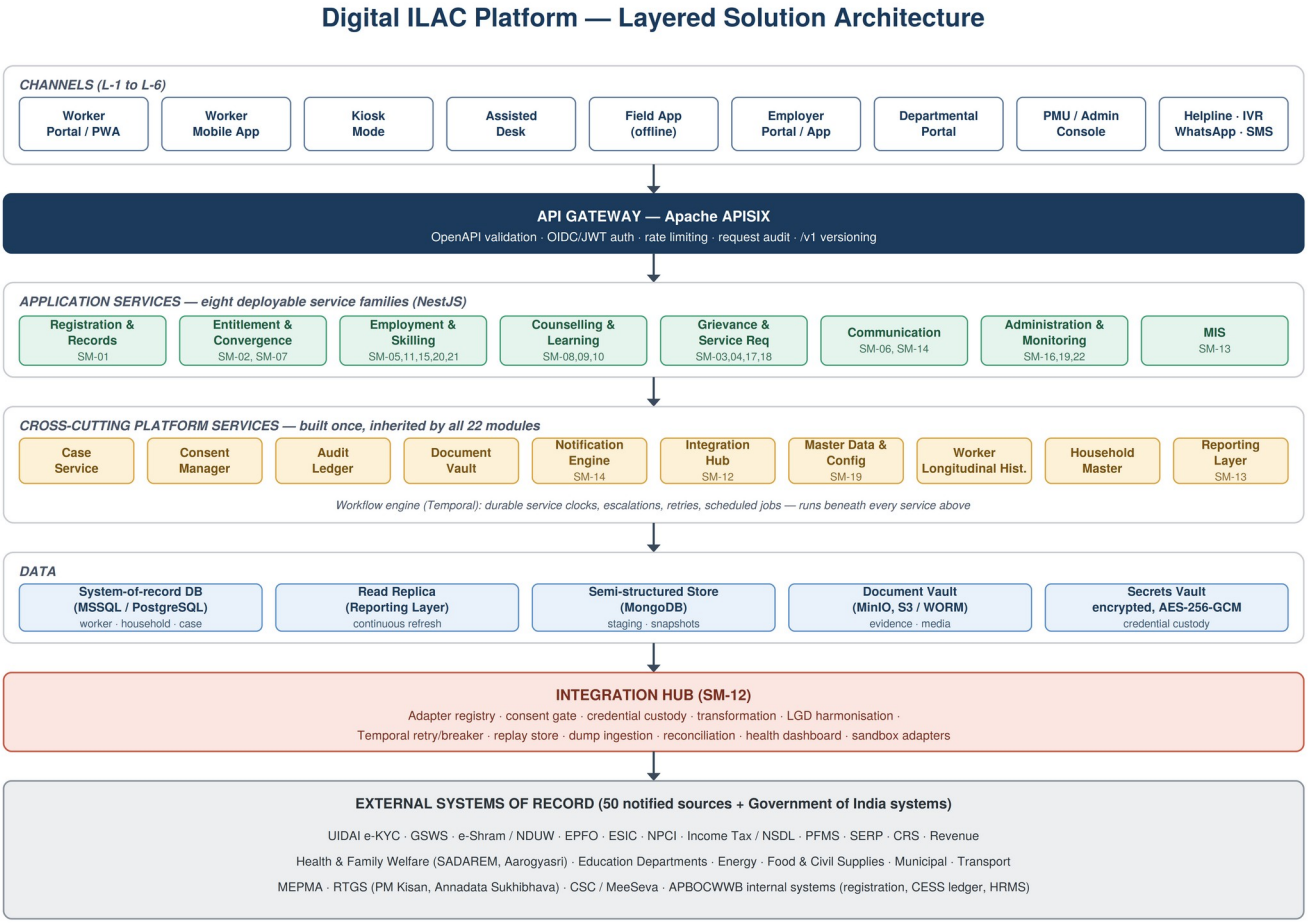
Technical Proposal

Solution Architecture, Delivery Plan, Integration, Security and Operations

Prepared for submission to APBOCWWB

PLATFORM AT A GLANCE

Digital ILAC — Layered Solution Architecture



One modular, API-first platform serving all six service loci — see Section 2 for the full architecture description

Detailed Introduction

This document sets out how the Digital ILAC Platform will be designed, built, integrated, secured, deployed, supported and handed over. It is organised so that a reader can go straight to the section that matters to their role — a programme officer to the delivery plan, a security reviewer to Section 8, a district field officer to Section 11 — without needing to read the whole document end to end. Every entry in the contents list below is a clickable link, in both the Word and PDF versions, that jumps straight to that page.

Table of Contents

1. Executive Summary and Understanding of the ILAC Programme 9

1.1 What ILAC is 9
1.2 The six service locations 9
1.3 The core mandate — "no worker turned away" 9
1.4 Dual-mode convergence — the platform works even when a department's system does not 10
1.5 What is being proposed — one platform, ten shared services, twenty-two modules 10
1.6 Key design decisions at a glance 11
1.7 Delivery in four months 11
1.8 How this proposal is organised 12
1.9 Programme scale and design parameters 12

2. Proposed Solution Architecture and Technology Stack 14

2.1 Architecture principles 14
2.2 Logical architecture 14
2.3 Data architecture 17
2.4 Workflow engine and service clocks 17
2.5 Identity, access and jurisdiction 18
2.6 Channel architecture 18
2.7 Deployment architecture 18
2.8 Technology stack 19
2.9 Non-functional design 20
2.10 Architecture decisions at a glance 20
2.11 Runtime units and deployment composition 21
2.12 Repository and package structure 22
2.13 API design standards 23
2.14 API gateway configuration and route classes 24
2.15 Internal structure of a service family 25
2.16 Events and asynchronous processing 26
2.17 Durable workflow catalogue 27
2.18 Caching and data-access rules 28
2.19 Configuration management model 29
2.20 Performance and capacity model 29
2.21 Component baseline and patch policy 30

3. Module-wise Compliance and Solution Approach (SM-01 to SM-22) 32

3.1 Module summary — SM-01 to SM-22 32
3.2 Module-by-module detail — SM-01 to SM-22 34

SM-01 — Worker Registration, Household Baseline Survey & Worker Records....35
SM-02 — Welfare-Scheme & Entitlement Module (Scheme Engine)....36
SM-03 — CSC & MeeSeva Service Request & Tracking....37
SM-04 — Grievance Redressal & Multi-Channel Complaint Tracking....38
SM-05 — Work Request, Job Matching & Daily-Wage Employment Tracking....40
SM-06 — Opportunity & Event Calendar, Mapping & Referral....41
SM-07 — Departmental Convergence, Register & Committee Record....42
SM-08 — Student & Education Counselling Management....43
SM-09 — Learning Management System & IEC Dissemination....44
SM-10 — Employment, Self-Employment & Enterprise Counselling....45
SM-11 — Work-Site Visit, Reskilling, Upskilling & RPL Scheduling....45
SM-12 — Common API & Systems Integration Framework (Integration Hub)....46
SM-13 — MIS, Data Analytics & Decision Support....48
SM-14 — Communication, Notification, Alert & Broadcast Management....49
SM-15 — Job Mela & Event Facilitation....50
SM-16 — Monitoring of ILAC Functions & Functionary Work Turnout....51
SM-17 — Para-Legal Services Facilitation at the ILAC....51
SM-18 — Common Agency Service Facilitation Interface....52
SM-19 — User, Role & Configuration Administration....53
SM-20 — Construction-Site Catalogue & Worker Grouping for Skilling....54
SM-21 — Construction Worker Livelihood Groups & Federation....55
SM-22 — Worker, Employer & Departmental Applications, Portals & Dashboards....56
3.3 How "no worker turned away" is enforced across every module....57
3.4 Cross-module event contracts....57
3.5 Module-to-phase allocation....57
3.6 Logical data model and schema ownership....58
3.7 Case Service design....60
3.8 Scheme Engine rule specification....61
3.9 Registration wizard step treatment....63
4. Four-Month Delivery Plan, Phase Plan, Milestones, Dependencies and Recovery Mechanism....65
4.1 Planning basis....65
4.2 Phase plan summary....65
4.3 What makes the window achievable....66
4.4 Milestone register....66
4.5 Inception — Weeks 1 and 2....67
4.6 Dependencies....67
4.7 Critical path....68
4.8 Recovery mechanism....68
4.9 Governance cadence during delivery....69
4.10 After Certified Final Go-Live....69
4.11 Sprint-level plan....69
4.12 Release train and environment promotion....70
4.13 Responsibility matrix for programme deliverables....71
4.14 Early-warning indicator bands....71
5. Integration Approach for Intra-Departmental, Government of India and Other AP Government Systems 73
5.1 Integration principles....73
5.2 Integration Hub (SM-12) architecture....73
5.3 Integration catalogue — summary....75

5.4 Aadhaar and UIDAI integration mechanics....76
5.5 Geography (LGD) harmonisation....76
5.6 Department onboarding process....77
5.7 Reconciliation and monitoring....77
5.8 Integration delivery plan by phase....77
5.9 Integration risks and mitigations....77
5.10 Adapter descriptor....78
5.11 Resilience parameters by integration pattern....78
5.12 Technical profile of principal sources....79
5.13 UIDAI OTP e-KYC call sequence....79
5.14 LGD resolution algorithm....80
5.15 Dump file contract....81
5.16 Onboarding evidence per source....81
6. Manual / Assisted Fallback Design and Later API Cut-Over Approach....83
6.1 Design principle — parity between manual and automated transactions....83
6.2 Six fallback classes....83
6.3 Fallback route per source....84
6.4 Fallback per module....85
6.5 The manual case record....85
6.6 Service clocks and escalation under fallback....85
6.7 Reconciliation workflow....85
6.8 Certification of fallback routes....85
6.9 API cut-over approach....86
6.10 Reverse cut-over on API degradation....86
6.11 Fallback governance....86
6.12 Fallback fields on the case record....86
6.13 Class selection and promotion rules....87
6.14 Cut-over promotion and rollback criteria....88
6.15 Fallback SOP template....88
6.16 Fallback key performance indicators....89
7. Data Migration, Golden Record, Data Quality and Reconciliation Plan....90
7.1 Objectives....90
7.2 Migration governance....90
7.3 Source inventory....91
7.4 Deduplication and matching....91
7.5 Golden record construction....91
7.6 Departmental dump ingestion....92
7.7 Consent and privacy in migration....92
7.8 Reconciliation plan....92
7.9 Migration execution plan....92
7.10 Ongoing data quality....93
7.11 Legacy-to-target mapping....93
7.12 Data quality rule catalogue....94
7.13 Matching algorithm parameters....95
7.14 Survivorship matrix....95
7.15 Production cut-over runbook timings....96

7.16 Reconciliation queries ... 96
8. Security, Privacy, Aadhaar/UIDAI, Audit, VAPT, Backup and Disaster Recovery Approach ... 98
8.1 Governing frameworks ... 98
8.2 Identity and access ... 98
8.3 Data protection and privacy ... 99
8.4 Application and API security ... 101
8.5 Audit and immutability ... 101
8.6 Security testing and VAPT ... 101
8.7 Security monitoring and incident response ... 101
8.8 Backup and recovery ... 102
8.9 Disaster recovery ... 102
8.10 Business continuity for centres ... 102
8.11 Compliance mapping summary ... 103
8.12 Threat model by trust boundary ... 103
8.13 Authentication flows and token policy ... 104
8.14 Cryptographic standards and key hierarchy ... 105
8.15 Aadhaar Data Vault operations ... 105
8.16 Network zones and flow matrix ... 106
8.17 Logging, retention and monitoring ... 107
8.18 OWASP ASVS 4.0 control mapping ... 107
8.19 Hardening baselines ... 108
8.20 Recovery objectives and backup schedule ... 108
8.21 Audit Ledger chain algorithm ... 109
9. Web, Mobile, Field, Kiosk and Assisted-Desk Approach ... 110
9.1 Channel-to-locus coverage ... 110
9.2 Design system and accessibility ... 110
9.3 Worker portal and mobile app ... 111
9.4 Employer portal and app ... 111
9.5 Kiosk mode ... 111
9.6 Assisted desk ... 111
9.7 Field application ... 111
9.8 Departmental portal and PMU console ... 112
9.9 Performance and device baseline ... 112
9.10 Screen inventory by surface ... 112
9.11 Progressive web application and caching strategy ... 112
9.12 Internationalisation and Telugu support ... 113
9.13 Performance budgets and accessibility checks ... 113
10. Offline Data Capture and Synchronisation Design ... 115
10.1 What completes fully offline ... 115
10.2 What cannot complete offline, and how it is deferred ... 115
10.3 Device-side architecture ... 116
10.4 Security of offline data ... 116
10.5 Synchronisation and conflict management ... 116
10.6 Clock, location and integrity ... 117
10.7 Operational modes and recovery ... 117
10.8 Testing ... 117

10.9 Local data model on the device ... 117
10.10 Synchronisation protocol ... 118
10.11 Resumable media upload ... 120
11. Technical Support Unit, Helpdesk and Seven District Field Support Cluster Model ... 121
11.1 The three tiers ... 121
11.2 Ticket levels ... 122
11.3 Seven District Field Support Clusters ... 122
11.4 Interfaces between tiers ... 122
11.5 Service levels and measurement ... 122
11.6 Knowledge management ... 123
11.7 Phase-wise ramp ... 123
11.8 Tooling ... 123
11.9 Severity matrix and proposed service targets ... 123
11.10 Ticket categories and routing ... 124
11.11 Helpdesk workload model and staffing basis ... 124
11.12 Cluster field kit and visit pattern ... 124
11.13 Runbooks for common operational events ... 125
12. Key Personnel and Governance Model ... 126
12.1 Team structure ... 126
12.2 Key personnel ... 126
12.3 Key personnel commitments ... 127
12.4 Three-tier governance structure ... 128
12.5 Escalation matrix ... 129
12.6 Department engagement governance ... 129
12.7 Quality governance ... 129
12.8 O&M governance ... 129
12.9 Role-to-permission examples ... 130
12.10 Change control procedure ... 130
12.11 RAID log structure ... 130
13. UAT, Testing, Training, Go-Live and Hypercare Plan ... 132
13.1 Test strategy ... 132
13.2 Functional testing ... 133
13.3 Integration testing ... 133
13.4 Non-functional testing ... 133
13.5 Data migration testing ... 133
13.6 User acceptance testing ... 133
13.7 Training plan ... 134
13.8 Go-Live plan ... 134
13.9 Hypercare ... 135
13.10 Documentation and knowledge transfer during testing and training ... 135
13.11 Defect severity definitions ... 135
13.12 Performance test scenarios ... 136
13.13 Training schedule and certification ... 136
13.14 Hypercare exit criteria ... 137
14. Source Code, Documentation and Exit-Management Plan ... 138

14.1 Principles 138
14.2 Repository structure and deposit mechanism 138
14.3 What is deposited 139
14.4 Documentation set 139
14.5 Knowledge transfer 140
14.6 Exit management plan 140
14.7 Vendor neutrality and reuse 141
14.8 Build pipeline, software bill of materials and reproducible build 141
14.9 Exit plan timeline 142
14.10 Documentation set contents 143
15. Compliance Matrix and Deviations 144
15.1 Compliance summary by RFP area 144
15.2 Deviations 144
15.3 Clarifications 145
15.4 Assumptions carried through this document 146
15.5 Cross-reference of RFP requirement areas to this proposal 146
15.6 Deviation impact summary 147
16. Hosting Landscape and Transition 148
16.1 Hosting topology 148
16.2 Year-1 recommended compute 149
16.3 Year-1 recommended storage, with 25% headroom 149
16.4 Year-1 recommended bandwidth 150
16.5 MeitY-empanelled CSP recommendation 151
16.6 Cloud-to-APSDC migration and transition approach 151
16.7 Ongoing capacity review 152
16.8 Kubernetes cluster design 152
16.9 Capacity workings behind the Year-1 figures 153
16.10 Disaster Recovery environment 154
16.11 Network and DNS design 154
16.12 Observability stack and retention 155
16.13 Infrastructure as code layout 156
Annex A. API Catalogue 157
Annex B. Data Dictionary Extract and Conventions 160
Annex C. Event Catalogue and Error Codes 162
Annex D. Configuration Reference, Reporting Marts and Glossary 165

1. Executive Summary and Understanding of the ILAC Programme

1.1 What ILAC is

The Integrated Labour Amenities Complex (ILAC) is a one-stop facilitation model for registered construction workers and their dependants. It brings together, at one counter and under one record, services a worker today has to seek from a dozen different offices: registration, welfare facilitation, employment and skilling support, documentation, education, health, grievance redressal and cross-departmental convergence.

The Digital ILAC Platform is the software that makes this model work — at every ILAC centre and labour adda, at every worksite and household a field officer visits, at every job mela and camp, on every worker's phone, and in the back office of the Andhra Pradesh Building and Other Construction Workers Welfare Board (APBOCWWB).

The 4E pathway. Every worker's journey on the Platform follows four stages — Enrollment, Entitlement, Empowerment and Employment. A worker registers once; the Platform then keeps their entitlements current, connects them to counselling, training and recognition of prior learning, and matches them to employers. The household — not the individual worker alone — is the unit of convergence: a child's school enrolment, a spouse's self-help-group membership and a household's ration card all decide which of the 38 catalogued welfare schemes a worker can draw on.

1.2 The six service locations

The Platform is designed to serve every point at which a worker or officer touches it, using the same worker record and the same rules everywhere:

Locus	Where	What happens there
L-1	ILAC centre & labour adda	Registration, assisted desk, welfare facilitation, grievance intake, counselling, kiosks, token/queue
L-2	Worksite	Worksite visits, vocational exposure, safety sessions, employer engagement, offline field capture
L-3	Household	Household baseline survey, profiling, dependant pathways, outreach
L-4	Job mela & camps	Registration, check-in, interviews, offers, placement follow-up, RPL, camp services
L-5	Remote & assisted channels	Web/mobile apps, helpline, IVR, WhatsApp, alerts, status tracking
L-6	Back office	MIS, dashboards, administration, configuration, API integration, department onboarding

1.3 The core mandate — "no worker turned away"

Every request made to the Platform must conclude in one of exactly two ways: a completed service, or a documented next step that names the responsible officer or department, the action required, a target date, the documents needed, and the follow-up route. A worker who leaves an ILAC centre without one

of these two outcomes represents a failure of the Platform — whatever the reason, including an unavailable departmental system.

The Platform also has a firm boundary: it facilitates services. It never performs statutory sanction, rejection, benefit disbursement, adjudication, credit approval or medical judgement — those decisions remain with the competent authority. The Platform routes the request, tracks the clock, and records the answer the authority returns.

1.4 Dual-mode convergence — the platform works even when a department's system does not

Departmental interfaces will be uneven for a long time — of the 50 notified data sources, 31 offer a live API, 16 supply periodic data dumps, one supplies both, and two are reachable only through the Real Time Governance Society's internal route. The Platform is designed around this reality from the first line of code: it uses an API where a stable, authorised interface exists, and a digitised manual or assisted process where it does not. No worker is ever denied, deferred or turned away because a departmental system is down. Manual and automated transactions carry the same case identifier, service clock, acknowledgement, document trail, audit trail and reconciliation workflow — this single design decision, detailed in Section 6, is what makes the Platform dependable on the day a department's system fails, which it eventually will.

1.5 What is being proposed — one platform, ten shared services, twenty-two modules

The Digital ILAC Platform is a single modular, API-first system. The 22 approved functional modules (SM-01 to SM-22) are feature packages built on ten cross-cutting platform services that are built once, in the first phase, and inherited by every module that follows:

Shared service	What every module gets from it
Case Service	The ILAC Case ID, the service clock, and the two permitted ways to close a case
Consent Manager	DPDP and biometric consent recorded before any external query is made
Audit Ledger	An append-only, hash-chained record of every change and every access to sensitive data
Document Vault	Encrypted, write-once storage for evidence and consent, linked to the Case ID
Notification Engine (SM-14)	SMS, WhatsApp, IVR and in-app dispatch with delivery tracking
Integration Hub (SM-12)	Adapters, credential custody, geography harmonisation, retries, replay and reconciliation for every source
Master Data & Configuration (SM-19)	Forms, scheme rules, service clocks and workflows as versioned configuration, never hard-coded
Worker Longitudinal History (WLH)	The immutable, lifelong timeline of every interaction a worker has with the Platform
Household Master	The household record that makes the household the unit of convergence

Shared service	What every module gets from it
Reporting Layer (SM-13)	Read-optimised reporting, governed by the same access rules as the live system

The 22 modules are grouped into eight independently deployable service families (Section 2.3) rather than being built as a fine-grained microservice estate. This is a deliberate choice for three reasons: it fits the four-month delivery window; it gives APBOCWWB a Platform that eight technical staff can operate after the Bidder exits, rather than a sprawling estate that only the original team understands; and it comfortably meets the required scale of at least 300 concurrent access nodes and unlimited registered users without needing finer-grained scaling.

1.6 Key design decisions at a glance

Decision	Why
Thin orchestration over departmental systems of record	The Platform never re-masters departmental data; it fetches, caches with source and timestamp, and always leaves the department as the authority
One Case Service with a two-outcome closure rule, enforced at the API	Makes "no worker turned away" a property verified once, not a promise repeated in 22 modules
Six fallback classes with certified per-source routes and controlled API cut-over	Turns a departmental outage from an emergency into a planned operating mode
Golden-record construction with human review on every probabilistic match	No worker's identity or entitlement is ever decided by a silent algorithm
Offline-first field application with both-version conflict retention	A field officer's work at a worksite or household is never lost to a lack of signal
QR card token on existing centre hardware, in place of a biometric smart card	Biometric smart-card hardware is a Part 2 capability outside this scope; the token gives the same tap-login and footfall capture on hardware centres already have
TypeScript end-to-end, open-source runtime throughout	One type system across backend, web and mobile reduces integration defects; every runtime component except the database is open-source and operable by the Authority after exit

1.7 Delivery in four months

The Platform is delivered across four phases inside a 17-week programme, with a Go-Live at the end of each phase and Certified Final Go-Live in Week 17. What makes this window realistic:

1. Fallback-first Go-Live — a module goes live on its certified manual route for any element whose departmental interface is not yet ready; the interface is cut over later without moving the phase gate.

2. Ten platform services built once, in Phase 1 — later phases add modules, adapters and channels; they do not add new platform plumbing.

3. Configuration over code — the 38-scheme catalogue, fallback routes, service clocks and notification templates are data, not programs, so they change without a release.

4. One type system, TypeScript end to end, removing a whole category of integration defects between the backend, the web application and the field application.

5. A Technical Support Unit operating from Week 1, so configuration, integration onboarding and data governance are moving before the first Go-Live.

The full phase plan, milestone register and recovery mechanism are in Section 4.

1.8 How this proposal is organised

Section 2 sets out the architecture and technology choices that everything else is built on. Section 3 works through every one of the 22 modules. Sections 4 to 14 cover delivery, integration, fallback, migration, security, channels, offline capture, support, governance, testing and exit — each the detailed "how" behind a commitment made here. Section 15 is the clause-by-clause compliance matrix, and Section 16 addresses hosting sizing and the transition to the AP State Data Centre.

1.9 Programme scale and design parameters

The parameters below are used consistently by the sizing, performance, support and testing sections of this proposal. A parameter taken from the RFP is marked as such. A planning assumption is confirmed against APBOCWWB volumes during inception (Section 4.5).

Parameter	Value	Basis
Concurrent access nodes	At least 300; 500 as the Year-1 growth case	RFP floor; Section 16.2
Registered worker records, Year 1	Up to 3,000,000	Planning assumption; Section 16
Households	About 800,000	Planning assumption; Section 16
ILAC centres	25 at start, scaling statewide	Planning assumption; Section 16
District Field Support Clusters	7, each covering 3 to 4 districts	Section 11.3
Functional modules	22 (SM-01 to SM-22)	RFP; Section 3
Shared platform services	10	Section 2.2.1
Deployable service families	8	Section 2.2.2
Notified data sources	50: 31 live API, 16 dump, 1 both, 2 RTGS-internal	RFP; Section 5.3
Catalogued welfare schemes	38, with 6 overlap rules	RFP; SM-02
Environments	5 (Development, Staging/QA, UAT, Production, DR)	RFP; Section 2.7
Interface languages	Telugu and English	RFP; Section 9.2
Fallback classes	6 (F1 to F6)	Section 6.2

Parameter	Value	Basis
Typed integration outcomes	8	Section 5.2.3
Case closure outcomes	2: Service Completed, Documented Next Step	Section 1.3
Delivery programme	17 weeks, 4 phases, 9 sprints, 14 milestones	Section 4

2. Proposed Solution Architecture and Technology Stack

This section describes the architecture the Digital ILAC Platform is built on and the technology stack it is built with. Two conventions apply throughout this document. First, where two equivalent open technologies satisfy the same requirement, both are named — the first as proposed, the second as an equivalent alternative acceptable to the Authority at no change in scope, schedule or function. Second, the system-of-record database is treated as a confirmed decision at inception (Section 15 records the recommended platform and the fallback position).

2.1 Architecture principles

Principle	How the Platform realises it
API-first	Every capability is a versioned REST API described by an OpenAPI 3.1 contract before any screen is built. Web, kiosk, assisted-desk, mobile and departmental clients all consume the same APIs through one gateway.
Modular	22 modules are feature packages on 10 shared services, grouped into 8 deployable families. A module can be added, replaced or removed without touching another module's code.
Configurable	Forms, scheme rules, service clocks, escalation rules, workflows and master data are versioned, effective-dated configuration — a configuration change is a release of data, not of code.
Vendor-neutral	Every runtime component other than the system-of-record database is open-source with an OSI-approved licence, operable by the Authority or any successor with no ongoing commercial relationship with the Bidder.
Auditable	Every create, update, status change, configuration change, consent event, integration call and access to sensitive data is written to an append-only, hash-chained Audit Ledger.
India-resident	All five environments, all data stores, all backups and the disaster-recovery site are hosted in India. No worker data leaves Indian jurisdiction.
Thin orchestration over systems of record	The Platform masters only worker, household and case data. Departmental data is fetched, cached with provenance, and reconciled — never re-mastered.
Scale without per-transaction cost	Stateless services scale horizontally; the database is sized for the full worker population with a read replica for reporting; no component charges per worker, user, record or API call.

2.2 Logical architecture

The diagram below shows the full layering of the Platform, from the six channel surfaces at the top, through the API gateway and eight application-service families, to the ten shared platform services, the data layer, the Integration Hub, and the external departmental and national systems it connects to.

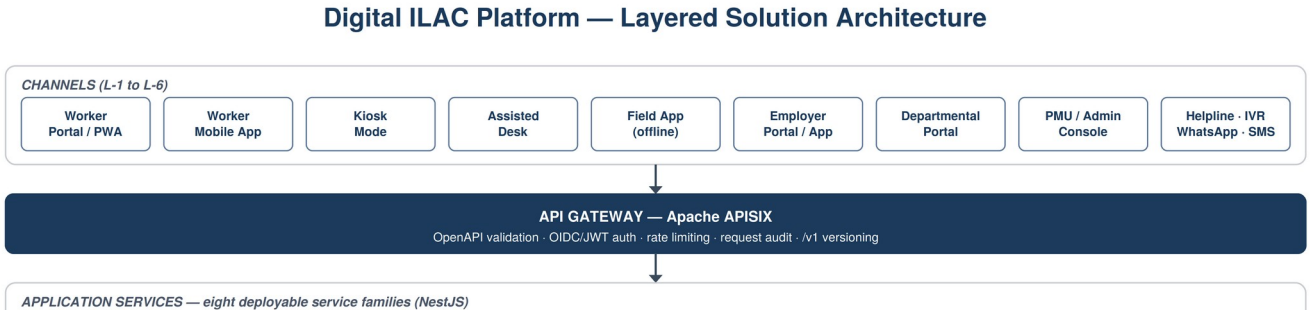


Figure 2.1 — Digital ILAC Platform layered architecture

A request flows top to bottom: a worker action at any channel becomes an authenticated API call at the gateway, is handled by exactly one application service, which composes the shared services (open a case, check consent, fetch through the Integration Hub, store a document, write an audit entry, raise a notification) and commits to the system-of-record database. Departmental systems are reached only through the Integration Hub — no module ever calls a department directly.

2.2.1 The ten cross-cutting platform services

These services are built once and inherited by all 22 modules — this is what allows the RFP's baseline requirements (assisted-mode parity, bilingual interfaces, configurable rules, role-based access, consent, audit, exportable reporting) to be verified as a platform property rather than promised separately by each of the 22 modules.

Service	Responsibility
Case Service	Issues the globally unique, channel-agnostic ILAC Case ID; runs the service clock from intake regardless of channel or mode; enforces the two-outcome closure rule.
Consent Manager	Records DPDP and biometric consent as separate, timestamped events with purpose codes; answers the consent gate in the Integration Hub before any outbound call.
Audit Ledger	Append-only table with a SHA-256 hash chain and a daily anchor written to write-once storage; the application account has insert-only rights — never update or delete.
Document Vault	Encrypted, S3-compatible object store with write-once retention for evidence, consent records and completion proofs, linked to the Case ID.

Service	Responsibility
Notification Engine (SM-14)	Accepts trigger events from any module, resolves recipients, dispatches through in-app, SMS, WhatsApp and IVR, tracks delivery and retries failures.
Integration Hub (SM-12)	The only path to any external system: adapter registry, consent gate, credential custody, geography harmonisation, retries, replay, reconciliation, health dashboard.
Master Data & Configuration (SM-19)	Versioned, effective-dated store for every configurable item across every module, with an approval workflow and full audit.
Worker Longitudinal History	Immutable, chronological per-worker timeline, fed by standardised events from every module, with role-scoped worker and staff views.
Household Master	Household record keyed independently of the worker, linking members and dependants and carrying the household baseline survey.
Reporting Layer (SM-13)	Reads from a replica on a defined refresh cadence; predefined and configurable reports and dashboards, under the same access rules as the live system.

Two further components underpin these services: the Identity and Access service (Keycloak or an equivalent custom implementation meeting the same controls, Section 8.2), and a durable workflow engine (Temporal) that turns every configured service clock, escalation rule and retry policy into a timer that survives process restarts and deployments — this is the mechanism, not just the policy, behind "no worker turned away."

2.2.2 Eight service families

The 22 modules are grouped into eight service families, each a separately deployable application with its own database schema and API contract, communicating only through published APIs and event contracts — no family reads another family's tables directly.

Service family	Modules	Primary loci
Registration & Records	SM-01	L-1, L-3, L-5
Entitlement & Convergence	SM-02 (Scheme Engine), SM-07 (convergence register)	L-1, L-5, L-6
Employment & Skilling	SM-05, SM-11, SM-15, SM-20, SM-21	L-1, L-2, L-4
Counselling & Learning	SM-08, SM-09, SM-10	L-1, L-5
Grievance & Service Requests	SM-03, SM-04, SM-17, SM-18	L-1, L-5, L-6
Communication	SM-06, SM-14	All
Administration & Monitoring	SM-16, SM-19, SM-22	L-6
MIS	SM-13	L-6

The Platform is deliberately built as a modular monolith of eight independently deployable applications, not as a fine-grained microservice estate. This is stated plainly because evaluators will look for it: at 300 concurrent access nodes, no single family needs independent scaling beyond horizontal replicas of the

whole family, and eight deployable units — each with a clear contract — can be operated, patched and extended by an in-house Authority team after exit, whereas a large microservice estate typically cannot be without the original team.

2.3 Data architecture

Store	Technology	Holds
System-of-record database	Relational database (Section 15 records the recommended engine)	Worker Master, Household Master, Case and case steps, Scheme Applications, CESS Ledger, Consent, Audit Ledger, configuration
Read replica	Same engine, readable secondary	Continuous replica used exclusively by the Reporting Layer and long-running exports — never by transactional writes
Semi-structured store	MongoDB replica set	Raw departmental payload snapshots, dump staging, mapping templates — never a system-of-record entity
Document Vault	MinIO (S3-compatible, self-hosted) or an approved India-region S3 service	Uploaded and fetched documents, live photographs, geo-tagged field media, generated certificates — with write-once retention on evidence
Secrets vault	Encrypted secrets table (AES-256-GCM) or HashiCorp Vault	Departmental API credentials, gateway keys, certificates — never readable without the master key, never logged
Cache	Redis (Valkey acceptable)	Session tokens, rate-limit counters, hot master data — loss is a performance event, never a data event

Aadhaar handling. The Aadhaar number is never stored in clear. Only a reference key derived through a keyed one-way function, plus the last four digits for display, is stored; every lookup uses the reference key. Bank account numbers are column-encrypted and masked by default. Full detail is in Section 8.3.4.

Immutable records. Worker Longitudinal History events, audit rows, closed transactional records and consent records are insert-only — a correction is always a new linked record, never an overwrite, so the history an auditor sees is never rewritten.

Spatial and fuzzy matching. Point-in-area address resolution (to assign a worker to the correct Service Area) and fuzzy name/address matching (for deduplication, Section 7.6) are delivered as functional requirements — the database platform recorded in Section 15 provides both through native geography types and full-text/trigram matching, without any GIS spatial-planning capability, which remains a Part 2 concern outside this scope.

2.4 Workflow engine and service clocks

Every long-running or time-bound behaviour in the Platform — a case lifecycle, a service clock, an integration retry, a scheduled re-evaluation — runs as a durable workflow on an open-source workflow engine (Temporal), self-hosted inside the Platform's own environments. This matters operationally: a service clock or a retry survives a server restart or a deployment, because its state is persisted by the engine, not held in application memory.

How the closure rule is technically enforced. The Case Service exposes exactly two closure operations — one requiring evidence of a completed service, the other requiring all five documented-next-step fields. There is no third closure path in the API, and no user interface, including the assisted desk and the field application, has a control that calls anything else. This is why "no worker turned away" is a system property, verified once in testing, rather than a policy that depends on 22 modules independently doing the right thing.

2.5 Identity, access and jurisdiction

Staff authenticate with username, password and mandatory multi-factor authentication (TOTP). Workers authenticate with mobile OTP, or a QR card-token scan at a kiosk or assisted desk. Employers authenticate with mobile OTP, gated behind APBOCWWB staff verification before job-posting access is activated.

Authorisation uses a four-dimension permission model — Module access, Action access (View/Create/Edit/Delete/Approve/Escalate), Centre scope, and Information visibility. A user's effective access is the union of their activated roles, scoped to their allocated centres; a centre-scoped Centre Data Entry Operator cannot see another centre's workers through any screen, report, export or API. This is enforced once, at the data-access layer, and applied identically by every module, including the Reporting Layer — no user interface is trusted to enforce access on its own.

2.6 Channel architecture

Every channel — worker portal, worker app, employer app, kiosk, assisted desk, field application, departmental portal, PMU console, helpline, IVR, WhatsApp and SMS — is a client of the same APIs, rendering the same form and workflow definitions. No channel has privileged access, and an assisted-mode screen is the self-service screen with an "acting on behalf of" context. A function added to self-service is available at the assisted desk on the same release — this is what makes assisted-mode parity structural rather than a separate development effort. Full channel detail is in Section 9.

2.7 Deployment architecture

2.7.1 Five environments

Environment	Purpose	Data	Runtime
Development	Continuous integration, developer testing, sandbox adapters	Synthetic only	Docker Compose / PM2
Staging / QA	Functional, regression, integration and performance testing	Synthetic and masked	PM2 or Kubernetes
UAT	Authority acceptance testing, training, fallback-route certification	Masked migration samples; full-volume migration mock	Kubernetes (proposed)
Production	Live service	Live	Kubernetes

Environment	Purpose	Data	Runtime
Disaster Recovery	Warm standby in a second Indian region	Continuously replicated	Kubernetes

Kubernetes is proposed for UAT, Production and DR because these environments must deliver load balancing, health-based restart, rolling deployment without downtime and regional failover — process management alone does not provide these. Full hosting topology, capacity sizing and the CSP recommendation are in Section 16.

2.8 Technology stack

Layer	Proposed	Equivalent alternative	Licence
Backend services	Node.js 22 LTS, NestJS, TypeScript (strict)	—	MIT / Apache-2.0
Workflow & service clocks	Temporal (self-hosted)	Database-backed state machine (Dev only)	MIT
API gateway	Apache APISIX	Kong Gateway (open-source)	Apache-2.0
Identity & access	Keycloak (OIDC, TOTP MFA)	Custom implementation on the same standards	Apache-2.0
System-of-record database	See Section 15 for the recommended engine and fallback	PostgreSQL/MSSQL 15+ with PostGIS and pg_trgm	As selected
Data access layer	Prisma ORM	TypeORM	Apache-2.0 / MIT
Semi-structured store	MongoDB (replica set)	JSON columns in the system-of-record database	SSPL
Document Vault	MinIO (S3 API, object lock)	India-region S3 on an approved cloud	AGPL-3.0
Secrets & credentials	Encrypted secrets table (AES-256-GCM)	HashiCorp Vault	As database / MPL-2.0
Cache & rate limiting	Redis	Valkey	BSD
Web, kiosk, assisted desk	React 18, TypeScript, Next.js (SSR, PWA)	React with Vite	MIT
Field mobile	React Native (Expo), (MSSQL)	Flutter with Drift	MIT
MIS & dashboards	Apache Superset (on the read replica)	Metabase	Apache-2.0 / AGPL-3.0
Container runtime	Docker; Kubernetes with Helm	PM2 on virtual machines (Dev/Staging)	Apache-2.0
Infrastructure as code	Terraform; Ansible	Either alone where the provider supports it	MPL-2.0 / GPL-3.0

Layer	Proposed	Equivalent alternative	Licence
Observability	OpenTelemetry, Prometheus, Grafana, Loki	—	Apache-2.0 / AGPL-3.0
Testing	Vitest, Supertest, Playwright, k6	—	MIT / AGPL-3.0

Every component other than the system-of-record database is open-source and self-hosted within the India-resident environments. APBOCWWB receives a perpetual, irrevocable, transferable, royalty-free licence to the bespoke Platform, and no component in the base scope carries a per-user, per-record, per-transaction or per-API-call charge.

2.9 Non-functional design

Quality	Design measures
Performance	Stateless services behind the gateway, connection pooling, read replica for all reporting, Redis caching of hot master data, cursor pagination, database indexes defined per query path.
Availability	Multiple replicas per service family, health/readiness endpoints, rolling zero-downtime deployments, database availability group with automatic in-region failover, DR in a second region.
Scalability	Horizontal scaling of every stateless component, family-level autoscaling, vertical database headroom plus a readable secondary, an expandable erasure-coded object store.
Security & privacy	Controls detailed in Section 8 — TLS 1.2+, encryption at rest, MFA, least privilege, jurisdiction scoping, consent gate, Aadhaar reference keys, masking, VAPT before every Go-Live.
Accessibility & language	WCAG 2.1 AA across all web surfaces, large-target kiosk layout, Telugu and English from a shared translation catalogue with database-level Telugu collation.
Maintainability	One language (TypeScript) end to end, contracts generated from code and validated in tests, strict compiler settings, architecture rules enforced in the monorepo.
Observability	Correlation IDs on every request, visible in the Audit Ledger, logs and the worker's own acknowledgement; per-service, per-adapter and per-service-clock metrics.

2.10 Architecture decisions at a glance

Decision	Chosen	Reason
Deployment granularity	Modular monolith, eight deployable families	Four-month delivery; operability by the Authority after exit; load profile does not require a finer split
Cross-cutting services	Ten shared platform services	The functional baseline is verified once, not 22 times; parity of manual and automated modes enforced in one place
Workflow engine	Temporal	Durable timers and retries are the actual

Decision	Chosen	Reason
		enforcement mechanism for service clocks and integration resilience
API gateway	Apache APISIX	Declarative OpenAPI validation, OAuth2/mTLS, rate limits and audit plugins out of the box
Identity	Keycloak (custom implementation acceptable)	Standards-based, audited, MFA out of the box; a smaller custom footprint is an accepted alternative
Document Vault	MinIO	Self-hosted, S3-compatible API, object lock for write-once evidence, no residency question
Field application	React Native (Expo) with MSSQL	Shared TypeScript types with the backend; mature, proven offline-sync library
Web framework	Next.js	Server-side rendering for low-end devices and full PWA support
Runtime for Production/DR	Kubernetes	Load balancing, health-based restart, rolling deploys and regional failover
Hosting	NIC / AP State Data Centre / MeitY-empanelled CSP, as approved	India residency with Authority approval; DR in a second Indian region (Section 16)

2.11 Runtime units and deployment composition

The eight service families, the platform worker pools and the supporting components below make up the complete runtime estate of one environment. Each family is built from one container image that starts in either API mode (serving HTTP) or worker mode (running Temporal workflows and activities), so a family scales its request handling and its background processing independently without becoming two applications.

Runtime unit	Replicas (Prod)	Mode	Notes
Family API pods (8 families)	3 to 8 each, autoscaled	api	Stateless. Readiness depends on database and Redis reachability, not on any departmental system.
Family worker pods (8 families)	2 to 6 each	worker	Temporal workers on a task queue named after the family. Scaled on task-queue backlog.
Integration Hub worker pool	4 to 10	worker	Adapters, dump ingestion and reconciliation. Egress only through fixed addresses.
Notification worker pool	2 to 6	worker	Dispatch, delivery tracking and broadcast fan-out for SM-14.
API gateway (APISIX)	3	data plane	Public subnet. Configuration held in version control and applied declaratively.

Runtime unit	Replicas (Prod)	Mode	Notes
Identity provider (Keycloak)	2	active-active	Backed by its own database schema. Realms for staff, worker and employer.
Temporal server	3 services, 2 replicas each	frontend, history, matching	Persistence in the system-of-record database engine under a separate schema.
Redis	3 nodes	primary plus 2 replicas, with sentinel	Cache and rate-limit counters only. No system-of-record data.
MongoDB	3 nodes	replica set	Raw payload snapshots and dump staging only.
MinIO (Document Vault)	4 or more nodes	distributed, erasure-coded	Object lock enabled on the evidence bucket.
Apache Superset	2	web plus worker	Connected to the read replica with a read-only login.
Observability stack	1 per environment	collector, Prometheus, Loki, Grafana	Retention in Section 16.12.

2.12 Repository and package structure

The whole Platform lives in one monorepo owned by APBOCWWB from the first commit (Section 14.2). Workspaces are managed with pnpm and task caching with Turborepo. The directory layout below is the structure the architecture rules enforce.

```
ilac-platform/
  apps/
    web-portal/                Next.js: worker, employer, kiosk, assisted desk, PMU console
    field-app/                 React Native (Expo) offline application
    worker-app/                React Native (Expo) worker and employer app
    gateway-config/            APISIX routes, plugins, OpenAPI bundles
  services/
    registration-records/      family 1: SM-01
    entitlement-convergence/   family 2: SM-02, SM-07
    employment-skilling/       family 3: SM-05, SM-11, SM-15, SM-20, SM-21
    counselling-learning/      family 4: SM-08, SM-09, SM-10
    grievance-requests/        family 5: SM-03, SM-04, SM-17, SM-18
    communication/             family 6: SM-06, SM-14
    administration-monitoring/ family 7: SM-16, SM-19, SM-22 (back end)
    mis/                       family 8: SM-13
  packages/
    platform/                  ten shared services as in-process NestJS modules
    contracts/                 OpenAPI 3.1 documents and generated TypeScript types
    domain-types/              enums, value objects, typed integration outcomes
    policy/                    policy decision point library
    sync-protocol/             change-set types shared by field app and server
    ui-kit/                    component library, Telugu and English catalogue
    testing/                   data generators, adapter conformance suite
  adapters/                    one package per source: descriptor, contract, mapping, mock, tests
  workflows/                   Temporal workflow and activity definitions
  db/                          migrations, seed data, data dictionary
  infra/                       terraform per environment, ansible, helm charts
  pipeline/                     CI/CD definitions, deposit manifest generator
  docs/                        architecture, runbooks, SOPs, manuals
```

Three rules are enforced in the build rather than by convention. A service package may import from `packages/*` and from itself, never from another service. Inside a service the dependency direction is api to application to domain, and domain imports nothing from the outer layers. An adapter package may import only `packages/domain-types`, `packages/contracts` and its own files. A violation of any of these fails the pull request, using dependency-cruiser rules held in the repository.

2.13 API design standards

Every endpoint in every family follows the same conventions, so a developer, a departmental integrator or a test author learns them once.

Aspect	Standard
Style	REST over HTTPS with UTF-8 JSON. Resource names are plural and kebab-case. An action that is not create, read, update or delete is a sub-resource, for example POST `/v1/cases/{id}/closures`.
Contract	One OpenAPI 3.1 document per family in `packages/contracts`. TypeScript types and request validators are generated from it, and a contract test fails the build if a running service diverges from the document.
Versioning	Major version in the path (`/v1`). Additive changes are not breaking. A breaking change creates `/v2`, and `/v1` stays available for at least two release cycles after consumers are notified.
Identifiers	A 26-character time-ordered `public_id` (Crockford base32) appears in URLs and payloads. Internal numeric keys never leave the database tier.
Errors	`application/problem+json` (RFC 9457) with `type`, `title`, `status`, `detail`, `instance`, a stable platform `code` (Annex C), `correlationId` and, for validation failures, `fieldErrors[]`. Messages are returned in the requested language.
Pagination	Cursor based (`limit`, `cursor`), default 50 and maximum 200, with a stable sort on `(created_at, id)`. Offset pagination is not offered on any list that can exceed 10,000 rows.
Filtering and sorting	`filter[field]=value` against an allow-list per resource and `sort=-created_at`. An unknown filter returns 400 and is never ignored.
Idempotency	The `Idempotency-Key` header is required on every POST that creates a case, application, ticket or enrolment. The key and a response hash are held for 24 hours; a replay returns the original response.
Concurrency	`ETag` on reads and `If-Match` on updates. A stale write returns 412. Field-application writes use the change-set protocol of Section 10.10.
Correlation	W3C `traceparent` is accepted and propagated. The gateway generates `X-Correlation-Id` when absent and returns it on every response. The same value appears in logs, Audit Ledger rows and the worker acknowledgement.
Time and locale	ISO 8601 in UTC on the wire and Indian Standard Time on screen. `Accept-Language` accepts `te` and `en`. Identifiers always use Western digits.
Rate limits	`RateLimit-Limit`, `RateLimit-Remaining` and `RateLimit-Reset` on every response. A refused call returns 429 with `Retry-After`.
File upload	Multipart upload to the Document Vault API with a size cap per document type, content-based type detection and an antivirus scan before the object becomes readable. The response carries `documentId` and the SHA-256 of the stored object.
Deprecation	`Deprecation` and `Sunset` headers on a deprecated route, with consumption by client reported in the Integration Health Dashboard.

A representative exchange follows. The request opens a case at an assisted desk and the second block shows the validation error returned when the closing step is incomplete.

```
POST /v1/cases
Authorization: Bearer <staff-jwt>
Idempotency-Key: 3f6c1a52-9d1e-4a7c-b0a4-1c6d9e2f7a10
Accept-Language: te
Content-Type: application/json

{ "workerId": "01JB8Z4Q2M7T9X3N5R1D0E6KHC", "moduleCode": "SM-04", "channel": "ASSISTED_DESK",
  "requestType": "GRIEVANCE", "centreId": "CTR-017", "mode": "assisted" }

201 Created
ETag: "1"   X-Correlation-Id: 7d1f0c5a-6a2e-4c0e-9a3f-2f4f7b1c9e11
{ "caseId": "ILC-26-0K3Q7M2V-8", "status": "OPEN", "clock": { "startedAt": "2026-11-03T05:42:11Z",
  "dueAt": "2026-11-10T11:30:00Z", "warnAt": "2026-11-08T07:30:00Z" } }
```

```
POST /v1/cases/ILC-26-0K3Q7M2V-8/closures        -> 422 Unprocessable Content
Content-Type: application/problem+json
{ "type": "https://ilac.ap.gov.in/problems/closure-incomplete",
  "title": "Documented next step is incomplete", "status": 422, "code": "CASE-4221",
  "detail": "Four of five required fields are present.",
  "correlationId": "7d1f0c5a-6a2e-4c0e-9a3f-2f4f7b1c9e11",
  "fieldErrors": [ { "field": "followUpRoute", "code": "REQUIRED", "message": "Select the follow-up
route" } ] }
```

2.14 API gateway configuration and route classes

The gateway is the only internet-facing surface (Section 8.3.7). Routes are declared in version control, generated in part from the OpenAPI documents, and applied by the pipeline. No route is edited by hand on a running gateway. Each route belongs to one class, and the class fixes authentication, limits and timeouts.

Class	Path pattern	Authentication	Limit (per minute)	Timeout
public-auth	`/v1/auth/otp/*`	None. OTP issue limited per mobile number and per source address.	5 per mobile per 10 min; 20 per address per 10 min	5 s
worker	`/v1/me/**`	Worker JWT from Keycloak	300 per user	15 s
staff	`/v1/**` (staff scopes)	Staff JWT carrying an MFA authentication claim	600 per user	30 s
employer	`/v1/employers/**`, `/v1/jobs/**`	Employer JWT	300 per user	15 s
inbound	`/v1/inbound/**`	Mutual TLS or OAuth2 client credentials	Per client contract	30 s
admin	`/v1/admin/**`	Staff JWT with administrator role, step-up MFA, VPN source range	120 per user	60 s
sync	`/v1/sync/**`	Staff JWT bound to a registered device	120 per device	120 s
health	`/healthz`, `/readyz`	Internal network only	None	2 s

```
routes:
  - id: reg-workers-read
    uri: /v1/workers/*
    methods: [GET]
    plugins:
      request-id:        { header_name: X-Correlation-Id, include_in_response: true }
      openid-connect:    { discovery: "https://idp.internal/realms/staff/.well-known/openid-
configuration",
                           bearer_only: true }
      limit-count:       { count: 600, time_window: 60, key_type: var, key: consumer_name,
                           policy: redis, rejected_code: 429 }
      request-validation: { header_schema: {...}, query_schema: {...} }   # generated from OpenAPI
      opentelemetry:     { sampler: { name: parent_base, options: { fallback: always_on } } }
    upstream:
      type: roundrobin
      discovery_type: kubernetes
      service_name: ilac-prod/registration-records-api:http
      timeout: { connect: 2, send: 30, read: 30 }
      checks: { active: { http_path: /readyz, healthy: { interval: 5, successes: 2 } } }
```

The gateway also carries a web application firewall rule set (OWASP Core Rule Set through the Coraza filter, or an equivalent accepted by the Authority), a response header policy (HSTS, `X-Content-Type-Options`, a strict `Content-Security-Policy` on web assets) and an access log in which every sensitive field is redacted before it leaves the gateway.

2.15 Internal structure of a service family

Every family follows the same four-layer structure. The layers, not the individual developers, decide where a given piece of code lives.

Layer	Holds	Rules
api	Controllers, DTOs, guards, OpenAPI decorators	Maps HTTP to use cases. Validates input. Calls the policy decision point. Contains no business rule and no data access.
application	Use cases, transaction boundaries, orchestration of shared services	One transaction per use case. Writes the audit row and the outbox row in the same transaction as the business change.
domain	Entities, value objects, state machines, rule evaluation	Plain TypeScript with no framework import and no I/O, so every rule is unit-testable in milliseconds.
infrastructure	Repositories, adapters, Temporal clients, object-store client	Implements the ports declared by the application layer. The only layer that imports Prisma or a network client.

Centre and jurisdiction scoping is applied once, in the data-access layer, so no screen, report, export or API can bypass it (Section 2.5). The scoping is a Prisma client extension that adds the principal's centre filter to every read, update and delete on a scoped model.

```
const READ_WRITE_OPS = new Set(['findMany','findFirst','findUnique','count','aggregate',
  'groupBy','update','updateMany','delete','deleteMany']);

export const scopedClient = (base: PrismaClient, ctx: RequestContext) =>
  base.$extends({
    query: {
      $allModels: {
        async $allOperations({ model, operation, args, query }) {
          if (SCOPED_MODELS.has(model) && READ_WRITE_OPS.has(operation)) {
            args.where = { AND: [args.where ?? {}, centreScope(model, ctx.principal)] };
          }
          return query(args);
        },
      },
    },
  });
```

A use case then performs the business change, the audit append and the event publication as one unit. If any of the three fails, none is committed, so there is no change without an audit row and no event without a change.

```
await this.prisma.$transaction(async (tx) => {
  const step = await this.cases.addStep(tx, caseId, input);          // domain rule enforced here
  await this.audit.append(tx, { actor, action: 'case.step.added', subject: caseId });
  await this.outbox.publish(tx, 'case.step.added', { caseId, stepId: step.publicId },
                            { aggregate: { type: 'case', id: caseId } });
});
```

2.16 Events and asynchronous processing

Modules coordinate through events written with the transactional outbox pattern. No message broker is introduced: the outbox relay and the Temporal task queues provide durable delivery, which keeps the operated component count within what the Authority's own staff can run (Section 2.2.2). Events carry identifiers and non-personal attributes only. A consumer that needs personal data fetches it from the owning family through its API under the consumer's own access rights.

```
{ "eventId": "01JB8Z6K4Q8M2V7T9X3N5R1D0E", "type": "case.closed.next_step", "version": 1,
  "occurredAt": "2026-11-03T05:42:11.482Z", "producer": "grievance-requests-api",
  "aggregate": { "type": "case", "id": "ILC-26-0K3Q7M2V-8", "sequence": 14 },
  "correlationId": "7d1f0c5a-6a2e-4c0e-9a3f-2f4f7b1c9e11",
  "actor": { "kind": "staff", "id": "usr_01JB8Z0M", "centre": "CTR-017" },
  "data": { "caseId": "ILC-26-0K3Q7M2V-8", "ownerRole": "DISTRICT_OFFICER", "targetDate": "2026-11-
10" } }
```

Property	Design
Atomicity	The outbox row is inserted in the business transaction. A rolled-back change publishes nothing.
Relay	A scheduled Temporal activity per family reads unpublished rows in `id` order in batches of 200 and writes each to the inbox of every subscribed consumer.
Delivery	At least once. Each consumer stores `eventId` in an inbox table under a unique key and discards a duplicate.
Ordering	Guaranteed per aggregate through the `sequence` number. A consumer that receives sequence 7 before 6 holds it for up to 60 seconds, then raises an alert.

Property	Design
Failure	Eight attempts with exponential backoff from 1 second to 10 minutes, then the event moves to a dead-letter table and an alert is raised. The TSU replays dead-lettered events from the operations console.
Retention	Published outbox rows are archived to the Document Vault in compressed batches after 30 days and retained under the data-class schedule of Section 8.3.5.
Schema evolution	Event types are versioned. Fields are only added within a version. A breaking change is a new type version, and producers publish both for one release cycle.

The full event catalogue, with producer and consumers for each type, is in Annex C.

2.17 Durable workflow catalogue

Every time-bound behaviour is a Temporal workflow. Workflow inputs carry identifiers only, and a payload codec encrypts anything else, so personal data does not sit in the Temporal history. Temporal history is held for 30 days. The business history that matters to an auditor is in the Platform's own tables and the Audit Ledger, never in Temporal. Workers are versioned by build identifier so that a deployment never changes the behaviour of a workflow already in flight, and every workflow is covered by a replay test in the pipeline.

Workflow	Started by	Timers and signals	Behaviour and retry policy
CaseLifecycle	`case.opened`	Warning at 75% of the clock, breach at 100%. Signals: `step.added`, `closure.requested`.	Owns the service clock. Starts Escalation on breach. Activity timeout 30 s, retries without limit from 1 s to 60 s.
Escalation	CaseLifecycle (child)	Next level after the configured interval. Signal: `acknowledged`.	Moves L1 to L2 to L3 to Board, notifying each owner. Full case history travels with each level.
DeferredFetch (F4)	Hub outcome SOURCE_UNAVAILABLE	Retry schedule per adapter. Promotion at 60% of the case clock.	Re-attempts the call. On success fills the value and clears the provisional flag. At the promotion point opens an F1 or F2 operator task.
AdapterSyncPull	Cron per source	Cadence from the adapter descriptor	Pull and delta sync for API sources. A missed window marks attributes stale.
DumpIngestion	File arrival	Validation timeout 2 h	Checksum, staging, validation, LGD mapping, identity match, load with lineage. A bad file is rejected whole.
Reconciliation	Daily per source, or on dump receipt	None	Compares held attributes with the source and classifies each difference (Section 5.7).
CutoverControl	Operator request	Evaluation window per state	Moves a source between Manual, Shadow, Canary and Automated primary only when the comparison metrics pass (Section 6.14).
SchemeReevaluation	Data-change events	Debounce of 30 s per household	Re-runs the Scheme Engine for the household and notifies on a changed result.
NotificationDispatch	`notification.requested`	Delivery-report wait per channel	Resolves recipient, channel and template. Sends, tracks the delivery report and retries or falls back to the next channel.

Workflow	Started by	Timers and signals	Behaviour and retry policy
BroadcastCampaign	Campaign approval	Rate window per provider	Fans a campaign out in batches within the SMS and WhatsApp provider throughput limits.
AuditAnchor	Cron, 00:05 IST	None	Seals the 16 ledger lanes and writes the anchor record to write-once storage.
Retention	Cron, nightly	None	Applies the retention schedule: tombstoning, archival and purge of expired temporary data.
SyncConflict	`sync.conflict.raised`	Resolution due date	Tracks the resolver queue and escalates an unresolved conflict.
ReportRefresh	Cron per mart	Cadence in Annex D	Refreshes reporting marts on the read replica and records the refresh time shown on every dashboard.

2.18 Caching and data-access rules

Redis holds only data whose loss is a performance event. The key patterns below are the complete list.

Key pattern	Content	TTL	Invalidation
`cfg:{type}:{key}:{ver}`	Published configuration item	Until superseded	`config.published` event evicts the previous version.
`lgd:{code}`	LGD hierarchy node and crosswalk	24 h	On master data publication.
`rbac:{userId}`	Policy snapshot for the principal	5 min	Role change increments `rbac:ver:{userId}`; the version is checked on every request, so a change applies at the next request.
`sess:{sid}`	Session and token metadata	Session lifetime	Logout, expiry or revocation.
`otp:{mobileHash}`	OTP state and attempt count	5 min	Successful verification.
`idem:{key}`	Idempotency record	24 h	None.
`rl:{route}:{principal}`	Rate-limit counter	Window length	Window expiry.
`scheme:cat:{ver}`	Compiled scheme-rule catalogue	Until superseded	Catalogue publication.

- **Connection pools.** Each API pod holds a pool of at most 20 connections and each worker pod at most 10. Replica maxima are set so that the sum stays below 70% of the database connection limit, leaving room for administration, replication and the reporting layer.
- **Statement timeouts.** 15 seconds by default, 30 seconds for administrator routes and 10 minutes for the reporting login only. A statement that exceeds its timeout is logged with its plan identifier.
- **No lazy loading.** Repositories return fully specified projections. Test suites assert the number of queries issued by each use case, so an N+1 pattern fails the build.

- **Reads and writes.** All transactional work uses the primary. The Reporting Layer, exports and Superset use the readable secondary only.
- **Index review.** Any new query on a table expected to exceed one million rows is submitted with its execution plan, and the index that serves it, in the same pull request.

2.19 Configuration management model

A configurable item is a versioned, effective-dated record in Master Data & Configuration (SM-19). The same lifecycle applies to every item type, so a change to a service clock is governed exactly like a change to a scheme rule.

Item type	Examples	Approval route
Form definition	Registration step fields, validation rules, labels in both languages	PMU Configuration Owner
Scheme rule	Eligibility criteria, benefit values, overlap rules	Scheme owner and PMU Head
Service clock	Targets, warning thresholds, escalation intervals, pause reasons	PMU Head
Escalation matrix	L1 to Board owners per module and district	PMU Head
Notification template	Message text per event, channel and language; DLT template identifier	PMU Configuration Owner
Fallback Register entry	Permitted classes per data element, evidence lists, cut-over state	PMU Head and TSU Head
Master data	LGD hierarchy, centres, service areas, document types, schemes	Data Steward

An item moves through five states: draft, in review, approved, published and superseded. Publication sets `effectiveFrom`, which may be in the future. A published version is immutable, and a rollback is the publication of the previous content as a new version. Environments are promoted by exporting a signed bundle (checksum and approver identity inside the bundle) from one environment and importing it into the next, so UAT-certified configuration is the configuration that reaches Production.

```
{ "type": "service_clock", "key": "grievance.l1", "version": 3, "effectiveFrom": "2026-12-01",
  "value": { "calendar": "ap-working-days", "target": { "amount": 5, "unit": "working_day" },
             "warnAtPercent": 75, "pauseOn": ["awaiting_worker_input"],
             "escalation": [ { "level": "L2", "atPercent": 100 }, { "level": "L3", "atPercent":
150 } ] },
  "approvedBy": "usr_01JB8Z0M", "bundleChecksum": "sha256:9c1b...e47a" }
```

2.20 Performance and capacity model

The load model below turns the 300-node floor into request rates the gateway and services are engineered for. The figures are design budgets. Response-time and throughput targets from the RFP service-level schedule replace them where the RFP states a number, and the performance tests of Section 13.12 verify them.

Step	Value	Derivation
Concurrent access nodes	300	RFP floor
Average interaction interval per node at peak	6 s	Planning assumption for assisted-desk and field use
Sustained interactive request rate	50 requests per second	300 / 6
Peak factor for the busiest 5 minutes	x 3	Morning opening, camp and mela surges
Peak interactive rate	150 requests per second	50 x 3
Background traffic (sync pushes, notifications, integration, reporting)	+ 40%	Planning assumption
Peak total at the gateway	210 requests per second	150 x 1.4
Engineered capacity	500 requests per second	About 2.4 times peak, matching the 500-node growth case

Interaction class	p95 server time	Note
Read: profile, status, list	400 ms	Served from the primary with indexed predicates; cursor pagination
Write: form step, case step, application	700 ms	Includes the audit append and outbox insert
Scheme evaluation for one household (38 schemes)	2.5 s	Compiled rules; fetched attributes read from cache and local tables
Platform overhead on an e-KYC call	300 ms	Excludes UIDAI processing time, which the Platform does not control
Dashboard widget on a pre-aggregated mart	3 s	Ad hoc exports run asynchronously and are delivered as files
Sync push of 100 change-sets	3 s	Server side; excludes media upload time
Registration wizard step transition on a 3G link	2 s end to end	Page weight budget in Section 9.13

2.21 Component baseline and patch policy

The baseline lines below are fixed at inception and recorded in the software bill of materials. The patch policy applies to every component in Production, including those the Authority will operate after exit.

Component	Baseline line	Upgrade rule
Node.js	22 LTS	Monthly patch releases. Move to the next LTS line inside the first year of operations.
NestJS, Prisma	Current stable major at inception	Minor updates monthly. A major update is a planned change through the Change Advisory Board.
System-of-record database	PostgreSQL 15 or higher, or SQL Server 2022 or higher, as confirmed at inception	Cumulative updates quarterly. Security fixes under the patch windows below.
Temporal, Keycloak, APISIX	Current stable release at inception	Quarterly review. Deployed through the staging and UAT environments first.

Component	Baseline line	Upgrade rule
Redis or Valkey, MongoDB, MinIO	Current stable release at inception	Quarterly review. Cluster nodes updated one at a time.
Kubernetes	A minor release supported by the upstream project	Stay within one minor release of the newest supported version.
React, Next.js, Expo SDK	React 18; current stable Next.js and Expo SDK at inception	Reviewed at each phase gate with the field-device test matrix.

Severity	Patch window	Scope
Critical (actively exploited or remote code execution)	48 hours	Emergency change. Gateway, identity provider, runtime and database first.
High	7 days	Standard change with accelerated approval.
Medium	30 days	Included in the next scheduled release.
Low	Next release train	Batched into the monthly dependency update.

3. Module-wise Compliance and Solution Approach (SM-01 to SM-22)

All 22 modules approved for the Digital ILAC Platform are delivered as feature packages on the ten shared platform services described in Section 2.2.1. Because those services carry the RFP's functional baseline — assisted-mode parity, Telugu and English interfaces, configurable rules, role-based and jurisdiction-based access, consent, immutable audit, exportable reporting — every module inherits that baseline automatically; it is verified once, not 22 times.

Sixteen of the 22 modules are built against an approved Module Specification. Where a specification was not supplied (SM-03, SM-07, SM-17, SM-18, SM-21, and parts of SM-01, SM-05, SM-15, SM-16, SM-20), a proposed design is presented below, built consistently on the same platform patterns, and marked for confirmation with APBOCWWB during the inception workshop. No phase commitment depends on that confirmation.

Compliance status. All 22 modules are marked Comply. SM-01 carries one qualifier: the face-match step of the Registration specification is a Part 2 AI-verification capability outside this scope, and is delivered in Part 1 as live photograph capture with logged operator verification instead (Section 15, Deviation D2). No other module carries a qualifier.

3.1 Module summary — SM-01 to SM-22

SM	Module	Phase	What it does	Key integrations
SM-01	Worker Registration, Household Baseline Survey & Worker Records	1	The 15-step registration journey; creates the lifelong Worker Master and linked Household Master; issues the BOCW Registration Number; annual renewal	UIDAI e-KYC, GSWS, e-Shram, EPFO, ESIC, NPCI, IT/NSDL, SADAREM, education departments
SM-02	Welfare-Scheme & Entitlement Module (Scheme Engine)	3	Evaluates a worker/household profile against all 38 catalogued schemes; applies 6 overlap rules; initiates, routes and tracks applications to Sanctioned/Rejected/Availed	GSWS scheme services, SERP, Food & Civil Supplies, Energy, Social Welfare, Health, WCD, RTGS, PFMS
SM-03	CSC & MeeSeva Service Request & Tracking	3	Raises citizen-service requests (certificates, Aadhaar updates, ration changes) on the worker's behalf; tracks to output document	MeeSeva/CSC where authorised; GSWS, CRS, Revenue for auto-fetch
SM-04	Grievance Redressal & Multi-Channel Complaint Tracking	1	Intake from every channel with one Case ID; PMU triage; L1/L2/L3/Board resolution levels; real-time requester tracking	SMS, WhatsApp, IVR, State grievance system where authorised
SM-05	Work Request, Job Matching & Daily-Wage	1	Employer registration & job posting; worker enrolment & engagement confirmation;	APBOCWWB cess-payer records, GST, e-Shram/NCS

SM	Module	Phase	What it does	Key integrations
	Employment Tracking		daily-wage/labour-adda tracking; centre assistance-request token queue	
SM-06	Opportunity & Event Calendar, Mapping & Referral	1	One calendar of meetings, training, melas and scheme windows; matches opportunities to worker profile; staff-initiated referrals with tracked outcome	Notification Engine; video-conferencing provider
SM-07	Departmental Convergence, Register & Committee Record	2, 3	Register of every outbound departmental request (API or manual); mandal/district/state convergence committee support with minutes and decisions	All 50 notified sources via the Integration Hub
SM-08	Student & Education Counselling Management	3	Organised counselling for workers' children — stream selection, scholarships, examinations — with follow-up cases to scheme applications	School/Intermediate/Technical/Higher Education, APCFSS
SM-09	Learning Management System & IEC Dissemination	2, 3	On-demand courses with assessments and certificates; targeted broadcast of Information, Education & Communication material	Video streaming (Document Vault), Notification Engine
SM-10	Employment, Self-Employment & Enterprise Counselling	3	Group counselling and one-to-one facilitation for wage employment, self-employment and enterprise formation, with tracked referrals	SM-02, SM-05, SM-11, SM-18, SM-21 internally
SM-11	Work-Site Visit, Reskilling, Upskilling & RPL Scheduling	1	Worksite visits with geo-tagged evidence; training/skill events; RPL assessment batches; per-worker skilling pathway	APSSDC / Skill India Digital Hub, Technical Education
SM-12	Common API & Systems Integration Framework (Integration Hub)	1, 2	The single path to every external system — adapters, credential custody, geography harmonisation, retries, reconciliation (see Section 5)	All 50 sources + Government of India systems
SM-13	MIS, Data Analytics & Decision Support	2	Predefined and configurable reports, role-scoped dashboards, the 4E programme funnel, threshold alerts, from a read-optimised replica	Internal — fed by every operational module
SM-14	Communication, Notification, Alert & Broadcast	1, 2	Single engine for every worker/staff/employer message across in-app, SMS,	DLT-registered SMS gateway, WhatsApp Cloud API, IVR

SM	Module	Phase	What it does	Key integrations
	Management		WhatsApp, IVR and helpline, plus targeted broadcasts	
SM-15	Job Mela & Event Facilitation	2	Pre-registration, on-site check-in, interview scheduling, offers and placement follow-up for job melas and camps, fully offline-capable	SM-01, SM-02, SM-04, SM-05, SM-11 internally
SM-16	Monitoring of ILAC Functions & Functionary Work Turnout	4	Staff attendance and derived work turnout from audited transactions; per-centre daily function checklist and scorecard	Internal; optional HRMS dump for staff master
SM-17	Para-Legal Services Facilitation at the ILAC	4	Legal-aid intake (wages, accidents, disputes), guidance from a rostered para-legal volunteer, referral to DLSA/legal-aid clinics	State legal-services/labour-department interfaces where notified
SM-18	Common Agency Service Facilitation Interface	1, 4	Generic facilitation for bank, insurer, post-office and other agency services with tracked outcomes	NPCI, e-Shram/NDUW, PMSBY/PMJJBY enrolment channels
SM-19	User, Role & Configuration Administration	1, 4	Users, roles, centres, jurisdictions; the configuration console for every configurable item across every module	HRMS dump for employee-master reconciliation
SM-20	Construction-Site Catalogue & Worker Grouping for Skilling	3	Catalogue of construction sites in each centre's jurisdiction; rule-based grouping of workers for training, RPL and camps	SM-05, SM-11, SM-15 internally; APSSDC batch export
SM-21	Construction Worker Livelihood Groups & Federation	3	Formation and functioning of worker livelihood groups and their mandal/district federations, with scheme and agency linkage	MEPMA and SERP SHG data for cross-reference
SM-22	Worker, Employer & Departmental Applications, Portals & Dashboards	2	The user-facing surfaces through which every module is used — worker/employer apps, departmental portal, PMU console, kiosk, assisted desk, field app	Through the API gateway only

The table above is a quick-reference index. Each module is detailed individually in Section 3.2.

3.2 Module-by-module detail — SM-01 to SM-22

Every module below inherits the platform baseline set out in Section 2 — Telugu/English interfaces, assisted-mode parity, role and jurisdiction-scoped access, consent capture, immutable audit, and

exportable reporting — so that baseline is not repeated 22 times. What follows for each module is its distinct purpose, its key functions, its main integrations, and how it behaves when an integration is unavailable.

SM-01 — Worker Registration, Household Baseline Survey & Worker Records

Purpose. Creates the single, lifelong Worker Master and the linked Household Master that every other module builds on, and issues the permanent BOCW Registration Number.

- A 15-step guided registration wizard, in Telugu or English, completing identity, address, employment, family, and disability/special-category details in one sitting.
- UIDAI e-KYC pulls Name, Date of Birth, Gender and Address automatically; the operator confirms rather than retypes them.
- Household Baseline Survey captured alongside registration, linking dependants, school-going children and household assets — the foundation for scheme convergence in SM-02.
- Annual renewal wizard that re-touches only the fields likely to change year on year, not a full re-registration.
- Live photograph capture with an operator Verified/Not Verified decision against the e-KYC photograph (see Deviation D2, Section 15.2, for why this replaces an automated face-match).

Integrations: UIDAI e-KYC, GSWS (caste/income/ration), e-Shram, EPFO, ESIC, NPCI (bank validation), Income Tax/NSDL, SADAREM, education departments, CRS.

Fallback: Any field that cannot be fetched is captured on the certified F1 (document-evidenced) route and re-verified automatically once the interface recovers (Section 6.3); registration is never blocked.

Technical specification. SM-01 owns the Worker Master and Household Master data and issues the BOCW Registration Number. It is the only module that writes identity data.

Aspect	Specification
Family and schema	Registration & Records family, schema `reg`. Aadhaar numbers are held only in the Aadhaar Data Vault (Section 8.15). Every other table holds the reference key and the last four digits.
Principal entities	• `worker`: Worker Master; the BOCW Registration Number is immutable once issued • `worker_identity_ref`, `worker_address` (LGD coded), `worker_bank_account` (account number column-encrypted), `worker_employment_profile`, `worker_nominee` • `household`, `household_member`, `household_survey` • `registration_application` (wizard draft and step state), `renewal`, `worker_alias_number`, `merge_event`
Key APIs	• `POST /v1/registrations` starts a draft and returns its identifier • `PUT /v1/registrations/{id}/steps/{n}` saves one step after server-side validation • `POST /v1/registrations/{id}/ekyc` runs OTP e-KYC through the Integration Hub • `POST /v1/registrations/{id}/submit` issues the Registration Number • `GET /v1/workers/{id}` and `GET /v1/workers/{id}/360` return the record and the 360-degree view • `POST /v1/workers/{id}/renewals` and `POST /v1/households/{id}/survey`

Aspect	Specification
Events	Emits `worker.registered`, `worker.updated`, `worker.renewed`, `household.updated`, `worker.merged`. Consumes `consent.granted` and `integration.attribute.fetched`.
Business rules and validations	● The Registration Number is assigned only on the server, from one sequence, inside the submit transaction. A device or client can never supply it. ● Eligibility checks come from configuration: minimum and maximum age on the application date and minimum days of construction work in the preceding 12 months, as set by the Act and Rules and confirmed at inception. ● The Aadhaar reference key is unique. A second record presenting the same key is routed to the merge review queue, not rejected. ● A mobile number used on more than a configured number of records raises a review flag. ● Biometric consent must exist before the photograph step. The operator records Verified or Not Verified with a reason (Deviation D2). ● A bank account is accepted as validated only on an NPCI response. Otherwise the field is held as provisional under F1 or F2. ● A worker always has a household, created as a single-member household where none exists.
Configuration items	Registration form definition per step; eligibility thresholds; document checklist per worker category; mobile reuse threshold; renewal field set; verification status vocabulary.
Non-functional design	● The wizard saves every step, so a dropped session resumes at the last saved step. ● One question per screen with a large-type read-back before submit. ● Step save p95 700 ms; Registration Number issue under 50 ms. ● Pre-registration is the only part of this module captured offline (Section 10.1).

SM-02 — Welfare-Scheme & Entitlement Module (Scheme Engine)

Purpose. Evaluates every registered worker and household against the full 38-scheme catalogue and turns eligibility into tracked applications.

- A configurable rule engine — schemes, eligibility criteria and benefit values are data, not code, so a new Government Order is a configuration change, not a release.
- Automatic re-evaluation whenever a worker's or household's underlying data changes (a birth, a school enrolment, an income-certificate update).
- Six overlap rules prevent duplicate or conflicting benefit claims across schemes with shared eligibility criteria.
- Application initiation, routing to the owning department, and status tracking through Applied → Sanctioned/Rejected → Availed.
- A worker-facing Eligible / Applied / Availed view so a worker always knows where they stand without asking an operator.

Integrations: GSWS scheme services, SERP, Food & Civil Supplies, Energy, Social Welfare, Health & Family Welfare, Women & Child Development, RTGS, PFMS.

Fallback: Where a department has no submission API, the application is packaged for manual desk submission (Class F3) and its status is entered manually until an API exists — the worker's Applied status and service clock are unaffected.

Technical specification. SM-02 is the most rule-intensive module. Its rules are data, compiled once per published catalogue version, and every result can be explained with the rule version that produced it.

Aspect	Specification
Family and schema	Entitlement & Convergence family, schema `ent`.
Principal entities	● `scheme`, `scheme_version`, `scheme_rule_version` (rule document), `overlap_rule` ● `scheme_evaluation` (one row per household or worker, scheme and rule version), `eligibility_explanation` ● `scheme_application`, `application_status_history`
Key APIs	● `GET /v1/schemes` and `GET /v1/schemes/{id}` ● `GET /v1/workers/{id}/entitlements` returns the Eligible, Applied and Availed view ● `POST /v1/scheme-evaluations` forces re-evaluation of a household ● `POST /v1/scheme-applications` initiates an application ● `POST /v1/scheme-applications/{id}/status` records a status returned by the department ● `GET /v1/scheme-applications/{id}/pack` produces the manual desk pack for F3 submission
Events	Emits `scheme.evaluated`, `scheme.application.submitted`, `scheme.application.status_changed`. Consumes `worker.updated`, `household.updated`, `worker.merged`, `integration.attribute.fetched`.
Business rules and validations	● Each scheme returns ELIGIBLE, NOT_ELIGIBLE or NEEDS_REVIEW, with the satisfied and failed criteria listed. ● NEEDS_REVIEW is returned when a required attribute is missing, stale beyond its freshness limit, or in conflict with the worker's declaration. ● The 6 overlap rules run after eligibility and are expressed with the rule types of Section 3.8. The six catalogue rules are validated with APBOCWWB at milestone M9. ● Application status moves Applied, then Sanctioned or Rejected, then Availed. Only the department can cause Sanctioned or Rejected; the Platform records the status it reads back. ● PM Kisan and Annadata Sukhibhava are evaluated from landholding data and held at NEEDS_REVIEW until RTGS data confirms. ● The two inactive schemes are never evaluated and remain for reconciliation.
Configuration items	Scheme catalogue (38); eligibility expressions; benefit values; overlap rule sets; application document checklists; owning department and clock per scheme; attribute freshness limits.
Non-functional design	● Evaluation p95 2.5 s for one household. The compiled catalogue is cached by version. ● Old rule versions are retained so a past result can be reproduced. ● Baseline evaluation of all migrated households runs as a batch on the worker pool and completes within 24 hours of the migration load. ● Re-evaluation is debounced for 30 seconds per household.

SM-03 — CSC & MeeSeva Service Request & Tracking

Purpose. Lets a Centre Data Entry Operator raise common citizen-service requests — certificates, Aadhaar updates, ration corrections — on a worker's behalf, without sending them to a separate CSC/MeeSeva counter.

- A catalogue of supported service types, each with its own document checklist and expected turnaround.

- Status tracking from Submitted through to the output document being available for the worker.
- Direct linkage to the worker's Case ID, so a certificate request shows up in the same timeline as everything else.

Integrations: MeeSeva/CSC where an authorised channel exists; GSWS, CRS and Revenue for auto-fetch of source documents.

Fallback: Manual desk submission (Class F3) with a physical or scanned acknowledgement, standard across services with no live submission API.

Technical specification. SM-03 treats every citizen service as a request with a catalogue entry, so a new certificate type is configuration and not code.

Aspect	Specification
Family and schema	Grievance & Service Requests family, schema `grv`.
Principal entities	• `service_request` (links to a Case ID), `service_catalogue_item`, `service_request_document`, `service_request_status_history`
Key APIs	• `GET /v1/service-catalogue` • `POST /v1/service-requests` • `GET /v1/service-requests/{id}` • `POST /v1/service-requests/{id}/documents` • `POST /v1/service-requests/{id}/status` • `POST /v1/service-requests/{id}/output` attaches the output document
Events	Emits `service_request.opened`, `service_request.status_changed`, `service_request.output_ready`. Consumes `consent.granted`.
Business rules and validations	• A request cannot be submitted until every mandatory document in its catalogue checklist is attached. • Source documents are auto-fetched from GSWS, CRS or Revenue where an interface and consent exist, and attached with source and fetch time. • Where no submission API exists the request is an F3 manual desk submission with a scanned or physical acknowledgement stored in the Document Vault. • A request closes only through the two closure operations of the Case Service.
Configuration items	Service catalogue entries with document checklist, expected turnaround and owning department; status vocabulary per service; clock per service.
Non-functional design	• Status entry is a staff action for services without a status API, and the same screen is used for both routes. • Documents are limited to 5 MB each and scanned before use.

SM-04 — Grievance Redressal & Multi-Channel Complaint Tracking

Purpose. Gives every worker one way to raise a complaint from any channel, and gives APBOCWWB one place to see and resolve it.

- Intake from the app, kiosk, assisted desk, helpline, WhatsApp, SMS or a field officer visit — every channel writes to the same ticket with the same Ticket ID.

- PMU triage assigns severity and routes to the right owner without the worker needing to know the internal structure.
- L1 / L2 / L3 / Board resolution levels (Section 11.2) with automatic escalation on breach of the configured service clock.
- Real-time status tracking for the requester by Ticket ID, on any channel, at any time.

Integrations: SMS, WhatsApp and IVR for intake and updates; the State grievance system, where APBOCWWB directs an interface to be built.

Fallback: A grievance raised on a paper register at an offline centre is keyed in as an assisted transaction with the clock back-dated to the paper intake time — never lost, never re-started.

Technical specification. SM-04 uses the same ticket structure for grievances and for support tickets, which is why the central helpdesk in Section 11 needs no separate product.

Aspect	Specification
Family and schema	Grievance & Service Requests family, schema `grv`.
Principal entities	• `ticket` (type GRIEVANCE or SUPPORT), `ticket_event` (append-only), `ticket_assignment`, `ticket_attachment`, `triage_rule`
Key APIs	• `POST /v1/tickets` • `GET /v1/tickets/{id}` and `GET /v1/tickets?filter[status]=...` • `POST /v1/tickets/{id}/assignments` • `POST /v1/tickets/{id}/escalations` • `POST /v1/tickets/{id}/comments` • `POST /v1/tickets/{id}/resolution` • `GET /v1/tickets/{id}/track` public status by Ticket ID and verified mobile number
Events	Emits `ticket.opened`, `ticket.assigned`, `ticket.escalated`, `ticket.resolved`, `clock.warning`, `clock.breached`. Consumes `notification.delivered`.
Business rules and validations	• Intake from SMS, WhatsApp, IVR, kiosk, app, assisted desk and field officer all create the same ticket with the same Ticket ID format. • A paper-register intake is keyed as an assisted transaction with `receivedAt` set to the paper time. The clock starts at that time, not at keying. • Triage assigns severity and owner from `triage_rule` without the worker supplying any internal category. • Levels L1, L2, L3 and Board escalate automatically on breach, and the ticket history moves with the escalation. • Resolution requires a resolution code and a note. A ticket reopened within a configured window continues the original clock record.
Configuration items	Ticket categories, severity matrix, triage rules, escalation matrix, clock per category and severity, resolution codes, channel templates.
Non-functional design	• Status lookup by Ticket ID is rate limited and returns only status and the next step. • Ticket events are insert-only. A correction is a new event. • Inbound WhatsApp and SMS messages are matched to a worker by verified mobile number, never by free text.

SM-05 — Work Request, Job Matching & Daily-Wage Employment Tracking

Purpose. Connects registered workers to employment — both formal job postings and day-to-day labour-adda work — and gives centres a token queue for walk-in assistance requests.

- Employer registration and verification, followed by job posting with role, wage, location and duration.
- Worker enrolment against a posting and employer-side engagement confirmation, closing the loop from posting to placement.
- Daily-wage and labour-adda attendance/engagement tracking for informal, day-to-day work.
- A Work Request token queue at the centre — a distinct object from a job posting — for a worker who simply needs assisted-desk help.

Integrations: APBOCWWB cess-payer records for employer cross-verification, GST for contractor validation, e-Shram/National Career Service for job-market reach.

Fallback: Engagement can be confirmed by an operator from a signed paper slip where an employer has no app access, preserving the same Registration-Number-linked record.

Technical specification. SM-05 separates two objects that the specifications both call a work request: the employer job posting and the worker assistance token.

Aspect	Specification
Family and schema	Employment & Skilling family, schema `emp`.
Principal entities	● `employer`, `employer_verification`, `job_posting`, `job_enrolment`, `engagement` ● `muster_entry` (daily wage and labour adda attendance) ● `work_request` (assistance token) with `token_queue_position`
Key APIs	● `POST /v1/employers` and `POST /v1/employers/{id}/verification` ● `POST /v1/jobs`, `GET /v1/jobs?filter[district]=...` ● `POST /v1/jobs/{id}/enrolments` ● `POST /v1/engagements` confirms engagement by Registration Number ● `POST /v1/musters` and `GET /v1/musters?date=...` ● `POST /v1/work-requests` and `POST /v1/work-requests/{id}/call-next`
Events	Emits `employer.verified`, `job.posted`, `job.enrolled`, `engagement.confirmed`, `muster.recorded`. Consumes `worker.registered`, `worker.updated`.
Business rules and validations	● An employer cannot post a job until APBOCWWB staff record a verification decision. The Platform records the decision and does not make it. ● Employer contact numbers of a worker are visible to the employer only after that worker enrols against the posting. ● Engagement is confirmed by Registration Number and is unique per worker, posting and date. ● Employer validation cross-checks cess-payer records and, for contractors, GST. ● A work request token is numbered per centre per day and the queue order is by arrival, with priority categories set by configuration.
Configuration items	Job roles and wage bands, token categories and priority, verification checklist, muster fields.

Aspect	Specification
Non-functional design	● A signed paper slip is accepted for engagement confirmation where an employer has no app access, with the slip scan attached. ● Posting search uses indexed district, trade and date filters. No free-text search on personal fields.

SM-06 — Opportunity & Event Calendar, Mapping & Referral

Purpose. One calendar for every meeting, training session, mela and scheme window, matched to the right workers and used to make and track staff referrals.

- A single calendar surface across all modules — no separate calendar per module or per district.
- Opportunity matching against a worker's profile (location, skill, eligibility) so outreach is targeted, not broadcast blindly.
- Staff-initiated referral between modules (for example, a counsellor referring a worker to SM-11 skilling) with a tracked outcome, not just a note.

Integrations: The Notification Engine (SM-14) for reminders; a video-conferencing provider where sessions are hosted remotely.

Fallback: Referrals and calendar entries can be created and read entirely offline through the field application (Section 10).

Technical specification. SM-06 owns the single calendar and the referral object that other modules create.

Aspect	Specification
Family and schema	Communication family, schema `com`.
Principal entities	● `calendar_event` (meeting, training, mela, scheme window), `event_audience`, `opportunity_match`, `referral`, `referral_outcome`
Key APIs	● `POST /v1/calendar-events` and `GET /v1/calendar-events?from=...&to=...` ● `GET /v1/workers/{id}/opportunities` ● `POST /v1/referrals` ● `POST /v1/referrals/{id}/outcome` ● `GET /v1/referrals?filter[status]=OPEN`
Events	Emits `calendar.event_published`, `referral.created`, `referral.outcome_recorded`. Consumes `worker.updated`, `scheme.evaluated`.
Business rules and validations	● Matching compares event audience rules (district, trade, scheme eligibility) with the worker profile and writes `opportunity_match` rows, so outreach lists can be reproduced. ● A referral names the target module, the receiving role and a due date. It cannot be closed without an outcome code. ● A referral to another module creates a case in that module and carries the Case ID back. ● A calendar event with attendance tracking must carry a venue or a meeting link.
Configuration items	Event types and audience rule templates, referral target catalogue, outcome codes.
Non-functional design	● Calendar reads and referral creation both work offline in the field application (Section 10.1). ● Reminders are raised through the Notification Engine and not by this module.

SM-07 — Departmental Convergence, Register & Committee Record

Purpose. The formal record of every outbound request the Platform makes to a department — whether by API or manually — and the support tooling for mandal, district and state convergence committee meetings.

- One register entry per departmental request, whatever the channel, so nothing sent to a department is untracked.
- Committee agenda preparation, attendance, minutes and decision recording at mandal, district and state levels.
- Decisions taken in committee link back to the worker cases that prompted them, closing the loop from meeting to outcome.

Integrations: All 50 notified sources, through the Integration Hub (Section 5) — SM-07 is the register of what the Hub does, not a separate integration path.

Fallback: Not applicable in the usual sense — SM-07 records both automated and manual departmental interactions identically, which is exactly its purpose.

Technical specification. SM-07 is the register of what the Integration Hub and the manual routes did. It is therefore populated by the Hub as well as by users.

Aspect	Specification
Family and schema	Entitlement & Convergence family, schema `ent`.
Principal entities	• `convergence_request` (one row per outbound departmental request, API or manual) • `committee`, `committee_meeting`, `agenda_item`, `attendance`, `minute`, `decision`, `decision_case_link`
Key APIs	• `GET /v1/convergence-requests?filter[department]=...` • `POST /v1/convergence-requests` for a manual (F3) request • `POST /v1/committees/{id}/meetings` • `POST /v1/meetings/{id}/minutes` • `POST /v1/meetings/{id}/decisions` • `GET /v1/decisions?filter[caseId]=...`
Events	Emits `convergence.request_logged`, `committee.decision_recorded`. Consumes `integration.call.completed`.
Business rules and validations	• Every Hub call and every manual departmental submission writes one register row with request time, mode, outcome code and the case it served. • A register row is immutable. A later status is a new linked row. • A committee decision that refers to a worker case is linked to the case and appears on the worker timeline. • Departmental users see only the register rows addressed to their own department.
Configuration items	Department master with nodal officers, committee types and levels (mandal, district, state), agenda templates.
Non-functional design	• Register writes are asynchronous from the Hub and never delay a worker-facing call. • Register volume is expected to exceed the case volume, so the table is partitioned by month.

SM-08 — Student & Education Counselling Management

Purpose. Organised counselling for the children of registered workers — stream selection, scholarships, competitive-exam guidance — connected to the scheme system rather than run as a separate exercise.

- Session scheduling and attendance for one-to-one or group education counselling.
- Structured guidance content by stage (post-10th, post-12th, competitive exams, higher education).
- Direct follow-up into SM-02 scheme applications where counselling identifies a scholarship or fee-support scheme the household is eligible for.

Integrations: School, Intermediate, Technical and Higher Education departments; APCFSS College Fee Database.

Fallback: Counselling itself is inherently an assisted, in-person or phone process — captured directly, no external dependency to fall back from.

Technical specification. SM-08 connects counselling to scholarship and fee-support schemes through a referral into SM-02 instead of a separate follow-up list.

Aspect	Specification
Family and schema	Counselling & Learning family, schema `cnl`.
Principal entities	• `counselling_session` (type EDUCATION), `counselling_attendance`, `student_profile`, `guidance_content`, `counselling_outcome`
Key APIs	• `POST /v1/counselling-sessions` • `POST /v1/counselling-sessions/{id}/attendance` • `POST /v1/students` and `GET /v1/students/{id}` • `GET /v1/guidance-content?filter[stage]=POST_10` • `POST /v1/counselling-sessions/{id}/outcome`
Events	Emits `counselling.session_held`, `counselling.outcome_recorded`, `referral.created`. Consumes `household.updated`.
Business rules and validations	• A student profile belongs to a household member under 18 or a dependant in education. It is created from the household survey and updated at each session. • An outcome that identifies a scholarship creates a referral to SM-02 with the student and household attached. • Group sessions record attendance per attendee, and an absent registered attendee is not counted as served.
Configuration items	Education stages, guidance content, stream and exam catalogues, outcome codes.
Non-functional design	• Student data is limited to what the counselling needs. Marks and school records are read from the education departments, with consent, and are not copied beyond the session record. • The college fee lookup uses the APCFSS database through the Hub (dump or API as notified).

SM-09 — Learning Management System & IEC Dissemination

Purpose. Gives workers on-demand courses with assessments and certificates, and gives APBOCWWB a way to push targeted Information, Education & Communication material to the right audience.

- A course library with video, document and quiz content, resumable across sessions and devices.
- Assessment and certificate issuance on course completion, visible on the worker's own profile.
- Targeted broadcast of IEC material by district, scheme, or worker category rather than a single undifferentiated blast.
- Offline playback of downloaded course content through the field and worker apps (Section 10).

Integrations: The Document Vault for video/content storage; the Notification Engine for course and broadcast alerts.

Fallback: Content already downloaded plays fully offline; new content and broadcast delivery queue for the next connected session.

Technical specification. SM-09 stores course media in the Document Vault and streams it with byte-range requests, so a worker on a weak connection can resume a lesson.

Aspect	Specification
Family and schema	Counselling & Learning family, schema `cnl`.
Principal entities	● `course`, `course_module`, `lesson` (video, document, quiz), `course_enrolment`, `lesson_progress` ● `assessment`, `assessment_attempt`, `certificate`, `iec_content`, `iec_audience_rule`
Key APIs	● `GET /v1/courses` and `GET /v1/courses/{id}` ● `POST /v1/courses/{id}/enrolments` ● `PUT /v1/course-enrolments/{id}/progress` ● `POST /v1/assessments/{id}/attempts` ● `GET /v1/certificates/{id}` returns a signed, verifiable certificate ● `POST /v1/iec-contents` and `POST /v1/iec-contents/{id}/publish`
Events	Emits `course.completed`, `certificate.issued`, `iec.published`. Consumes `worker.registered`.
Business rules and validations	● Progress is stored per lesson with a position marker, and the position merges by the highest value when the same worker learns on two devices. ● A certificate is issued only when every required lesson is complete and the assessment score meets the configured pass mark. ● A certificate carries a verification code resolvable at a public verification URL that returns validity and course name only. ● IEC publication to an audience goes through the Notification Engine as a broadcast and respects consent and language preference.
Configuration items	Courses, pass marks, certificate template, IEC categories, audience rules, supported video renditions.
Non-functional design	● Video is delivered in two renditions (low-bandwidth and standard) with byte-range support and optional download for offline playback. ● Downloaded content in the apps is stored in the encrypted local store.

SM-10 — Employment, Self-Employment & Enterprise Counselling

Purpose. Group and one-to-one counselling that turns a worker's employment interest into a tracked referral — to wage employment, self-employment, or enterprise formation.

- Group counselling session scheduling and attendance, alongside one-to-one facilitation for more involved cases.
- Structured pathways toward wage employment, self-employment or small-enterprise formation, each with its own guidance content.
- Tracked referral into SM-02 (schemes), SM-05 (jobs), SM-11 (skilling), SM-18 (agency services) or SM-21 (livelihood groups), so a counselling outcome is never just a conversation with no record.

Integrations: Internal only — SM-02, SM-05, SM-11, SM-18 and SM-21.

Fallback: Not applicable — an in-person counselling function with no external interface to depend on.

Technical specification. SM-10 produces referrals and nothing else. Every counselling outcome ends in a tracked hand-off.

Aspect	Specification
Family and schema	Counselling & Learning family, schema `cnl`.
Principal entities	• `counselling_session` (types GROUP and ONE_TO_ONE), `pathway` (WAGE, SELF_EMPLOYMENT, ENTERPRISE), `pathway_step`, `counselling_outcome`
Key APIs	• `POST /v1/counselling-sessions` • `POST /v1/counselling-sessions/{id}/pathway` • `POST /v1/counselling-sessions/{id}/outcome` • `GET /v1/workers/{id}/pathways`
Events	Emits `counselling.session_held`, `referral.created`. Consumes `scheme.evaluated`, `job.posted`.
Business rules and validations	• An outcome must carry at least one referral target (SM-02, SM-05, SM-11, SM-18 or SM-21) or a documented reason for none. • A pathway step has an owner, a due date and a status, and open steps appear on the worker timeline. • A worker can belong to one active pathway of each type at a time.
Configuration items	Pathway templates and guidance content, referral target catalogue, outcome codes.
Non-functional design	• Internal module only. It has no external interface and so no fallback class. • Group sessions record the attendee list from a scanned QR token or an operator selection.

SM-11 — Work-Site Visit, Reskilling, Upskilling & RPL Scheduling

Purpose. Manages field officer worksite visits and the full skilling pathway — reskilling, upskilling, and Recognition of Prior Learning (RPL) — for individual workers.

- Worksite visit scheduling and geo-tagged, photo-evidenced visit records captured through the offline field application.

- Training and skill-event scheduling, attendance and outcome recording.
- RPL assessment batch management, from nomination through assessment to certification.
- A per-worker skilling pathway view, combining visits, training and RPL history into one picture.

Integrations: APSSDC / Skill India Digital Hub, Technical Education department.

Fallback: Worksite visits, training attendance and RPL nomination all capture fully offline (Section 10.1); only the final certification sync depends on connectivity.

Technical specification. SM-11 produces the geo-tagged evidence trail for worksite visits, and its capture path is the offline field application.

Aspect	Specification
Family and schema	Employment & Skilling family, schema `emp`.
Principal entities	• `site_visit`, `visit_media`, `skilling_event`, `skilling_enrolment`, `rpl_batch`, `rpl_nomination`, `rpl_assessment`, `skilling_pathway`
Key APIs	• `POST /v1/site-visits` (also through `POST /v1/sync/push`) • `POST /v1/site-visits/{id}/media` • `POST /v1/skilling-events` and `POST /v1/skilling-events/{id}/enrolments` • `POST /v1/skilling-events/{id}/attendance` • `POST /v1/rpl-batches` and `POST /v1/rpl-batches/{id}/nominations` • `GET /v1/workers/{id}/skilling-pathway`
Events	Emits `visit.recorded`, `skilling.attended`, `rpl.nominated`, `rpl.certified`. Consumes `worker.registered`, `group.formed`.
Business rules and validations	• A visit record requires a location fix, which the field application treats as mandatory for evidence media. Location status is recorded as ok, low-accuracy, unavailable or permission-denied. • Each photograph and video is hashed on capture and the hash is verified on receipt at the Document Vault. • A worker can be nominated to one RPL batch per trade at a time. • Certification results are read from APSSDC or the Skill India Digital Hub where an interface exists and are otherwise entered with the certificate scan attached.
Configuration items	Trade and skill catalogue, visit types, RPL batch parameters, attendance thresholds for completion.
Non-functional design	• Visit and attendance capture complete entirely offline. Only certificate synchronisation needs connectivity. • Media is uploaded after record data and in resumable chunks (Section 10.11).

SM-12 — Common API & Systems Integration Framework (Integration Hub)

Purpose. The single, shared path through which every one of the other 21 modules reaches an external system — described in full in Section 5.

- Adapter registry, credential custody, consent gating and LGD geography harmonisation for all 50 notified sources.
- Retry, circuit-breaker and replay handling so a department's outage degrades gracefully rather than propagating as an error to a worker.

- Scheduled reconciliation and a live integration-health dashboard used by the helpdesk and District Field Support Clusters.

Integrations: By definition, every external system the Platform touches — see the full catalogue in Section 5.3.

Fallback: SM-12 is itself the mechanism that decides, per call, whether the automated or the manual/assisted route is used (Section 6).

Technical specification. SM-12 is the single route to every external system. Its design is described in Section 5. This table adds the data and interface detail.

Aspect	Specification
Family and schema	Shared service, released with the Integration Hub worker pool. Operational data in schema `core`; raw payload snapshots in the MongoDB staging collections.
Principal entities	● `adapter_descriptor` and `adapter_version`, `integration_call` (outcome, latency, source, consent reference), `fetched_attribute` (value, source, as-of time) ● `reconciliation_item`, `mismatch_case_link`, `cutover_state`, `dump_file`, `dump_batch`, `lgd_crosswalk`, `credential_ref`
Key APIs	● `GET /v1/admin/adapters` and `PUT /v1/admin/adapters/{id}/cutover-state` ● `GET /v1/admin/integration-health` ● `POST /v1/admin/reconciliation/{sourceId}/runs` ● `GET /v1/admin/reconciliation-items?filter[status]=NEEDS_REVIEW` ● `POST /v1/admin/dumps` registers a received file ● `POST /v1/inbound/workers/verify` and `POST /v1/inbound/scheme-status` for departments
Events	Emits `integration.call.completed`, `integration.attribute.fetched`, `reconciliation.mismatch_raised`, `source.degraded`, `source.recovered`. Consumes `consent.granted`, `consent.withdrawn`.
Business rules and validations	● A call without a consent reference for its purpose code returns CONSENT_REQUIRED and no network call is made. ● Every outcome is one of the eight typed outcomes of Section 5.2.3. A module never receives a departmental error body. ● Responses are validated against the contract at runtime. A shape change returns CONTRACT_DRIFT and the response is quarantined. ● Credentials are resolved at call time from the secrets vault and never appear in logs, payload snapshots or errors. ● Changing a cut-over state needs the Integration Lead role and writes an audit record.
Configuration items	Adapter descriptors, timeout, retry and breaker policy per source, consent purpose mapping, LGD crosswalks, reconciliation schedule, cut-over state.
Non-functional design	● Circuit breaker, bulkhead and rate-limit parameters are in Section 5.11. ● Payload snapshots in MongoDB are hashed, encrypted at rest and expire under the retention schedule. ● Egress uses the fixed addresses published to departments for allow-listing.

SM-13 — MIS, Data Analytics & Decision Support

Purpose. Turns every module's operational data into the reports and dashboards APBOCWWB, the PMU and departments actually need, without slowing down the live system.

- Predefined reports covering registration, scheme uptake, grievances, employment, skilling and fallback share.
- Configurable, role-scoped dashboards, including the 4E programme funnel (Enrollment → Entitlement → Empowerment → Employment).
- Threshold-based alerts (for example, a district's fallback share crossing an agreed level) rather than requiring someone to go looking for a problem.
- Exports in standard formats for offline analysis, respecting the same jurisdiction-scoped access as the live system.

Integrations: Internal only — fed by every operational module through the read replica (Section 2.3).

Fallback: Not applicable — a reporting function with no external dependency.

Technical specification. SM-13 reads from the replica only and shows the time of last refresh on every report and dashboard.

Aspect	Specification
Family and schema	MIS family, schema `mis` on the readable secondary, built from `core` and family schemas by scheduled refresh.
Principal entities	• Marts: `mart_registration_daily`, `mart_scheme_funnel`, `mart_ticket_sla`, `mart_employment`, `mart_skilling`, `mart_fallback_share`, `mart_integration_health`, `mart_data_quality` • `report_definition`, `dashboard_definition`, `alert_rule`, `export_job`
Key APIs	• `GET /v1/reports` and `POST /v1/reports/{id}/runs` • `GET /v1/dashboards/{id}` with `?scope=district:...` • `POST /v1/exports` and `GET /v1/exports/{id}` • `GET /v1/alerts` and `POST /v1/alert-rules`
Events	Emits `alert.raised`, `export.ready`. Consumes the refresh schedule only; it does not subscribe to business events.
Business rules and validations	• Row access in every mart is filtered by the same role and centre scope as the live system, applied in the data-access layer and in the Superset row-level security configuration. • The 4E funnel counts a worker once per stage, using the first date on which the stage condition became true. • Threshold alerts evaluate after each mart refresh and raise one alert per breach window. • Exports are generated asynchronously, stored in the Document Vault for 7 days and listed in the audit trail.
Configuration items	Report and dashboard definitions, alert thresholds, mart refresh cadence (Annex D), export size limits.
Non-functional design	• Refresh cadence ranges from every 15 minutes for operational marts to nightly for the data-quality mart. • Dashboard queries run against pre-aggregated marts. Row-level exports run through the export job and never through a dashboard widget.

SM-14 — Communication, Notification, Alert & Broadcast Management

Purpose. The single engine behind every message the Platform sends — to a worker, an employer, or staff — across every channel, plus targeted broadcast campaigns.

- Trigger-based dispatch — any module raises an event, SM-14 resolves the recipient, channel and template.
- In-app, SMS (DLT-registered), WhatsApp and IVR delivery, with delivery-status tracking and automatic retry on failure.
- Targeted broadcasts by district, scheme, worker category or centre, rather than a single blast to everyone.

Integrations: A DLT-registered SMS gateway, the WhatsApp Business Cloud API, and an IVR/telephony provider.

Fallback: In-app and WhatsApp delivery continue even where SMS DLT template approval is still pending for a given event type (Section 4.6.1).

Technical specification. SM-14 is event driven. A module never sends a message; it raises `notification.requested` and the Notification Engine resolves the rest.

Aspect	Specification
Family and schema	Communication family, schema `com`; dispatch in the Notification worker pool.
Principal entities	• `notification` (one per message), `notification_attempt`, `delivery_report`, `template`, `template_version`, `channel_preference`, `campaign`, `campaign_audience`, `dlt_template_registration`
Key APIs	• `POST /v1/notifications` (internal, called by other families) • `GET /v1/notifications?filter[workerId]=...` • `POST /v1/campaigns` and `POST /v1/campaigns/{id}/approve` • `PUT /v1/me/channel-preferences` • `POST /v1/inbound/delivery-reports` for provider callbacks
Events	Emits `notification.delivered`, `notification.failed`. Consumes every `notification.requested` event.
Business rules and validations	• The recipient is resolved from the verified mobile number and channel preference, never from free text in the request. • SMS templates must carry a DLT template identifier. A template without one cannot be activated for SMS, and the in-app and WhatsApp legs continue. • A failed attempt retries on the same channel up to the configured count and then falls back to the next channel in the preference order. • A campaign requires approval by a second user before it can start. • Quiet hours (default 21:00 to 07:00) delay non-urgent messages.
Configuration items	Templates in Telugu and English, DLT identifiers, channel order, retry counts, quiet hours, campaign approval rule.
Non-functional design	• Dispatch runs as a Temporal workflow per message. Provider throughput limits are configured per provider and enforced by the campaign workflow. • Message bodies containing personal data are stored encrypted and purged after the configured period, leaving the delivery metadata.

SM-15 — Job Mela & Event Facilitation

Purpose. Runs the full lifecycle of a job mela or camp — from pre-registration through on-site check-in to placement follow-up — fully usable with no connectivity on the day.

- Worker and employer pre-registration ahead of the event.
- On-site check-in, interview scheduling and offer recording, captured through the offline field application.
- Placement follow-up tracking after the event, feeding back into SM-05 employment records.

Integrations: Internal — SM-01, SM-02, SM-04, SM-05 and SM-11.

Fallback: Designed offline-first from the outset (Section 10.1) — a mela's connectivity on the day is never assumed.

Technical specification. SM-15 treats connectivity on the day of a mela as unavailable. Everything an operator does on the day is captured on the device first.

Aspect	Specification
Family and schema	Employment & Skilling family, schema `emp`.
Principal entities	• `mela_event`, `mela_registration` (worker and employer), `mela_checkin`, `interview_slot`, `interview_result`, `offer`, `placement_followup`
Key APIs	• `POST /v1/mela-events` • `POST /v1/mela-events/{id}/registrations` • `POST /v1/mela-checkins` (also through sync) • `POST /v1/mela-events/{id}/interview-slots` • `POST /v1/offers` and `POST /v1/offers/{id}/response` • `POST /v1/placement-followups`
Events	Emits `mela.checked_in`, `offer.made`, `placement.confirmed`. Consumes `worker.registered`, `job.posted`.
Business rules and validations	• Check-in is idempotent per worker per event. A second tap returns the original check-in. • A worker not yet registered is pre-registered on the device and receives a provisional identifier, completed on sync. • An offer links to a job posting or an employer requisition, and a placement confirmed within the follow-up window creates an engagement in SM-05. • Interview slot capacity cannot be exceeded, and a device that overbooks offline raises a sync conflict, not a silent overwrite.
Configuration items	Event types, check-in categories, slot lengths, follow-up window, offer response codes.
Non-functional design	• A device is seeded before the event with the registered worker list for the catchment, a delta of the employer and posting list, and the form definitions. • Check-in sustained rate planned at 30 per minute per desk.

SM-16 — Monitoring of ILAC Functions & Functionary Work Turnout

Purpose. Gives supervisors and the PMU a factual, audit-trail-derived view of what each centre and each staff member actually did, day by day.

- Staff attendance capture, cross-checked against audited transaction activity rather than self-reported alone.
- A per-centre daily function checklist confirming the centre opened, was staffed, and delivered its core services.
- A functionary scorecard built from real transaction counts — registrations processed, grievances resolved, visits completed.

Integrations: Internal; an optional HRMS dump for staff-master reconciliation (Section 5.3).

Fallback: Attendance can be recorded by a supervisor on paper during an outage and keyed in once connectivity returns.

Technical specification. SM-16 derives turnout from the audit trail rather than from self-reporting, so the figures cannot be edited by the person being measured.

Aspect	Specification
Family and schema	Administration & Monitoring family, schema `adm`.
Principal entities	• `attendance_record`, `centre_daily_checklist`, `checklist_item_result`, `functionary_activity_summary`, `scorecard`, `staff_master_ref`
Key APIs	• `POST /v1/attendance` (also through sync) • `POST /v1/centre-checklists` and `GET /v1/centre-checklists?filter[centreId]=...&date=...` • `GET /v1/scorecards?filter[centreId]=...&period=...` • `POST /v1/attendance/paper-entries`
Events	Emits `attendance.recorded`, `checklist.submitted`. Consumes audit-derived activity summaries from the nightly job.
Business rules and validations	• `functionary_activity_summary` is computed nightly from the Audit Ledger, counting registrations, tickets resolved, visits completed and similar transactions per user. • An attendance record with no audited activity in the same period is shown as a discrepancy on the supervisor view, not corrected automatically. • A paper attendance sheet keyed after an outage keeps its original date and records the keying time and user separately.
Configuration items	Checklist items per centre type, scorecard weights, working calendar, discrepancy threshold.
Non-functional design	• Scorecards are computed on the replica and cached for the day. • Staff master reconciliation with the HRMS dump runs weekly and raises mismatches for the administrator.

SM-17 — Para-Legal Services Facilitation at the ILAC

Purpose. Gives a worker facing a wage dispute, workplace accident or other legal issue a structured entry point to guidance and formal legal aid.

- Intake of legal-aid needs by category (wages, accident, dispute, other), with the same Case ID and service-clock discipline as any other request.
- Guidance from a rostered para-legal volunteer at the centre, with session notes attached to the case.
- Structured referral to the District Legal Services Authority or a legal-aid clinic, with the referral tracked to a documented next step.

Integrations: State legal-services or labour-department interfaces, where and if notified.

Fallback: The module is assisted by design — intake and referral are always operator-mediated, so there is no automated route to fall back from.

Technical specification. SM-17 is assisted by design. Every step is performed by a person, so the module records and tracks and never advises automatically.

Aspect	Specification
Family and schema	Grievance & Service Requests family, schema `grv`.
Principal entities	• `legal_aid_case` (linked to a Case ID), `legal_aid_category`, `paralegal_roster`, `paralegal_session`, `legal_referral`
Key APIs	• `POST /v1/legal-aid-cases` • `POST /v1/legal-aid-cases/{id}/sessions` • `POST /v1/legal-aid-cases/{id}/referrals` • `GET /v1/paralegal-roster?filter[centreId]=...` • `POST /v1/legal-aid-cases/{id}/closure`
Events	Emits `legal_aid.opened`, `legal_aid.referred`, `legal_aid.closed`. Consumes `ticket.escalated` for grievances tagged as legal.
Business rules and validations	• Session notes are visible only to the para-legal volunteer, the case owner and designated supervisors. They are classed as Class 6 data and excluded from general exports. • A referral to the District Legal Services Authority or a clinic must carry the documents list and a follow-up date. • The case closes through the standard two-outcome rule. A completed outcome needs the referral confirmation or the settlement record.
Configuration items	Legal-aid categories, roster rules, DLSA and clinic directory, follow-up intervals.
Non-functional design	• The roster shows volunteer availability per centre per day. A centre with no rostered volunteer routes the intake to the nearest centre with a documented next step.

SM-18 — Common Agency Service Facilitation Interface

Purpose. A generic facilitation pattern for agency services that do not warrant their own module — bank account opening, insurance enrolment, post-office services — with the same tracked-outcome discipline as everything else.

- A configurable catalogue of agency service types, so a new agency service is a configuration addition, not a new module.
- Request capture, submission (where an interface exists) or manual handoff, and outcome tracking through to completion.

Integrations: NPCI, e-Shram/NDUW, and PMSBY/PMJJBY insurance-enrolment channels.

Fallback: Manual desk submission (Class F3) is the default for any agency service without a live interface.

Technical specification. SM-18 is a catalogue-driven pattern. A new agency service is a catalogue row and a checklist, never a new module.

Aspect	Specification
Family and schema	Grievance & Service Requests family, schema `grv`.
Principal entities	• `agency_service_type`, `agency_request`, `agency_request_document`, `agency_request_status_history`
Key APIs	• `GET /v1/agency-services` • `POST /v1/agency-requests` • `POST /v1/agency-requests/{id}/handoff` • `POST /v1/agency-requests/{id}/status` • `POST /v1/agency-requests/{id}/outcome`
Events	Emits `agency_request.opened`, `agency_request.completed`. Consumes `consent.granted`.
Business rules and validations	• A service type defines its document checklist, whether a submission interface exists, and the outcome codes. • Bank account opening cross-checks the account number through NPCI when an account is reported as opened. • e-Shram and insurance enrolment record the enrolment number returned by the agency and link it to the worker. • Manual desk submission (F3) is the default for a service type without an interface.
Configuration items	Agency service catalogue, checklists, outcome codes, interface flag per service.
Non-functional design	• Request volumes are low and bursty, so the module has no dedicated scaling rules and shares the Grievance & Service Requests family pods.

SM-19 — User, Role & Configuration Administration

Purpose. The console through which APBOCWWB administers every user, role, centre and jurisdiction, and every configurable item across every one of the other 21 modules — described in Section 2.5 and Section 8.2 for its access-control role.

- User, role, centre and jurisdiction management, including the four-dimension permission model.
- The single configuration console for forms, scheme rules, service clocks, escalation rules, notification templates and the Fallback Register.
- A full change-approval workflow and audit trail on every configuration change, so a rule change is itself governed, not a silent edit.

Integrations: An HRMS dump for employee-master reconciliation.

Fallback: Not applicable — an internal administration function.

Technical specification. SM-19 is both the administration console and the configuration store described in Section 2.19.

Aspect	Specification
Family and schema	Administration & Monitoring family, schema `adm` for users and organisation; schema `core` for versioned configuration.
Principal entities	• `user_account`, `role`, `role_permission` (module, action, centre scope, visibility), `user_role_assignment`, `centre`, `service_area`, `lgd_location` • `config_item`, `config_version`, `config_bundle`, `fallback_register_entry`
Key APIs	• `POST /v1/users`, `PUT /v1/users/{id}/roles`, `POST /v1/users/{id}/deactivate` • `POST /v1/roles` and `PUT /v1/roles/{id}/permissions` • `POST /v1/centres` and `PUT /v1/centres/{id}/service-areas` • `GET /v1/config-items?filter[type]=...` • `POST /v1/config-items/{key}/versions` and `POST /v1/config-versions/{id}/publish` • `POST /v1/config-bundles/export` and `POST /v1/config-bundles/import`
Events	Emits `user.role_changed`, `user.deactivated`, `config.published`. Consumes `hrms.dump_loaded`.
Business rules and validations	• A user's effective access is the union of active roles, scoped to the allocated centres. A role change applies at the next request. • Deactivation blocks login at once and keeps historical actions attributed to the user. • A user cannot approve a configuration change that the same user drafted. • Importing a bundle checks the signature, the checksum and the target environment before any change is applied. • Every configuration publication writes an audit row with the approver.
Configuration items	The configuration item types of Section 2.19, role templates, centre and jurisdiction masters.
Non-functional design	• Permissions are cached per principal with a version stamp (Section 2.18). • The console is available only over the administrator route class with step-up MFA.

SM-20 — Construction-Site Catalogue & Worker Grouping for Skilling

Purpose. Maintains a live catalogue of construction sites in each centre's jurisdiction and groups workers by site, trade or need for training, RPL and camp participation.

- Site catalogue capture and update through field officer surveys.
- Rule-based worker grouping (by site, trade, skill gap, or geography) driving targeted invitations to SM-11 training and SM-15 events.

Integrations: Internal — SM-05, SM-11 and SM-15; batch export to APSSDC for skilling coordination.

Fallback: Site survey capture is fully offline through the field application (Section 10.1).

Technical specification. SM-20 keeps the site catalogue fresh through field surveys and turns it into groups that SM-11 and SM-15 use for invitations.

Aspect	Specification
Family and schema	Employment & Skilling family, schema `emp`.
Principal entities	• `construction_site`, `site_survey`, `site_contractor`, `site_worker_link`, `worker_group`, `worker_group_member`, `grouping_rule`

Aspect	Specification
Key APIs	• `POST /v1/sites` and `PUT /v1/sites/{id}` • `POST /v1/site-surveys` (also through sync) • `POST /v1/grouping-rules` and `POST /v1/grouping-rules/{id}/run` • `GET /v1/worker-groups/{id}/members` • `POST /v1/worker-groups/{id}/invitations`
Events	Emits `site.surveyed`, `group.formed`, `group.invited`. Consumes `worker.updated`, `engagement.confirmed`.
Business rules and validations	• A site is identified by an LGD location and a normalised name. A new survey that matches an existing site by name and a 250 metre radius updates it rather than creating a second record. • Grouping rules combine site, trade, skill gap and geography. A group run is stored with the rule version so a past invitation list can be reproduced. • A group invitation goes through the Notification Engine with the worker's language preference.
Configuration items	Grouping rule templates, site categories, matching radius, invitation templates.
Non-functional design	• Site survey capture is fully offline. Matching and grouping run on the server after sync. • Distance matching uses the native geography type of the database engine.

SM-21 — Construction Worker Livelihood Groups & Federation

Purpose. Supports the formation and ongoing functioning of worker livelihood groups and their mandal and district-level federations, linking members to schemes and agency services as a group.

- Group formation, membership management and meeting records, captured through the field application where groups meet at a worksite or household.
- Federation-level structures above individual groups, with their own record of meetings and decisions.
- Group-level linkage into scheme applications (SM-02) and agency services (SM-18), where a scheme or facility is more naturally accessed collectively.

Integrations: MEPMA and SERP self-help-group data, for cross-reference and to avoid duplicate group registration.

Fallback: Group and meeting records capture fully offline; cross-reference against MEPMA/SERP data runs on next sync.

Technical specification. SM-21 models a livelihood group as a group of workers and a federation as a group of groups, with the same meeting and decision structure at each level.

Aspect	Specification
Family and schema	Employment & Skilling family, schema `emp`.
Principal entities	• `livelihood_group`, `group_membership`, `group_meeting`, `meeting_attendance`, `federation`, `federation_member_group`, `federation_decision`, `group_scheme_link`

Aspect	Specification
Key APIs	• `POST /v1/livelihood-groups` and `POST /v1/livelihood-groups/{id}/members` • `POST /v1/group-meetings` (also through sync) • `POST /v1/federations` and `POST /v1/federations/{id}/groups` • `POST /v1/federations/{id}/decisions` • `POST /v1/livelihood-groups/{id}/scheme-links`
Events	Emits `group.formed`, `group.meeting_held`, `federation.decision_recorded`. Consumes `worker.updated`.
Business rules and validations	• A worker belongs to at most one active group of a given type. • Group registration is cross-checked against MEPMA and SERP self-help-group data on sync, to avoid duplicate registration of an existing group. • Federation levels are mandal and district. A group belongs to one federation at each level. • A scheme link records a group-level application in SM-02 and the members covered.
Configuration items	Group types, minimum membership, federation levels, meeting frequency expectations.
Non-functional design	• Meeting and membership capture works offline. Cross-reference with MEPMA and SERP data runs on the next sync and flags matches for review.

SM-22 — Worker, Employer & Departmental Applications, Portals & Dashboards

Purpose. The user-facing layer through which every other module is actually used — described in full in Section 9.

- Worker and employer mobile apps and web portals, the kiosk client, and the assisted-desk console.
- The departmental portal and the PMU/administration console.
- Every surface renders the same underlying module APIs — SM-22 has no logic of its own beyond presentation and session handling.

Integrations: Exclusively through the API gateway — SM-22 never calls a department or a shared service directly.

Fallback: Kiosk and assisted-desk surfaces degrade to on-screen tokens and local buffering as described in Section 9.5–9.6.

Technical specification. SM-22 has no business rules of its own. The table describes the surfaces, their authentication and the single API surface they use.

Aspect	Specification
Family and schema	Administration & Monitoring family for the back-end session and presentation APIs; front ends in `apps/web-portal`, `apps/worker-app` and `apps/field-app`.
Principal entities	• `ui_session`, `kiosk_session`, `device_registration`, `portal_content`, `dashboard_layout`

Aspect	Specification
Key APIs	● `GET /v1/me` and `GET /v1/me/home` for the worker and employer apps ● `POST /v1/kiosk-sessions` and `DELETE /v1/kiosk-sessions/{id}` ● `POST /v1/devices` registers a field device ● `GET /v1/portal-content?filter[page]=...` ● `GET /v1/departmental/home` and `GET /v1/pmu/inbox`
Events	Emits `session.started`, `session.ended`, `device.registered`. Consumes `user.deactivated` to end sessions.
Business rules and validations	● A kiosk session holds no data beyond the session and revokes its tokens on timeout, sign-out or completion. ● A field device is registered once per user and can be revoked remotely. Revocation destroys the wrapping key for local data. ● Every surface calls the same module APIs through the gateway. No surface calls a shared service or a department directly.
Configuration items	Portal content blocks in both languages, kiosk timeout values, session limits, dashboard layouts per role.
Non-functional design	● Surface behaviour, performance budgets and the device baseline are in Section 9. ● The web portal is server rendered for first paint on low-end devices and installable as a PWA.

3.3 How "no worker turned away" is enforced across every module

Every request opened in any module, at any location, through any channel, receives an ILAC Case ID and a service clock from the shared Case Service (Section 2.2.1). A case can close in exactly two ways: Service Completed, with evidence attached, or Documented Next Step, with five mandatory fields — responsible officer or department, required action, target date, document requirements, and follow-up route. No screen in any module — including the assisted desk and the offline field application — has a control that can close a case any other way.

3.4 Cross-module event contracts

Every worker-facing module writes to the Worker Longitudinal History and raises notification triggers to the Notification Engine, rather than keeping its own history or sending its own messages. This is what gives a worker one continuous timeline regardless of which of the 22 modules they interacted with, and gives APBOCWWB one place to see everything that happened to a worker, ever.

3.5 Module-to-phase allocation

Phase	Modules delivered
Phase 1 (Weeks 1–5)	SM-01 (with Deviation D2), SM-04, SM-05, SM-06, SM-11, and the core of SM-12, SM-14, SM-18, SM-19
Phase 2 (Weeks 6–9)	SM-07 (core), SM-09 (core), SM-13, SM-15, SM-22, plus completion of SM-12 and SM-14

Phase	Modules delivered
Phase 3 (Weeks 10–13)	SM-02, SM-03, SM-08, SM-10, SM-20, SM-21, plus completion of SM-07 and SM-09
Phase 4 (Weeks 14–17)	SM-16, SM-17, plus completion of SM-18 and SM-19

3.6 Logical data model and schema ownership

The system-of-record database holds one schema per service family plus a `core` schema for the shared services. A family writes only to its own schema and to `core` through the shared-service interfaces. References between schemas are by `public_id` and are validated through the owning family's API, so no foreign key crosses a schema boundary and a family can be changed, restored or moved without a cross-schema constraint stopping it. The read replica carries a further `mis` schema built by scheduled refresh.

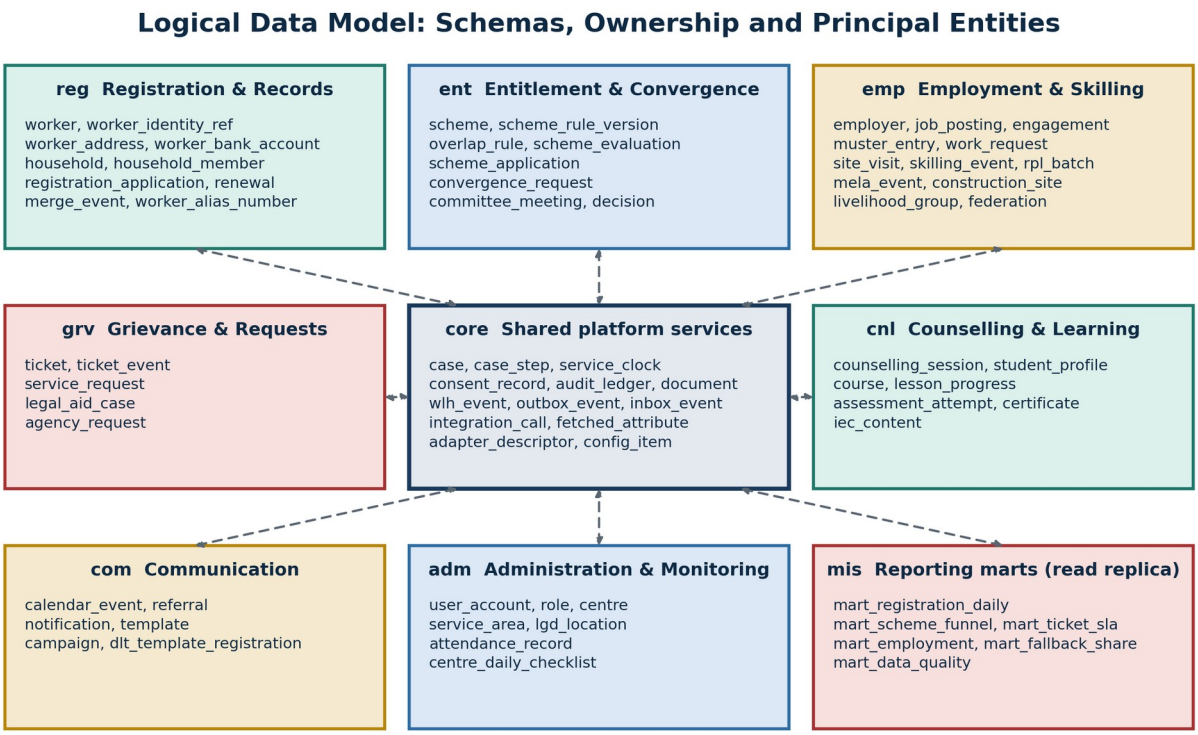


Figure 3.1 — Logical data model: schemas, ownership and principal entities

The volumes below are planning figures for the Year-1 population of Section 1.9. They drive index design, partitioning and the storage sizing of Section 16.3.

Entity group	Year-1 rows (order)	Growth driver	Partitioning
`worker`, `household`, `household_member`	3.0 M, 0.8 M, 3.5 M	Registration and migration	None; indexed on LGD location and centre
`case`, `case_step`	6 M, 25 M	About 2 cases per worker per year and 4 steps per case	`case_step` by month

Entity group	Year-1 rows (order)	Growth driver	Partitioning
`audit_ledger`	120 M	About 40 audited events per worker per year	By month and by lane (16 lanes)
`wlh_event`	12 M	Worker-facing events	By year
`integration_call`, `fetched_attribute`	30 M, 20 M	Registration, re-verification, scheduled pulls	`integration_call` by month
`notification`, `notification_attempt`	25 M, 35 M	Events times channels	By month
`scheme_evaluation`	8 M current rows	One per household and scheme, replaced by version	None; latest-version index
`ticket`, `ticket_event`	1.5 M, 8 M	Grievances and support tickets	`ticket_event` by month

The rules below apply to every table in every family. They are checked by a schema linter in the pipeline, which fails the build when a migration breaks one.

- **Keys.** A surrogate `bigint` identity key for joins and a 26-character `public_id` for the outside world. Natural keys such as the Registration Number are unique constraints, not primary keys.
- **Standard columns.** `created_at`, `created_by`, `updated_at`, `updated_by` and `row_version` on every mutable table. Insert-only tables carry `created_at` and `created_by` only.
- **No hard deletes.** Personal data is tombstoned by overwriting the personal fields. Ledger and closed transactional rows are never updated or deleted.
- **Naming.** Lower-case snake_case, singular table names, foreign key columns named `<table>_id`, boolean columns prefixed `is_` or `has_`.
- **JSON columns.** Permitted only for a validated snapshot of a submitted form or a configuration document, never for a field that is filtered or joined on.
- **Encryption.** Class 1 and Class 2 columns are encrypted at the column level. The Aadhaar number is not in the system-of-record database at all (Section 8.15).
- **Indexes.** Every foreign key column and every API filter on a large table has an index. Unused indexes are reported monthly from the engine statistics and removed after review.

The data-access layer supports two engines, so each logical type has a defined physical mapping. This is the mapping used by the migration scripts in `db/migrations`.

Logical type	SQL Server 2022 or higher	PostgreSQL 15 or higher
Surrogate key	`bigint IDENTITY`	`bigint GENERATED ALWAYS AS IDENTITY`
Public identifier	`char(26)` with a unique index	`char(26)` with a unique index
Text, Telugu and English	`nvarchar(n)` with an Indic_General_100 collation	`text` in UTF-8 with an ICU collation for `te-IN`
Timestamp (UTC)	`datetimeoffset(3)`	`timestamptz`
Money	`decimal(14,2)`	`numeric(14,2)`

Logical type	SQL Server 2022 or higher	PostgreSQL 15 or higher
Boolean	`bit`	`boolean`
Validated JSON	`nvarchar(max)` with an `ISJSON` check	`jsonb`
Point and distance	`geography` (SRID 4326)	`geography(Point,4326)` with PostGIS
Fuzzy name and address match	Full-text index plus a phonetic and trigram function in the data-access layer	`pg_trgm` with GIN indexes
Optimistic concurrency	`rowversion` mapped to `row_version`	Integer `row_version` incremented by trigger
Insert-only ledger	`DENY UPDATE, DELETE` on the application login	`REVOKE UPDATE, DELETE` plus a guard trigger
Read replica	Readable secondary replica in an availability group	Streaming replica with hot standby

3.7 Case Service design

The Case Service is the mechanism behind "no worker turned away" (Section 3.3). It exposes two closure operations and no third. The figure shows the states, and the table gives each transition with the condition that guards it.

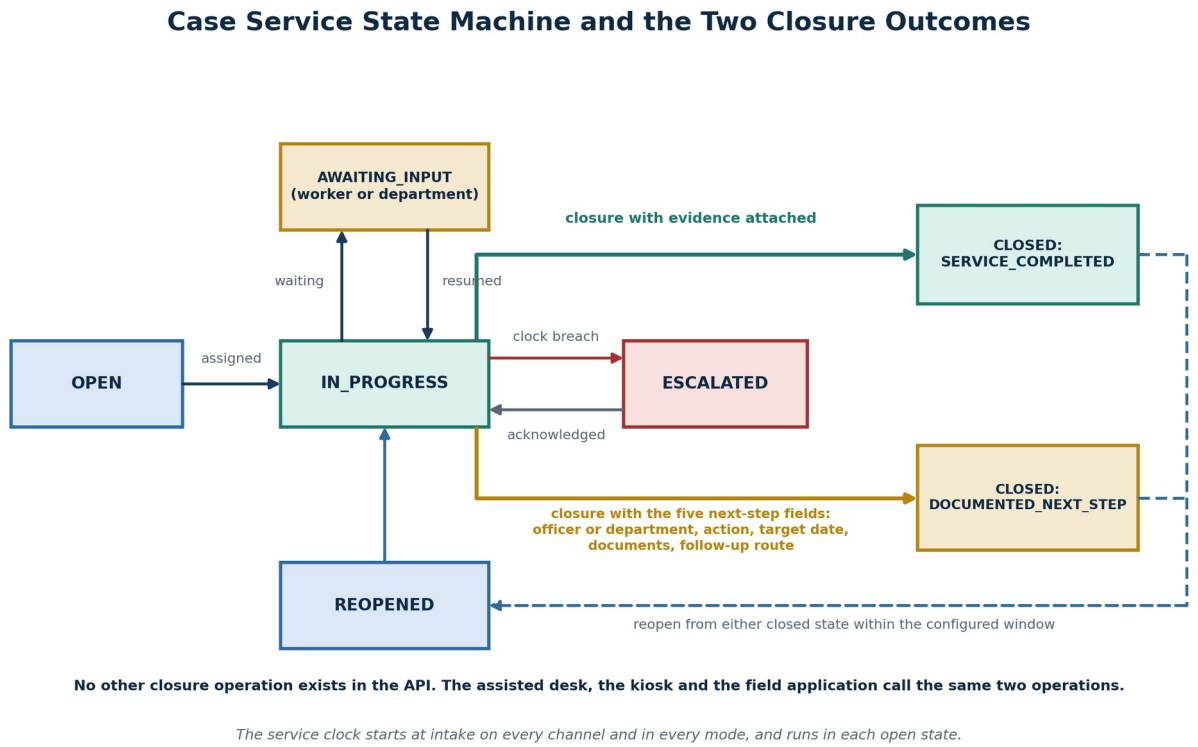


Figure 3.2 — Case Service state machine and the two closure outcomes

From	Trigger	To	Guard and effect
OPEN	Assignment or first step	IN_PROGRESS	An owner role exists. The service clock keeps running.

From	Trigger	To	Guard and effect
IN_PROGRESS	Step of type WAIT_WORKER or WAIT_DEPARTMENT	AWAITING_INPUT	A reason code is required. The clock pauses only if the configured `pauseOn` list contains the reason, and never for an internal reason such as an interface outage.
AWAITING_INPUT	Input received or status read back	IN_PROGRESS	Records who supplied the input and when.
IN_PROGRESS, AWAITING_INPUT	Clock breach	ESCALATED	The Escalation workflow starts and notifies the next-level owner. The full case history moves with the escalation.
ESCALATED	Acknowledgement by next-level owner	IN_PROGRESS	The owner field changes. The original clock record continues.
IN_PROGRESS, AWAITING_INPUT, ESCALATED	`POST /closures` with type COMPLETED	CLOSED: SERVICE_COMPLETED	An evidence document identifier and an outcome code are required.
IN_PROGRESS, AWAITING_INPUT, ESCALATED	`POST /closures` with type NEXT_STEP	CLOSED: DOCUMENTED_NEXT_STEP	All five fields are present: `responsibleParty`, `requiredAction`, `targetDate`, `documentsRequired[]`, `followUpRoute`. The target date cannot be in the past and the responsible party must exist in master data.
CLOSED (either)	Reopen request within the configured window	REOPENED	Supervisor role and a reason are required. The original clock record continues.
REOPENED	Automatic	IN_PROGRESS	The owner is the previous owner unless changed.

Five invariants hold for every case in every module and are tested in the pipeline with property-based tests:

- A committed `case_step` is never updated or deleted. A correction is a new linked step.
- Every step carries `mode` (automated, assisted or manual), the fallback class and reason code when one applies, the operator identity and role, and the evidence document references.
- The service clock is owned by the CaseLifecycle workflow and uses server time. A device or a client cannot supply a start time, except that a paper intake carries a `receivedAt` that must not be later than the keying time.
- The Case ID is assigned by the server in the form `ILC-` followed by a two-digit year, an eight-character Crockford base32 sequence and a check character. A device holds a provisional client identifier until synchronisation returns the server value.
- Every closed case has exactly one closure record, and the closure record is the only source of the outcome shown in MIS.

3.8 Scheme Engine rule specification

The Scheme Engine (SM-02) evaluates a rule document per scheme. A rule document is configuration, validated against a JSON schema when it is saved, compiled when it is published and cached by version. The example below uses illustrative values. The real criteria and amounts come from the 38-scheme catalogue validated with APBOCWWB at milestone M9.

```
{ "schemeCode": "SCH-014", "version": 4, "effectiveFrom": "2026-12-01", "owner": "Social Welfare",
  "subject": "household",
  "eligibility": { "all": [
    { "attr": "worker.status", "op": "eq", "value": "REGISTERED_ACTIVE" },
    { "attr": "household.income.annual", "op": "lte", "value": 120000, "freshness": "P12M", "source":
"GSWS" },
    { "any": [ { "attr": "member.child.age", "op": "between", "value": [5, 17] },
               { "attr": "member.child.education.stage", "op": "in", "value": ["SCHOOL",
"INTERMEDIATE"] } ] } ] },
  "benefit": { "type": "CASH", "amount": 15000, "per": "child", "period": "YEAR" },
  "overlap": ["OV-02", "OV-05"],
  "needsReviewWhen": ["MISSING_REQUIRED_ATTRIBUTE", "STALE_ATTRIBUTE", "CONFLICT_WITH_DECLARATION"] }
```

Evaluation of one subject follows six steps. The same code path serves a live evaluation, a re-evaluation after a data change and the baseline batch run after migration.

Step	Name	What happens
1	Build context	Collect worker and household attributes with their provenance (declared, verified, fetched with source and as-of time, migrated-unverified).
2	Select versions	Choose the active version of every scheme on the evaluation date. Inactive schemes are skipped.
3	Evaluate criteria	Each criterion returns PASS, FAIL or UNKNOWN using three-valued logic. A missing attribute, an attribute older than its freshness limit, or one that conflicts with the worker's declaration returns UNKNOWN.
4	Combine	ELIGIBLE only when every required criterion passes. NOT_ELIGIBLE when a criterion fails with certainty. Otherwise NEEDS_REVIEW, and the explanation names each unknown criterion and what would resolve it.
5	Apply overlap rules	Overlap rules run in priority order. A suppressed scheme stays visible with the rule that suppressed it, so the worker and operator can see why.
6	Persist and publish	Store the result, the explanation and the rule version. Emit `scheme.evaluated` only when the result differs from the previous one.

The six overlap rules of the catalogue are written with the following rule types.

Rule type	Meaning	Typical use
MUTUAL_EXCLUSION	Only one scheme of a named set can be availed at a time.	Two schemes funding the same expense
PRECEDENCE	Where a worker is eligible for several schemes, the higher-precedence one is offered first and the others are shown as alternatives.	Central and state schemes covering the same benefit
HOUSEHOLD_CAP	A limit on beneficiaries or total value per household in a period.	Per-household assistance caps
LIFETIME_ONCE	The benefit is available once to a person.	One-time grants
PERIOD_ONCE	The benefit is available once per financial year or event.	Annual assistance

Rule type	Meaning	Typical use
AGE_BRACKET	Eligibility moves between schemes as a member crosses an age boundary.	Education and pension schemes

```
{ "subject": "household:01JB8Z4Q2M7T9X3N5R1D0E6KHC", "evaluatedAt": "2026-12-04T06:10:02Z",
  "results": [
    { "scheme": "SCH-014", "ruleVersion": 4, "result": "ELIGIBLE",
      "satisfied": ["worker.status", "household.income.annual", "member.child.age"] },
    { "scheme": "SCH-022", "ruleVersion": 2, "result": "NEEDS_REVIEW",
      "unknown": [ { "attr": "household.ration.card", "reason": "STALE_ATTRIBUTE",
                     "resolve": "Refresh from GSWS or accept the ration card copy (F1)" } ] },
    { "scheme": "SCH-031", "ruleVersion": 1, "result": "ELIGIBLE", "suppressedBy": "OV-02" } ] }
```

3.9 Registration wizard step treatment

The table gives the technical treatment of each of the 15 registration steps in SM-01: what is captured, how it is validated and where it comes from. Step titles are working titles and are aligned with the approved Module Specification at inception. The treatment holds whatever the final labels are.

No.	Step	Validation	Source and route	Fallback
1	Mobile verification	OTP of 6 digits, 3 attempts, valid 5 minutes. Mobile reuse threshold applied.	SMS gateway	WhatsApp OTP, then operator attestation (F2) with supervisor countersign
2	Consent	Affirmative action. Notice version, channel and time logged. Read aloud at the desk.	Consent Manager	None. Registration cannot proceed without consent.
3	Identity (e-KYC)	UIDAI response signature verified. Reference key created in the Aadhaar Data Vault.	UIDAI OTP e-KYC through the Hub	F1 with physical Aadhaar copy; automatic re-verification
4	Photograph	JPEG, minimum 640 x 480 pixels, hash stored. Operator Verified or Not Verified with reason.	Live capture	Operator decision is the control (Deviation D2)
5	Personal details	Name in Telugu and English, spouse or parent name, marital status. Date of birth consistent with e-KYC.	e-KYC, operator entry	F2 where e-KYC is unavailable
6	Address	LGD code mandatory at village or ward level. PIN consistent with LGD. Service Area assigned.	e-KYC address, LGD service	Manual LGD selection
7	Family and dependants	Relationship vocabulary, age, education status. Dependants under 18 linked to a student profile.	Declaration; ration card via GSWS	F4, then F1, then F6
8	Employment	Trade from catalogue. Days worked in the preceding 12 months within 0 to 366 and against the Act threshold.	Declaration plus evidence	F1 with employer or union certificate
9	Employer or contractor	GST checksum. Cess-payer lookup.	GST and cess-payer register through the Hub	F1, then F2

No.	Step	Validation	Source and route	Fallback
10	Bank account	IFSC format. Name match score against NPCI response above the configured threshold.	NPCI validation through the Hub	F1 with passbook scan, then F2
11	Nominee	Shares sum to 100. Relationship vocabulary.	Declaration	Not applicable
12	Social category and income	Category and income band against certificate data.	GSWS integrated and income certificate	F4, then F1, then F6
13	Disability and special category	Percentage read-only when fetched.	SADAREM through the Hub	F1 with certificate
14	Other coverage and documents	EPFO UAN, ESIC number, e-Shram UAN. Document checklist by category, 5 MB per file.	EPFO dump, ESIC and e-Shram through the Hub	F5, then F2 with supervisor countersign
15	Read-back and submit	All mandatory steps complete. Duplicate check on the Aadhaar reference key. Large-type read-back.	Server issues the Registration Number	Not applicable; submit is always on the server

4. Four-Month Delivery Plan, Phase Plan, Milestones, Dependencies and Recovery Mechanism

4.1 Planning basis

The Platform is delivered across a 17-week programme from contract signing, with a Phase Go-Live at the end of each phase and Certified Final Go-Live in Week 17. This is planned as nine two-week sprints, each phase closing with a hardening sprint devoted to defect closure, UAT support, fallback-route certification and repository deposit — not new feature work. The exact calendar dates will be aligned to the RFP's mandatory delivery schedule at contract signing; the phase content, milestone evidence and dependency logic in this section stand regardless of which calendar dates are ultimately fixed.

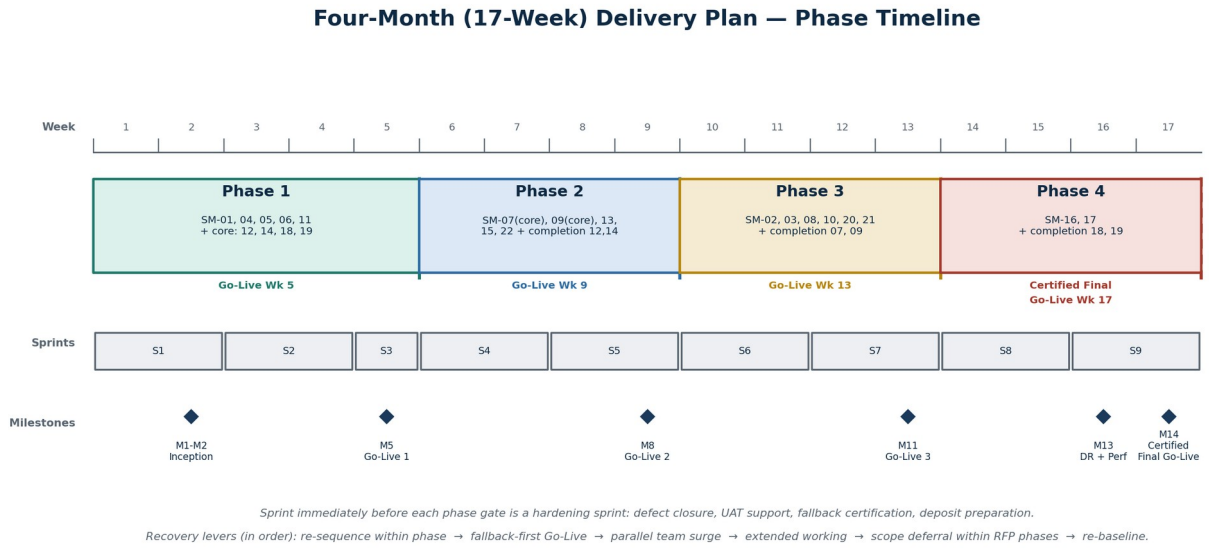


Figure 4.1 — Four-phase, 17-week delivery timeline with sprints and milestones

4.2 Phase plan summary

	Phase 1	Phase 2	Phase 3	Phase 4
Weeks	1–5	6–9	10–13	14–17
Go-Live	End Wk 5	End Wk 9	End Wk 13	End Wk 17 — Certified Final
Key integrations live via API	UIDAI e-KYC, GSWS, e-Shram, EPFO, ESIC, NPCI, CRS, Marriage, SADAREM, GST, APBOCWWB internal	Cut-over of Phase 1 fallback sources; 12 dump sources; Webland, e-Panta, SERP, PM-JAY, PFMS	Scheme-department APIs, RTGS route, CSC & MeeSeva	HRMS refresh; legal-aid referral interface where available
Support	TSU from Week 1; helpdesk live at Go-Live; clusters staffed	All seven District Field Support Clusters operational	Cluster coverage extended to scheme desks	Full O&M support model

	Phase 1	Phase 2	Phase 3	Phase 4
	for Phase 1 districts			

4.3 What makes the window achievable

1. Fallback-first Go-Live. A module goes live on its certified manual route for any data element whose departmental API is not yet ready; the interface is cut over later (Section 6.10) with no phase gate depending on a department's own timetable.

2. Ten platform services built once, in Phase 1. Later phases add modules, adapters and channels — they do not add new plumbing.

3. Configuration over code. The scheme catalogue, fallback routes and notification templates are configured data, versioned in the Master Data & Configuration Service, not programmed and released.

4. Typed end to end. One TypeScript type system from database to mobile screen removes a whole class of integration defects between backend, web and field application.

5. TSU from Week 1, with the inception workshop in Weeks 1–2 confirming positions on modules without a supplied specification, the department onboarding list, and the DLT template registrations that gate Phase 1.

4.4 Milestone register

ID	Milestone	Week	Depends on
M1	Contract signing, kick-off, Steering Committee constituted	1	Contract
M2	Inception report accepted; requirement walkthrough complete; database platform decision recorded	2	M1; Authority access to specifications and repositories
M3	Solution design baseline approved; UAT environment live	3	M2
M4	Platform core (10 shared services) demonstrable in UAT; Production & DR provisioned; first monthly deposit	4	M3; hosting provisioned
M5	Phase 1 Go-Live	5	M4; wave-1 adapters, wave-1 migration, training
M6	Phase 1 hypercare exit	7	M5
M7	Phase 2 UAT entry; second monthly deposit	8	M6
M8	Phase 2 Go-Live	9	M7
M9	Phase 3 UAT entry; 38-scheme catalogue validated with APBOCWWB	11	M8; scheme-owner participation
M10	Migration wave 2 complete; third monthly deposit	12	M9
M11	Phase 3 Go-Live	13	M10
M12	Phase 4 UAT entry	15	M11
M13	DR drill and 300-node performance test passed	16	M12; DR environment
M14	Certified Final Go-Live; O&M starts	17	M13

Monthly source-code deposits fall in Weeks 4, 8, 12 and 17; phase-gate deposits fall in Weeks 5, 9, 13 and 17. A deposit is a milestone condition, not a courtesy — Section 14 details the deposit mechanism.

4.5 Inception — Weeks 1 and 2

Inception fixes the requirement baseline before build volume starts. Its key activities: a requirement-by-requirement walkthrough against this proposal, with a signed variance list where positions differ; confirmation of the database platform decision (Section 15); department onboarding kick-off with letters and data-sharing-agreement drafts issued to every notified department; service-clock and escalation-rule configuration workshops; DLT SMS template registration start; and legacy data-access arrangements for migration profiling. No phase commitment depends on the outcome of inception — proposed designs in Section 3 stand as the baseline until formally varied.

4.6 Dependencies

4.6.1 External dependencies (selected)

Dependency	Needed by	Mitigation if late
Hosting provisioned in an Authority-approved facility for five environments	Weeks 1–4	Provisioning requested at signing; environment build automated so late provisioning costs days, not weeks
Authority-controlled repository access for source deposit	Week 1	Interim deposit to an Authority-nominated escrow location with transfer on access
UIDAI AUA/KUA arrangement through APBOCWWB	Sandbox Wk 3; production Wk 5	Identity step runs on the certified Class F1 route (Aadhaar document + OTP) until enabled
Data-sharing agreements and credentials for wave-1 departments	Sandbox Wk 3; production Wk 5	Fallback-first Go-Live is the design; cut-over runs when credentials arrive
Departmental dump files with agreed schema	From Week 6 as available	Dump pipeline accepts files in any order; no module blocks on a missing dump
Legacy registration, renewal, CESS ledger and claims extracts	Wave-1 extract Wk 1; wave-2 extract Wk 10	Extract request at signing; profiling starts on partial extracts
DLT template approvals for every notification event type	Phase 1 templates Wk 4	Submitted from Week 1; in-app and WhatsApp legs continue even if SMS approval lags
UAT participants nominated by APBOCWWB for each phase	Two weeks before each UAT window	UAT calendar fixed at inception; daily facilitation and prepared scripts and data
Empanelled VAPT auditor appointment	Week 3	Auditor engaged at signing; internal security testing runs in parallel from Week 2
Centre hardware for kiosk, token print, QR scan and camera (excluded from this scope)	Phase 1 centres, Wk 5	Software degrades gracefully to on-screen token and operator-entered visit record

4.6.2 Internal dependency rules

- Identity, RBAC and Master Data & Configuration (SM-19 core) complete before any module screen is built.
- The four shared services — Case Service, Audit Ledger, Consent Manager, Document Vault — complete in Sprint 1 before module build begins in Sprint 2.
- The Integration Hub core is in place before any adapter is built; adapters are built against the Hub contract and mocks, never as point-to-point calls.
- Migration wave-1 sign-off (Section 7) completes before Phase 1 Go-Live — masters are loaded before operators serve workers.
- No module goes live with an uncertified fallback route, and no phase gate is approved with an open Critical or Major VAPT finding.

4.7 Critical path

The critical path to Certified Final Go-Live runs through Phase 1, because Phase 1 carries the platform core, the identity-bearing registration module, and the first wave of departmental integrations that every later phase builds on: hosting provisioning → the ten shared services → the SM-01 registration wizard (including UIDAI e-KYC and the first integration wave) → migration wave 1 → VAPT round 1 → Phase 1 UAT sign-off → Phase 1 Go-Live. From Phase 2 onward the critical path shifts to Scheme Rule Catalogue validation and migration wave 2 ahead of Phase 3, then to final VAPT closure, the DR drill and the 300-node performance test ahead of certification.

Activities with float — portal visual polish, LMS content loading, site-catalogue seeding, and the cut-over of any individual source from fallback to API — sit off the critical path, because a certified fallback route is by design an acceptable state at certification.

4.8 Recovery mechanism

Eight early-warning indicators — sprint-completion rate, integration readiness, defect density and ageing, UAT pass rate, dependency ageing, migration reconciliation exceptions, VAPT closure, and team availability — are reviewed weekly by the Programme Manager and reported to the PMU. Any indicator outside its agreed band becomes a RAID entry with an owner and a recovery action.

Six recovery levers are applied, strictly in this order:

Order	Lever	What it does
1	Re-sequence within the phase	Move sprint content between sprints of the same phase; pull forward items with no dependency
2	Fallback-first Go-Live	A module goes live on its certified fallback route for the affected element; the API is cut over later, with no worker impact
3	Parallel team surge	Additional build/test capacity deployed to the lagging workstream from the Bidder's bench

Order	Lever	What it does
4	Extended working	Bounded extended shifts, with key-personnel rotation to protect availability
5	Scope deferral within RFP phases	A module's completion scope (for modules spanning two phases) moves to its later listed phase — never out of its first-listed phase
6	Re-baseline	Only if levers 1–5 cannot hold a gate; goes to the Steering Committee with a formal re-baseline paper

Descoping of any approved module, or of any core obligation, is explicitly not a recovery lever and is never proposed.

Worked example. A three-week delay in GSWS API credentials was absorbed in the reference plan without moving the Phase 1 gate: the affected registration fields went live on the certified F1 document-evidenced route with automatic re-verification queued; when credentials arrived in Week 6, the adapter entered Shadow mode, moved to Canary in districts with active field-support presence, then to statewide Automated primary by Week 8 — with every value captured under fallback automatically reconciled against the live API on cut-over.

4.9 Governance cadence during delivery

Cadence	Forum	Content
Daily	Stand-up	Blockers, sprint progress, dependency status
Weekly	Status report and meeting	Milestone status, early-warning indicators, RAID, dependency register
Fortnightly	Sprint review and planning	Demonstration of completed scope, acceptance of sprint outputs, next sprint plan
At each phase gate	Phase-gate review	Exit-criteria evidence, Go-Live decision, conditions
Monthly	Steering Committee	Programme health, escalations, re-baseline decisions, change control

4.10 After Certified Final Go-Live

Hypercare continues for an agreed number of weeks with daily defect review and on-site cluster presence. The support model of Section 11 becomes the operating steady state; O&M SLA reporting starts from the certification date. Any source still on a certified fallback route at certification continues to be cut over under Section 6.10, with monthly status reported to APBOCWWB. Monthly source deposits continue throughout operations, and knowledge transfer (Section 14.5) continues on a rolling schedule so that exit, whenever it happens, is a formality rather than an event.

4.11 Sprint-level plan

The sprint plan below follows the timeline in Figure 4.1 and assigns the platform and module work to each sprint. Sprints S3, S5, S7 and S9 are the hardening sprints that close each phase.

Sprint	Weeks	Build and delivery focus	Exit evidence
S1	1 to 2	Inception outputs. Monorepo, pipeline and Development and Staging environments. Identity realms and RBAC. Master Data & Configuration core. Case Service, Audit Ledger, Consent Manager and Document Vault.	Four shared services running in Staging. Contract baseline `v0` for Phase 1 modules. Database platform decision recorded (M2).
S2	3 to 4	Integration Hub core with the UIDAI sandbox adapter and wave-1 adapters built against mocks. Worker Longitudinal History, Household Master and Notification Engine core. SM-01 wizard and SM-04 intake.	UAT environment live (M3). Platform core of 10 services demonstrable in UAT, Production and DR provisioned (M4).
S3	5	Hardening. Completion of SM-05, SM-06 and SM-11. Migration wave-1 reconciliation. Fallback route certification and VAPT round 1 retest.	Phase 1 UAT sign-off, zero Critical and Major defects, Phase 1 Go-Live (M5).
S4	6 to 7	SM-07 core, SM-09 core, SM-13 reports and SM-22 surfaces. Phase 1 fallback sources moved to Shadow. Dump ingestion online for the first 12 dump sources.	Phase 1 hypercare exit (M6). Phase 2 UAT entry (M7).
S5	8 to 9	Hardening. SM-15 mela operations, completion of SM-12 and SM-14, field pilot for mela flows, Phase 2 UAT cycles.	Phase 2 Go-Live (M8).
S6	10 to 11	SM-02 Scheme Engine with the 38-scheme catalogue, SM-03, SM-08 and SM-10. Migration wave 2.	38-scheme catalogue validated with APBOCWWB (M9). Wave 2 complete (M10).
S7	12 to 13	SM-20 and SM-21. Completion of SM-07 and SM-09. Scheme-department APIs, RTGS route and CSC and MeeSeva. Hardening and Phase 3 UAT.	Phase 3 Go-Live (M11).
S8	14 to 15	SM-16 and SM-17. Completion of SM-18 and SM-19. HRMS refresh and legal-aid referral interface where available.	Phase 4 UAT entry (M12).
S9	16 to 17	Hardening. DR drill, 300-node performance test, final VAPT closure, reproducible-build witness, final deposit.	DR and performance evidence (M13). Certified Final Go-Live (M14).

4.12 Release train and environment promotion

Environment	Receives	Frequency	Promotion gate
Development	Every merge to the main branch	Continuous	Unit, contract and lint checks pass.
Staging / QA	Build candidate from main	Daily	Integration suite, end-to-end suite and adapter conformance suite pass. Static, dependency, secrets and container scans clean of Critical and High findings.
UAT	Release candidate	End of each sprint	Regression pass, DAST run, performance smoke test, release notes and configuration bundle approved by the QA Lead.
Production	Certified release	At each phase gate; hotfix on demand	UAT sign-off certificate, VAPT closure, rollback plan rehearsed, Steering Committee Go decision.

Environment	Receives	Frequency	Promotion gate
Disaster Recovery	The Production artefacts	With Production	Same image digests and manifests as Production. Verified by the DR drill.

A release is a set of image digests, a configuration bundle and a database migration set. All three are promoted together and the same digests are used in every environment, so what UAT tested is what Production runs. Database migrations are written expand-then-contract: a release first adds structures that both the old and new code can use, and a later release removes the old structures. This is what allows a rolling deployment with no downtime and a rollback that does not need a database restore.

4.13 Responsibility matrix for programme deliverables

R is responsible, A is accountable, C is consulted and I is informed. The Bidder Project Director is accountable for every Bidder-side deliverable and is omitted from the columns for brevity.

Deliverable	Programme Manager	Solution Architect	APBOCWWB PMU	Dept. nodal officer	TSU
Inception report and requirement baseline	R	C	A	I	C
Solution design baseline	C	R	A	C	C
Module build and release	A	R	I	I	I
Department onboarding case	A	C	C	R	R
Fallback route certification	R	C	A	C	R
Migration gates G1 to G6	A	C	A	I	C
UAT scripts and sign-off	R	C	A	C	C
VAPT and closure evidence	A	C	I	I	C
Training and DEO certification	R	I	A	I	C
Go/No-Go decision	R	C	A	I	C
Source deposit and reproducible build	A	R	I	I	C

4.14 Early-warning indicator bands

The eight early-warning indicators of Section 4.8 have defined bands. A reading in the amber band is discussed at the weekly status meeting. A reading in the red band becomes a RAID entry with an owner and a recovery action. The bands are proposed here and agreed with the PMU at inception.

Indicator	Green	Amber	Red
Sprint-completion rate	90% or more	80% to 89%	Below 80%
Integration readiness (sources with sandbox and credentials against plan)	90% or more of plan	70% to 89%	Below 70%
Defect ageing (Critical or Major open for more than 5 working days)	None	1 or 2	3 or more
UAT first-pass rate	90% or more	80% to 89%	Below 80%

Indicator	Green	Amber	Red
Dependency ageing (dependencies past their need-by date)	None	Late by up to 3 working days	Late by more than 3 working days
Migration reconciliation exceptions (unresolved, as share of loaded records)	Below 0.1%	0.1% to 0.5%	Above 0.5%
VAPT closure (Critical and High findings)	Closed within 5 working days	Closed within 10 working days	Open beyond 10 working days
Team availability (key personnel)	95% or more	90% to 94%	Below 90%

5. Integration Approach for Intra-Departmental, Government of India and Other AP Government Systems

This section describes how the Digital ILAC Platform connects to APBOCWWB's own systems, to Government of India systems, and to the other Andhra Pradesh Government systems notified for the programme — 50 sources in total, plus the national systems (UIDAI, e-Shram, EPFO, ESIC, NPCI, Income Tax/NSDL, PFMS) the Registration module depends on.

5.1 Integration principles

Principle	What it means in the design
Thin orchestration over systems of record	The Platform never becomes the system of record for departmental data. It fetches, caches with an as-of timestamp, and links — the department remains the authority. Every fetched value carries source, fetch time and consent reference.
Dual-mode convergence	Every integration has two certified routes — an API route and a digitised manual/assisted route — producing the same Case ID, service clock, acknowledgement, document linkage, audit trail, reconciliation entry and MIS visibility (Section 6).
Consent before any external query	The Integration Hub refuses any outbound call that does not carry a valid consent reference for that purpose.
No worker turned away	An integration failure never blocks a worker's request; the Case Service records a documented next step and the request continues on the fallback route.
Facilitation only	No integration performs sanction, rejection, disbursement, adjudication, credit approval or medical judgement — submissions are requests; decisions are read back from the competent authority.
API-first with published contracts	Every interface has an OpenAPI 3.1 specification in the Authority-controlled repository, built against a mock, then a department sandbox, before going live.
Geography-harmonised	Every location attribute from every source is resolved to Local Government Directory (LGD) codes before it is stored or used.
Everything reconciled, everything observable	Each source has a scheduled reconciliation job, a mismatch queue and a health panel — silent drift is treated as a defect.
India-resident processing	All adapters, queues, staging stores and payload snapshots run inside the five India-resident environments.

5.2 Integration Hub (SM-12) architecture

The Integration Hub is the single component through which all departmental data enters or leaves the Platform — no functional module calls a department directly. This gives one place for credential custody, consent enforcement, retries, replay, reconciliation and monitoring, and is what makes the dual-mode guarantee enforceable rather than aspirational.

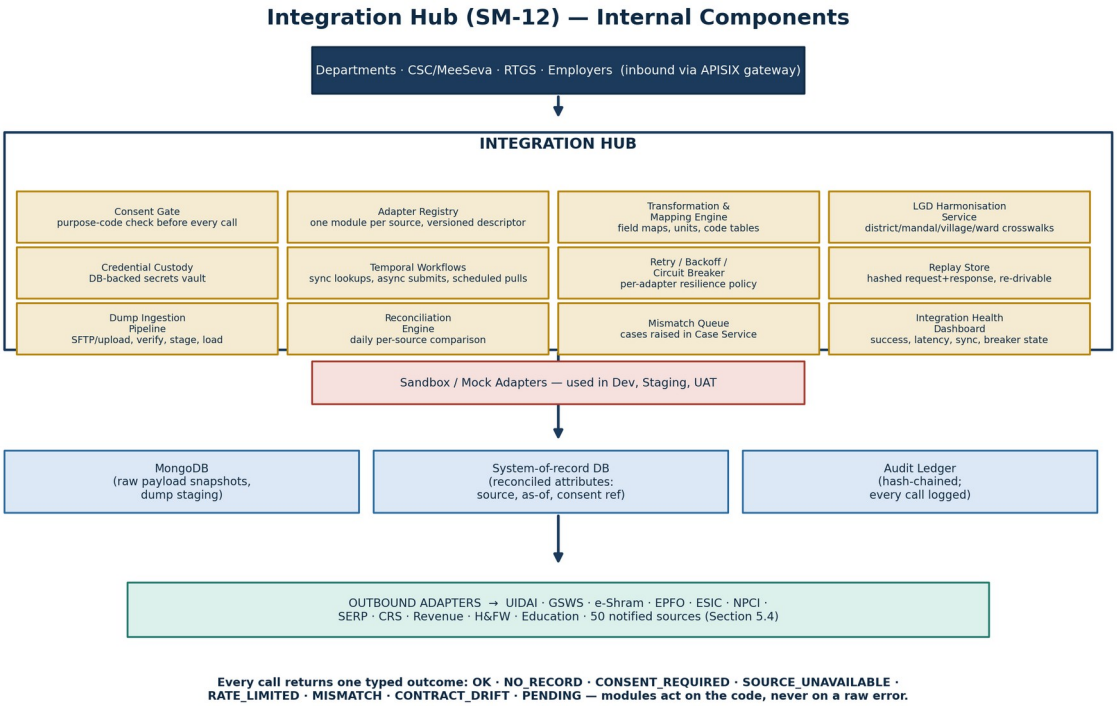


Figure 5.1 — Integration Hub internal components

5.2.1 Adapter descriptor

Each of the 50 sources is declared by a configuration descriptor, not hard-coded: source ID, department, integration pattern, contract reference, authentication method and credential reference (resolved at call time, never logged), consent purpose code, timeout/retry/circuit-breaker policy, LGD crosswalk reference, fallback class (Section 6), reconciliation schedule, and the modules that consume it. Adding a source, or changing how an existing one behaves, is a configuration and adapter change — never a platform release.

5.2.2 Integration patterns

Pattern	Used for	How it behaves on failure
(a) Synchronous real-time lookup	UIDAI e-KYC, GSWS caste/income/ration lookup, NPCI validation, SADAREM disability percentage — a worker is waiting	On timeout or breaker-open, returns SOURCE_UNAVAILABLE; the module records a documented next step and continues on the fallback class; an automatic re-attempt fills the value later
(b) Asynchronous request & callback	Scheme application submission, status tracking, PFMS disbursement confirmation	Where no submission API exists or it is unavailable beyond budget, the application is packaged for manual departmental submission (Class F3)
(c) Scheduled pull & delta sync	Any API source with a daily/monthly/yearly/seasonal cadence — re-verification and bulk enrichment	A missed window is flagged and affected attributes marked stale; no worker-facing operation is blocked
(d) Bulk dump ingestion	16 dump sources plus the dump leg of SERP death data	Receipt → checksum → staging → validation → LGD harmonisation → identity matching → load with lineage; a

Pattern	Used for	How it behaves on failure
		bad file is rejected whole with a report, never partially loaded
(e) RTGS-internal data request	PM Kisan, Annadata Sukhibhava — held by the Real Time Governance Society	Evaluated from Webland/e-Panta landholding data and classified Needs Review until RTGS data confirms
(f) Inbound APIs exposed by the Platform	Worker registration verification, scheme status callback, MIS extract for departments	Rate-limited, versioned, audited under the same gateway controls as every other endpoint

5.2.3 A fixed set of typed outcomes

Functional modules never see departmental error formats. The Hub returns one of eight typed outcomes, and every module handles integration failure the same way — this is what makes "no worker turned away" implementable once rather than 50 times:

Outcome	Module behaviour
OK	Value returned and stored with source and as-of date; continue
NO_RECORD	Department answered definitively that no record exists; treated as a business result, no retry
CONSENT_REQUIRED	Route to the consent step; the source is never called
SOURCE_UNAVAILABLE	Documented next step recorded; continue on the fallback class; automatic re-attempt scheduled
RATE_LIMITED	As SOURCE_UNAVAILABLE, with re-attempt deferred by the department's own retry-after value
MISMATCH	Manual-review flag raised; neither the fetched nor the declared value overrides the other
CONTRACT_DRIFT	Response quarantined; treated as unavailable for the worker; TSU alerted
PENDING	Case remains open with its service clock running; status view shows Under Review

5.3 Integration catalogue — summary

The 50 notified sources, plus the Government of India systems the Registration module depends on, are catalogued in full in the Integration Adapter Guide (Section 14.4), one guide per source, each with its contract, field mapping, credential procedure and certified fallback route. The table below summarises the catalogue by category; the complete 50-row register with per-source data elements, sync cadence and fallback class is held in the Compliance Matrix annex (Section 15) and the Department Onboarding Registry.

Category	Count	Representative sources	Predominant pattern
Identity & citizen data	5	UIDAI e-KYC, GSWS (Integrated Certificate, Income Certificate), CRS, Revenue (marriage)	(a) synchronous

Category	Count	Representative sources	Predominant pattern
			lookup
Formal-sector & financial	6	e-Shram/NDUW, EPFO, ESIC, NPCI, Income Tax/NSDL, PFMS	(a)/(d) lookup + dump
Welfare & scheme sources	19	GSWS scheme services (Talliki Vandanam, Shaadi Tohfa, Matsyakara Bharosa, P4), SERP, Food & Civil Supplies (AePDS, Deepam), WCD (Bal Sanjeevani), Housing, Minority Welfare	(c)/(d) scheduled pull / dump
Health & disability	3	SADAREM, Aarogyasri / PM-JAY	(a)/(d)
Land, property & agriculture	9	Webland, e-Panta, Urban Property & Property Tax, CCRC, Vahan, crop-sale data (Markfed, AP State Civil Supplies)	(d) dump
Education	6	School, Intermediate, Technical, Higher Education, APCFSS College Fee Database	(a)/(d)
Employment schemes	2	MGNREGA (GSWS-held and PRD-held copies)	(d) dump
Central RTGS-routed schemes	2	PM Kisan, Annadata Sukhibhava	(e) RTGS-internal
Internal APBOCWWB / HR	3	Legacy registration database, CESS ledger, cess-payer register, HRMS	(a)/(d)
Channel & utility	5	DLT SMS gateway, WhatsApp Business Cloud API, IVR/helpline telephony, meeting-link provider, email	Not a data source

Across the 50 notified sources: 31 offer a live API, 16 supply periodic dumps, one (SERP death data) supplies both, and two (PM Kisan, Annadata Sukhibhava) are RTGS-internal only. Sync cadence ranges from daily to yearly to one-time; two catalogued schemes are currently inactive and are retained for reconciliation only.

5.4 Aadhaar and UIDAI integration mechanics

The Platform operates under APBOCWWB's AUA/KUA registration, or as a sub-AUA as APBOCWWB directs, with OTP-based authentication and e-KYC only — no biometric device dependency, consistent with the exclusion of biometric hardware from this scope. Only the four e-KYC fields (Full Name, Date of Birth, Gender, Permanent Address) are consumed and stored, as read-only fetched attributes with source and timestamp; corrections are directed to UIDAI. The Aadhaar number itself is never stored in clear — a reference key is used for every subsequent lookup, and the number is masked in every interface, log, export and printed document. Every authentication and e-KYC transaction is logged with its UIDAI transaction ID, timestamp, outcome and consent reference. Where OTP fails or UIDAI is unavailable, registration continues on the certified F1 route (operator verification of the physical Aadhaar copy) with automatic re-verification when the interface recovers — the worker is never turned away.

5.5 Geography (LGD) harmonisation

Every source describes location differently — GSWS secretariat codes, Revenue village codes, Municipal ward numbers, PRD panchayat codes. The Master Data & Configuration Service holds the LGD hierarchy

for Andhra Pradesh (State, District, Mandal, Village/Town, Ward) with names in English and Telugu, plus a crosswalk table per department mapping that department's own codes to LGD codes. A single resolution service converts any incoming location tuple to an LGD code with a confidence level — deterministic code matches take priority, and unmapped codes are queued for resolution rather than silently dropped. This feeds jurisdiction assignment at registration, Service Area definitions, cross-department reconciliation, and geographic targeting for broadcasts.

5.6 Department onboarding process

Each source goes through an eight-step onboarding process, itself tracked as a case on the Platform so its status is visible on the Integration Health Dashboard: nodal officer nomination → data-sharing agreement → field-mapping workshop → credential issuance into the secrets vault → sandbox testing against the adapter contract → UAT with departmental participation, including a demonstration of the fallback route → production enablement with monitoring live → post-Go-Live review after the first full sync cycle.

5.7 Reconciliation and monitoring

Every source with a cadence has a scheduled reconciliation job comparing the Platform's held attributes against the latest source response or dump — daily for daily/custom sources, on receipt for dumps, within one day of sync for monthly/yearly sources. Every difference is classified (source-changed, platform-changed, conflict-with-declaration, unmatched-identity) and handled accordingly; a conflict with a worker's own declaration is never silently overridden — it becomes a case in the Needs Review queue with a service clock and an owner. Integration health — success rate, latency, circuit-breaker state, last successful sync, fallback volume — is exported through OpenTelemetry to dashboards used by the central helpdesk and the seven District Field Support Clusters (Section 11) to explain a delayed verification to a worker.

5.8 Integration delivery plan by phase

Integration delivery is sequenced so that each module has, at its phase, either a live API or a certified fallback route, and so that the Integration Hub core is available from Phase 1. Phase 1 brings UIDAI e-KYC and the first wave of departmental sources live; Phase 2 cuts over the Phase 1 fallback sources and brings dump ingestion online; Phase 3 brings scheme-department APIs, the RTGS route, and CSC/MeeSeva; Phase 4 completes HRMS refresh and legal-aid referral interfaces. The full phase-by-phase source list is in Section 4.2.

5.9 Integration risks and mitigations

Risk	Mitigation
Departmental API not authorised or data-sharing agreement delayed	Fallback route certified for every source from Phase 1; onboarding case visible in governance; APBOCWWB escalation path
API unavailable, degraded or rate-limited in production	Circuit breaker short-circuits to fallback; automatic re-attempt workflow; delta sync fills values later without

Risk	Mitigation
	worker action
Departmental interface contract changes without notice	Runtime schema validation raises CONTRACT_DRIFT before a bad value reaches a worker record; alert to TSU

5.10 Adapter descriptor

Each source is declared by a descriptor like the one below and loaded by the Hub at start-up and on configuration publication. The adapter code implements the descriptor's contract, and the policies in the descriptor decide how it behaves. The values shown are illustrative.

```
{ "sourceId": "SRC-017", "name": "GSWS Integrated Certificate", "department": "GSWS",
  "pattern": "SYNC_LOOKUP", "contract": { "ref": "adapters/gsws/openapi.yaml", "version": "1.3" },
  "auth": { "type": "OAUTH2_CLIENT_CREDENTIALS", "credentialRef": "vault:gsws/prod/client" },
  "consentPurpose": "SOCIAL_CATEGORY_VERIFICATION",
  "policy": { "timeoutMs": 8000, "retries": 1, "breaker": { "failureRatePct": 50, "minCalls": 10,
              "windowSec": 60, "openSec": 30, "halfOpenProbes": 3 }, "bulkhead": 40 },
  "lgdCrosswalk": "xwalk/gsws-secretariat-to-lgd@2026-10",
  "fallback": { "classes": ["F4", "F1", "F6"], "promoteAtClockPct": 60 },
  "reconcile": { "schedule": "DAILY", "keys": ["aadhaarRef", "certificateNo"] },
  "consumers": ["SM-01", "SM-02", "SM-03"], "cutoverState": "AUTOMATED_PRIMARY" }
```

A descriptor change is a configuration publication with the same approval and audit as any other (Section 2.19). A change to the contract version or the pattern also requires the adapter conformance suite to pass in UAT before the new descriptor is published to Production.

5.11 Resilience parameters by integration pattern

The defaults below apply to each pattern and are overridden per source in its descriptor. All delays carry a random jitter of plus or minus 20% so that many workers retrying together do not arrive at a recovering department at the same moment.

Pattern	Timeout	Retries	Circuit breaker	Isolation
(a) Synchronous lookup	Connect 2 s, read 8 s	One immediate retry, for idempotent reads only	Opens at 50% failures over 10 or more calls in 60 s. Half-open after 30 s with 3 probe calls.	Bulkhead of 40 concurrent calls per source
(b) Asynchronous submit	15 s per call	Five attempts by workflow, 30 s rising to 30 min	As above	Idempotency key on every submission
(c) Scheduled pull	30 s per page	Three per page, resume from the last cursor	Opens on three failed windows	One pull per source at a time
(d) Bulk dump	Validation limit 2 h per file	None. A rejected file is re-requested from the department.	Not applicable	One file in validation per source
(e) RTGS-internal	File based	None	Not applicable	Treated as a dump
(f) Inbound APIs	30 s at the gateway	Not applicable	Rate limit per client	Per-client quotas in the gateway

A department that returns a `Retry-After` value is obeyed. The Hub records the time and does not call the source again until it expires, and the outcome returned to the module is RATE_LIMITED.

5.12 Technical profile of principal sources

The table shows the expected integration profile of the principal sources. The authentication shown is the mechanism expected for each class of source. The contract and credentials issued by each department at onboarding (Section 5.6) are what govern, and the Integration Adapter Guide records them per source.

Source	Pattern	Expected authentication	Data consumed	Cadence	Fallback order
UIDAI e-KYC	(a)	AUA/KUA licence, signed requests	Name, date of birth, gender, address, photograph	Real time	F1
GSWS certificates and scheme services	(a), (c)	OAuth2 or API key with allow-listed addresses	Caste, income, ration card, scheme status	Daily	F4, F1, F6
e-Shram / NDUW	(a), (d)	API credentials as notified	UAN, registration status	On demand and periodic	F5, F2
EPFO	(d)	SFTP key pair	Membership status, UAN	Periodic dump	F5, F2 with countersign
ESIC	(a)	As notified	Insurance number, coverage	On demand	F4, F1
NPCI	(a)	As notified	Account validation, name match	Real time	F1, F2
Income Tax / NSDL	(a)	As notified	PAN validation only	On demand	F2
SADAREM	(a)	API key	Disability type and percentage	Daily	F4, F1
CRS and Revenue (marriage)	(a)	As notified	Birth, death and marriage records	Daily or on demand	F4, F1
Webland and e-Panta	(d)	SFTP	Landholding, crop booking	Periodic dump	F5, F1
SERP	(a), (d)	As notified	Death data, self-help-group membership	Daily	F4, F5, F1
Food & Civil Supplies	(a), (d)	As notified	Ration card, family members	Daily or dump	F4, F5, F1
Aarogyasri / PM-JAY	(d)	SFTP	Coverage status	Periodic dump	F5, F1
PFMS	(b)	As notified	Disbursement status	Daily status	F3 manual status
Education departments, APCFSS	(a), (d)	As notified	Enrolment, stage, fee data	Per term	F5, F1
MGNREGA (two copies)	(d)	SFTP	Job card, days worked	Periodic dump	F5, F1
PM Kisan, Annadata Sukhibhava	(e)	RTGS extract	Beneficiary status	Per RTGS extract	F5, F1
DLT SMS, WhatsApp Cloud API, IVR	Provider APIs	API key or OAuth token	Send, delivery report, call events	Real time	Next channel in preference order

5.13 UIDAI OTP e-KYC call sequence

The UIDAI adapter is the only component deployed inside the Aadhaar Data Vault boundary. It receives a reference key, retrieves the number inside the boundary and never returns the number to a caller. The sequence below shows each step, the component that performs it and the data that moves.

No.	Component	Action
1	Client to gateway	The operator or worker enters the Aadhaar number. It travels over TLS 1.2 or higher and is held in memory only, never written to a log or local store.
2	Registration family	Asks the Consent Manager for a valid consent with purpose AADHAAR_EKYC. If none exists the flow returns to the consent step and no further call is made.
3	Aadhaar Data Vault	Exchanges the number for a reference key (keyed one-way function) and stores the number encrypted inside the vault. The reference key and the last four digits replace the number from this point.
4	Integration Hub	Checks the consent gate, resolves the credential reference from the secrets vault and calls the UIDAI adapter with the reference key and the purpose code.
5	UIDAI adapter	Retrieves the number inside the vault boundary and requests an OTP through the AUA/KUA channel. Records the UIDAI transaction identifier.
6	Worker and operator	The OTP reaches the Aadhaar-linked mobile. The operator enters it. It is held in memory only and discarded after use.
7	UIDAI adapter	Submits the OTP with the transaction identifier. UIDAI returns the signed e-KYC response.
8	UIDAI adapter	Verifies the signature. Extracts the four permitted fields and the photograph. Discards everything else. Stores the photograph in the Document Vault.
9	Integration Hub	Writes the `integration_call` row with UIDAI transaction identifier, timestamp, outcome and consent reference. Returns the typed outcome OK with attributes carrying source and as-of time.
10	Registration screen	Shows the fetched values as read-only. The operator confirms. A timeout or UIDAI error returns SOURCE_UNAVAILABLE and the step continues on the F1 route.

5.14 LGD resolution algorithm

The resolution service converts any incoming location tuple to an LGD code with a confidence value. It is deterministic: the same input and the same crosswalk version always give the same result.

```
resolve(dept, tuple):                    # tuple = (code?, state?, district?, mandal?,
village_or_ward?)
  1. if tuple.code and crosswalk[dept][tuple.code]:        return (lgd, 1.00, "CODE")
  2. path = normalise(tuple names)                          # NFC, case fold, strip suffixes, ward -> int
     m = exact_path_match(path, lgd_names_te_en)            # Telugu and English names
     if unique(m):                                          return (m.lgd, 0.95, "NAME_PATH")
  3. parent = deepest_resolved_ancestor(path)
     c = trigram_candidates(path.leaf, children_of(parent), min_sim=0.85)
     if unique_best(c, margin=0.05):                        return (c.lgd, 0.80 + 0.14*c.sim, "FUZZY")
  4. queue_for_steward(dept, tuple)                         return (parent.lgd, 0.00, "UNRESOLVED")
```

Confidence	Handling	Where it applies
0.95 and above	Accepted	Stored without review. Used for jurisdiction assignment and reconciliation.
0.80 to 0.94	Accepted with a review flag	Stored with the flag. Not used to assign a Service Area until the steward confirms.
Below 0.80	Unresolved	The nearest resolved ancestor is stored. The tuple joins the steward queue and the source row is marked for retry after the next crosswalk publication.

Normalisation removes zero-width joiners from Telugu text, unifies common transliteration variants (for example a and aa, v and w), strips administrative suffixes such as rural and urban markers, and converts ward numbers to integers. The normalisation rules are configuration and are versioned with the crosswalk.

5.15 Dump file contract

Every dump source follows one contract so that ingestion is the same code for every department. The contract is recorded in the schema registry entry of the source and signed off at onboarding.

Element	Specification
Landing	SFTP or secure upload into a per-source landing area reachable only over the VPN. One directory per source. Files are write-once.
Naming	`{sourceId}_{yyyymmdd}_{sequence}.{csv|jsonl}.gz` with a companion `.manifest.json`.
Encoding	UTF-8 with a header row. Dates in ISO 8601. Delimiter, quoting and null representation declared in the registry.
Manifest	Source identifier, extract time, row count, SHA-256 of the data file, schema version and the department's own file sequence number.
Receipt	The checksum is verified on receipt. A mismatch rejects the file and raises an alert. A gap in the sequence number raises a warning.
Validation	Header against the registered schema, row count against the manifest, column types, mandatory columns, and a reject threshold per source. A file above the threshold is rejected whole with a report.
Load	Staged in MongoDB, then LGD mapping, identity matching and attachment to the Worker Master with the as-of date and full lineage.
Retention	Raw files are kept in the Document Vault under the retention schedule. Prior values on the worker record are retained, not overwritten.

```
{ "sourceId": "SRC-034", "extractedAt": "2026-12-01T21:00:00+05:30", "rows": 1842211,
  "sha256": "9f2c1b0e...a7d4", "schemaVersion": "2.1", "sequence": 118,
  "file": "SRC-034_20261201_0118.csv.gz" }
```

5.16 Onboarding evidence per source

Each of the eight onboarding steps of Section 5.6 produces an artefact that is stored with the onboarding case. The case is not marked complete until every artefact exists.

Step	Activity	Evidence artefact	Owner
1	Nodal officer nomination	Nomination letter and contact details	Department, PMU
2	Data-sharing agreement	Signed agreement with purpose, fields and retention	PMU, Department
3	Field-mapping workshop	Signed field map and code tables	Integration Lead, Department
4	Credential issuance	Credential reference in the secrets vault, rotation date	Integration Lead, Security Lead
5	Sandbox testing	Conformance suite report against the adapter contract	Integration Lead

Step	Activity	Evidence artefact	Owner
6	UAT with the department	Test script results and fallback route demonstration	QA Lead, Department
7	Production enablement	Cut-over state record and monitoring dashboard link	TSU Head
8	Post-Go-Live review	Review note after the first full sync cycle	Integration Lead

6. Manual / Assisted Fallback Design and Later API Cut-Over Approach

This section describes how the Platform guarantees that no worker or dependant is ever denied, deferred, suspended or turned away because an external interface is absent or failing, and how each manual or assisted route is later replaced by an automated interface without loss of case history.

6.1 Design principle — parity between manual and automated transactions

The Platform treats fallback as a first-class operating mode of every module, not as an exception branch. The same Case Service, the same case schema and the same downstream services are used regardless of how a data element was obtained, and every manual or automated transaction shares seven identical properties: the same case identifier, the same service clock, the same acknowledgement, the same document linkage, the same audit trail, the same reconciliation workflow, and the same MIS visibility.

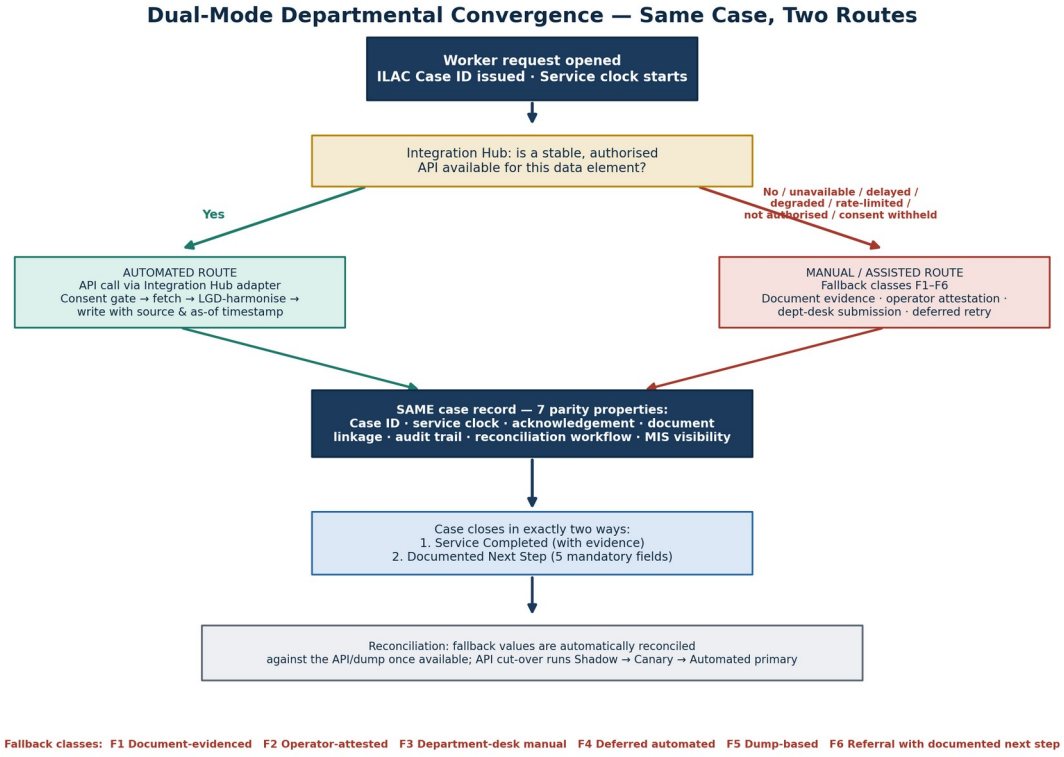


Figure 6.1 — Dual-mode convergence: one case, two routes, one outcome

6.2 Six fallback classes

Every situation in the RFP, the module specifications and the 50-source list is covered by one of six fallback classes. Class selection is not left to individual judgement — the Master Data & Configuration Service holds, for every data element and request type, the ordered list of permitted classes, and the operator screen offers only those classes in that order.

Class	When used	Who may perform it	Reconciliation
F1 Document-evidenced	A physical or previously digitised document exists that the worker can present	Worker (self-upload) or Centre Data Entry Operator (assisted)	Compared with the fetched value once the API/dump becomes available

Class	When used	Who may perform it	Reconciliation
F2 Operator-attested	No document is available and the element is needed to continue the service	Centre Data Entry Operator, with supervisor countersignature on high-impact elements	Mandatory — the element is marked provisional until reconciled
F3 Department-desk manual submission	An application must reach a department but no submission API exists, or is not authorised	Centre Data Entry Operator or APBOCWWB staff	Status entered manually until an API exists; each entry is a reconciliation item
F4 Deferred automated	The interface exists and is authorised but is unavailable, delayed, degraded or rate-limited right now	System (Integration Hub) — automatic retry	Automatic — the retry replaces the placeholder and the Scheme Engine re-evaluates if needed
F5 Dump-based	The department supplies data as periodic dumps rather than a live API	System (Integration Hub)	Value marked "as of [dump date]"; reconciled automatically on the next dump
F6 Referral with documented next step	The request cannot be completed at this contact, location or role	Any authorised operator, counsellor or field officer	Case stays open with its clock running until the next step is completed

6.3 Fallback route per source

Each of the 50 notified sources and each Government of India system has a defined, certified fallback route — the primary mode, the permitted fallback classes in order, the accepted evidence document, the attesting role, and the trigger for reconciliation. Representative examples:

Source	Primary mode	Fallback path	Evidence / reconciliation
GSWS (caste, income, ration)	API, daily	F4 → F1 → F6	Integrated / Income Certificate, ration card; next daily sync
UIDAI e-KYC	API, real-time	F1 (physical Aadhaar copy + OTP)	Automatic re-verification workflow
EPFO membership status	Dump	F5 → F2 (supervisor countersign mandatory)	Next dump or live API when authorised
Scheme application submission (no dept. API)	F3, manual desk	—	Status entered manually; reconciled on API cut-over
SERP death data	API and dump, daily	F4 → F5 → F1	Death certificate; next daily sync
PM Kisan / Annadata Sukhibhava	RTGS-internal	F5 → F1	Reconciled on RTGS extract

The full 50-row Fallback Register is maintained as a live configuration dataset and is available to APBOCWWB in the Compliance Matrix annex and through the Reporting Layer.

6.4 Fallback per module

Every module has a stated fallback pattern reusing the same six classes: assisted registration proceeds by document upload when an external fetch fails (SM-01); scheme applications route to a manual desk pack when no submission API exists (SM-02); grievances raised on any channel, including a paper register at an offline centre, share one Ticket ID (SM-04); engagement confirmations can be recorded by an operator from a signed slip where an employer has no app access (SM-05); every field function at a worksite, household or camp captures fully offline with deferred synchronisation (SM-11, SM-15, Section 10).

6.5 The manual case record

Manual and assisted cases use exactly the same case schema as automated cases — mode (automated/assisted/manual), fallback class, reason code, the operator's identity and role, the evidence document references, the reconciliation status, and the service clock are all fields on the same record, not a separate register. A committed step is never edited; a correction is always a new linked step, so the record an auditor sees is never silently rewritten.

6.6 Service clocks and escalation under fallback

The service clock starts at intake on every channel and in every mode — a grievance received by the helpline and keyed by an operator has exactly the same clock as one lodged through the app. The clock does not pause for internal reasons such as an interface outage; a deferred (F4) request that has been pending too long is automatically promoted to F1 or F2, prompting the operator to obtain the element directly rather than let the worker wait indefinitely. Escalation on breach follows the same L1/L2/L3/Board pattern used for support tickets (Section 11), with the full case history moving with the escalation.

6.7 Reconciliation workflow

A scheduled job runs at least daily, checking every pending fallback item against the now-available source. A match clears the provisional flag; a mismatch goes to a Needs Review queue for a human decision — neither the fetched value nor the worker's declaration is ever silently overwritten. Reconciliation status, by source, fallback class, reason code, district, centre and operator, is a standard report available to APBOCWWB through the Reporting Layer.

6.8 Certification of fallback routes

A module is never proposed for Go-Live until every fallback route it depends on is certified — a repeatable process, not a one-time document: a written Standard Operating Procedure per route in Telugu and English; a UAT test script executed with the interface simulated as absent, unavailable, delayed, degraded and rate-limited, demonstrating all seven parity properties; an entry in the Fallback Register with SOP version and test evidence; formal sign-off by APBOCWWB; and a training pack with an assessed and certified operator register.

6.9 API cut-over approach

Cut-over is the controlled replacement of a fallback route by a live API, run per source and, where useful, per district, through four states:

State	Behaviour
Manual only	All requests use the certified fallback route
Shadow	The API is called in parallel with the fallback route; results are compared but never used for any case decision — no worker-facing change
Canary	The API becomes primary for a selected subset of districts, chosen where field-support presence is active; comparison continues elsewhere
Automated primary	The API is primary statewide; the fallback route remains certified and is entered only on the triggers of Section 6.1

On cut-over, a one-time backfill reconciliation runs against every case step still carrying a provisional value for the newly automated element — matches are confirmed silently, mismatches go to Needs Review, and any resulting eligibility change is notified to the worker. At any state, if comparison criteria fail or a department withdraws an interface, the source returns to its previous state the same working day by a configuration change — no code deployment is needed.

6.10 Reverse cut-over on API degradation

Degradation is detected automatically by the circuit breaker on each adapter. First response: new requests are queued as deferred (F4) with the clock still running. If the outage persists towards the service-clock breach threshold, the Case Service opens operator tasks offering the certified F1/F2 route. If the outage persists further, the source can be set to "Fallback active, API suspended" so every new request uses the fallback route directly, with no F4 queue, until recovery is confirmed — this is a design principle stated plainly: a degrading department never keeps a worker waiting.

6.11 Fallback governance

The Fallback Register — one row per source and data element, one row per module scenario — holds primary mode, permitted classes, evidence lists, cut-over state, certification date and approver, all versioned and audited. A monthly review with APBOCWWB covers fallback share per source and district, reconciliation ageing, the Needs Review backlog, degradation episodes, and the cut-over pipeline, tracked against key performance indicators including fallback share, deferred share, reconciliation ageing, mismatch rate, next-step completion rate, and cumulative cut-overs completed.

6.12 Fallback fields on the case record

There is no separate manual register. The fields below are columns on `case_step` and apply to every step in every mode, which is what allows a single report to compare automated, assisted and manual work.

Field	Type	Meaning
`mode`	Enumeration	AUTOMATED, ASSISTED or MANUAL.

Field	Type	Meaning
`fallback_class`	Enumeration or null	F1 to F6 when a fallback route produced the step.
`reason_code`	Code	Why the route was used: SOURCE_UNAVAILABLE, NO_API, NOT_AUTHORISED, DUMP_ONLY, WORKER_DOCUMENT, REFERRAL and similar.
`operator_id`, `operator_role`	Reference	The person who performed the step, and the role they held at the time.
`evidence_refs`	Document identifiers	Documents in the Document Vault supporting the step, with their SHA-256.
`countersigned_by`	Reference or null	Supervisor who countersigned a high-impact F2 element. Mandatory where the Fallback Register requires it.
`is_provisional`	Boolean	True until reconciliation confirms the value.
`reconciliation_status`	Enumeration	PENDING, MATCHED, MISMATCH_REVIEW, RESOLVED or NOT_REQUIRED.
`reconcile_after`	Timestamp	Earliest time the reconciliation job will compare the value.
`received_at`	Timestamp	Original intake time for a paper or offline capture. The clock starts here.
`promoted_from`	Enumeration or null	The class from which the step was promoted, for example F4 promoted to F1.
`sop_version`	Text	Version of the SOP the operator followed.

6.13 Class selection and promotion rules

The operator screen offers only the classes the Fallback Register permits for the data element, in the register's order. The rules below apply the same way in every module.

Situation	Rule	Timing
Hub returns SOURCE_UNAVAILABLE or RATE_LIMITED	The value is queued as F4 and the request continues on the next permitted class if the element is needed to proceed.	Immediate
An F4 request is still pending	The case opens an operator task to obtain the element by F1 or F2. The operator may decline and keep waiting for F4 only where the Register allows.	At 60% of the case clock
Source set to "Fallback active, API suspended"	F4 queueing is skipped and the first non-automated class is offered directly.	From the time of the setting
F1 chosen	A document of an accepted type from the Register is attached and hashed. The operator confirms the visible values match the document.	At capture
F2 chosen for a high-impact element	A supervisor countersignature is required before the element can be used in a decision. High-impact elements are identity, bank account, income, disability and nominee.	At capture

Situation	Rule	Timing
No submission API (F3)	The Platform produces a desk pack: the application, the document checklist, a cover sheet carrying the Case ID and a QR reference. The operator submits it and records the acknowledgement.	At submission
Dump source (F5)	The value is shown with the dump date. It is never described as current.	On every display
Request cannot be completed here (F6)	A Documented Next Step is recorded with the five mandatory fields and the case stays open with its clock running.	At closure

6.14 Cut-over promotion and rollback criteria

The CutoverControl workflow (Section 2.17) moves a source from one state to the next only when its criteria are met and recorded. The agreement rate is the share of compared values that match between the API and the certified fallback route, excluding comparisons where both returned NO_RECORD.

Transition	Entry criteria	Evaluation window	Rollback trigger
Manual only to Shadow	Adapter passes the conformance suite. Credentials issued. Consent purpose mapped. Monitoring panel live. Department informed in writing.	None	Not applicable
Shadow to Canary	Agreement rate of 98% or more over at least 2,000 comparisons (or all calls where volume is lower). No CONTRACT_DRIFT in the last 5 working days. p95 latency inside the descriptor budget.	At least 10 working days	Agreement below 95% on any 2 consecutive days
Canary to Automated primary	Canary districts stable for at least 10 working days. Error rate below 2%. Fallback share in canary districts below 5%. No unresolved backfill mismatch older than 5 working days. Cluster Lead confirms.	At least 10 working days	Breaker open for more than 4 hours in a working day, or a department withdrawal
Any state back to the previous state	Operator with the Integration Lead role sets the state with a reason.	Same working day	Not applicable

6.15 Fallback SOP template

Every certified route has one SOP in Telugu and English built from the same template, so an operator reading any SOP finds the same sections in the same order.

Section	Content
Purpose and scope	The data element or request type, the module, and the situations in which the route applies.
Trigger	The exact condition that offers the route, including the outcome code the operator will see.
Who may perform it	Roles permitted, with the countersignature rule where it applies.
Evidence	Accepted document types with an example of each and the checks the operator makes against the original.
Steps	Numbered screen-by-screen steps with the field names exactly as shown on the screen.

Section	Content
What the worker is told	The wording used for the acknowledgement, the next step and the expected time.
Reconciliation	What happens when the source recovers, and what the operator does with a mismatch.
Escalation	Who to call and which Ticket category to use if the route does not work.
Records	The fields the Platform records and where the evidence is kept.
Version and approval	SOP version, approver, certification date and the test script that proved it.

6.16 Fallback key performance indicators

Indicator	Definition	Reported by
Fallback share	Steps with a fallback class divided by all steps for the same element, in the period.	Source, district, centre, module
Deferred share	F4 steps divided by all fallback steps.	Source, district
Reconciliation ageing	Median and 90th percentile age of steps still PENDING.	Source, fallback class
Mismatch rate	Steps in MISMATCH_REVIEW divided by steps reconciled in the period.	Source
Next-step completion rate	Documented Next Steps completed on or before their target date divided by all due in the period.	Module, district, owner role
Cut-overs completed	Cumulative count of sources in Automated primary against the plan.	Phase
Needs Review backlog	Open Needs Review items and the age of the oldest.	Source, owner

7. Data Migration, Golden Record, Data Quality and Reconciliation Plan

7.1 Objectives

- Establish one lifelong Worker Master and one Household Master for every registered construction worker, with the BOCW Registration Number preserved as the immutable permanent identifier.
- Bring every legacy address, centre and jurisdiction onto LGD codes so that Service Area matching and departmental exchange work from day one.
- Remove duplicates and resolve conflicting values through documented survivorship rules — never by silent overwrite.
- Tag every migrated record with its consent status, so no external departmental query is ever made for a worker without a logged DPDP consent.
- Seed the Worker Longitudinal History with a migrated Registration event per worker, so the timeline is complete from day one of Platform operation.
- Prove the migration with control totals, field-level reconciliation and a CESS ledger balance reconciliation, before any cut-over is certified.

Departmental systems of record are not migrated and are not replaced — values fetched or dumped from a department remain owned by that department, with source and lineage preserved. No statutory sanction, rejection or disbursement decision is inferred during migration, and no biometric templates of any kind are migrated or created.

7.2 Migration governance

A Data Governance Committee, chaired by an APBOCWWB Data Owner and including the Migration Lead, an APBOCWWB Data Steward and the Security Officer, meets weekly during migration and monthly thereafter. It approves cleansing rules, matching thresholds, survivorship rules, reconciliation tolerances, and each sign-off gate, recording every decision in a numbered decision log deposited with each phase gate.

Gate	What it proves
G1 Source inventory & profiling accepted	Every source profiled for completeness, validity, uniqueness, consistency and referential integrity
G2 Rules accepted	Cleansing rules, deduplication rulebook and survivorship matrix approved by the Data Governance Committee
G3 Mock migration 1 accepted	Control totals and rule-hit statistics on a representative subset, in Development
G4 Mock migration 2 accepted	Full reconciliation at full volume, in UAT, including APBOCWWB's own verification
G5 Dress rehearsal accepted	Timed runbook executed end to end, including a tested rollback
G6 Cut-over certified	Live load reconciled, CESS balance reconciled, WLH and consent counts reconciled

7.3 Source inventory

In scope: the legacy worker registration database, the CESS contribution ledger, legacy scheme claim and disbursement records, the centre/district/mandal/village lists, the staff and HRMS records, the 38-scheme catalogue and Government Orders, and 16 departmental dumps (Aarogyasri, HRMS, Urban Property, CCRC, Housing, MGNREGA from two sources, Vahan from two sources, EPFO, university enrolment, Need Assessment Survey, Benefit for Imams and Mouzzains, historical vehicle registration, Sunna Vaddi, pension transaction data). Every source is profiled for completeness, validity (format rules), uniqueness (duplicate rate on Registration Number, Aadhaar reference, mobile, UAN), consistency and referential integrity before a single cleansing rule is written.

7.4 Deduplication and matching

Matching runs in three tiers. Deterministic matches on the Aadhaar reference key, the BOCW Registration Number, or the e-Shram UAN merge automatically. A composite deterministic match (mobile + date of birth + normalised first name) also auto-merges, subject to a review trigger if one mobile number appears on an unusually large number of records. Probabilistic matches — on normalised name, date of birth, relationship name, village and gender, using phonetic and trigram similarity — are never auto-merged on identity-critical fields; every probabilistic match goes to a human review queue. Every merge is reversible: pre-merge records are retained with the merge decision as a visible event on the worker's timeline.

Household clustering. A household cluster forms when records share a ration card number, the same LGD-coded address, or a spouse or parent-child link. Clusters spanning more than one legacy registration are confirmed through the review queue; every worker who has no cluster still gets a single-member household, so that every worker has a household record consistent with the household being the unit of convergence.

Survivorship rules. When a field has more than one candidate value, a documented rule decides the surviving value — for example, identity fields prefer a UIDAI-verified value over a legacy declaration; the BOCW Registration Number keeps the earliest-issued number, with later numbers recorded as aliases; Days Worked is never summed across sources, only the most recent 12-month declaration is used. Non-surviving values are retained in lineage, never deleted.

7.5 Golden record construction

A ten-stage pipeline runs for every source: extract with checksum → stage raw → validate against format rules → cleanse against the approved rule catalogue → match and cluster → apply survivorship → load to the system of record within one transaction per household → seed the Worker Longitudinal History → run a baseline Scheme Engine evaluation so eligibility results are populated before Go-Live → tag consent status.

Every golden field carries lineage — source, source reference, extract timestamp, rule IDs applied, and a verification status of Verified, Migrated-unverified, or Under review. A migrated-unverified value is

displayed to operators with a visible marker and is confirmed or corrected at the worker's first visit, renewal or login after Go-Live, triggering the standard OTP/e-KYC/NPCI verifications at that point — migration does not claim to verify what it has not verified.

7.6 Departmental dump ingestion

Each of the 16 dump sources follows the same pipeline: schema registration and sign-off → secure receipt with checksum → staging → LGD mapping → identity matching against the Worker Master → as-of dating (a value is never presented as current without its date) → attachment with full lineage → refresh on the next extract, with prior values retained, not overwritten. A dump value that conflicts with a worker's own declaration produces a Needs Review outcome, never an automatic downgrade of eligibility.

7.7 Consent and privacy in migration

Legacy records are inspected for any recorded consent; where evidence exists it is scanned into the Document Vault and a Consent Record is created with its original date. Where none exists, the record is tagged "consent required" and the Platform makes no external query for that worker until consent is captured at first contact — migration itself performs no external calls; it only loads what APBOCWWB already holds and what departments have already supplied under their own authority.

7.8 Reconciliation plan

- Control totals for every source and run: extracted = loaded + rejected + merged-away + deferred. Any imbalance blocks the gate.
- Field-level reconciliation with sampled value-by-value comparison, zero tolerance on identity-critical fields.
- CESS ledger reconciliation to the legacy ledger, countersigned by APBOCWWB accounts; unlinked entries are retained, never dropped.
- Scheme-status reconciliation, counting applications per scheme per status before and after migration.
- WLH and consent reconciliation — the number of migrated Registration events equals the number of loaded Worker Masters.

7.9 Migration execution plan

Cycle	Environment	Purpose	Gate
Profiling	Staging	Full extracts profiled; rules designed	G1
Mock migration 1	Development	Representative subset; pipeline and rules proved	G3
Mock migration 2	UAT	Full volume; full reconciliation and APBOCWWB UAT	G4
Dress rehearsal	UAT (production-like)	Timed runbook, rollback test, go/no-go checklist	G5
Cut-over	Production	Live load plus final delta	G6

From an agreed freeze point, the legacy system accepts no new registrations; urgent transactions are captured on a running paper register and entered as assisted transactions with full Case ID discipline immediately after cut-over. A database backup is taken and point-in-time recovery verified before every load; rollback is tested end to end in the dress rehearsal.

Worker Master, Household Master, Centre and Service Area masters, staff and consent tagging are loaded for the Phase 1 Go-Live. Scheme catalogue, claims history and the Scheme Engine baseline evaluation are completed for the Phase 3 Go-Live of SM-02. Departmental dump ingestion runs per source as departments supply files, across Phases 1 to 3.

7.10 Ongoing data quality

A data-quality dashboard in the Reporting Layer tracks completeness, verification rate, duplicate rate, open review items, LGD coverage, dump freshness and consent coverage — every dashboard shows the data's last-refresh timestamp. New review items generated after Go-Live route to the same TSU-operated stewardship workflow used during migration, and a monthly data-quality review with APBOCWWB covers the dashboard, rule-hit trends, and any proposed rule changes.

7.11 Legacy-to-target mapping

The table shows how each legacy source maps to the target entities. Field-level maps are produced during profiling (gate G1) and signed at gate G2. The mapping below fixes the entity-level decisions.

Legacy source	Target	Principal transformation
Worker registration database	`worker`, `worker_address`, `worker_employment_profile`, `worker_nominee`, `household`	Names normalised in Telugu and English. Dates parsed from every legacy format to ISO 8601. Addresses resolved to LGD codes. Registration Number preserved as the immutable identifier. Households clustered as in Section 7.4.
Renewal records	`renewal`	Renewal period normalised to financial year. Gaps are kept, not filled.
CESS contribution ledger	`cess_ledger_entry`	Amounts loaded as exact decimals. An entry that cannot be linked to a worker or payer is retained with an `unlinked` flag, never dropped.
Scheme claims and disbursements	`scheme_application`, `application_status_history`	Legacy status mapped to Applied, Sanctioned, Rejected or Availed. No status is inferred where the legacy value is ambiguous; it is loaded as Under review.
Centre, district, mandal and village lists	`centre`, `service_area`, `lgd_location`, `lgd_crosswalk`	Legacy codes mapped to LGD. Unmapped codes go to the steward queue.
Staff and HRMS records	`user_account`, `staff_master_ref`	Accounts created inactive. Activation follows the user onboarding procedure.
38-scheme catalogue and Government Orders	`scheme`, `scheme_version`, rule drafts	Rule documents are authored in the configuration console from the Orders, not converted mechanically. Each is validated with the scheme owner at M9.
16 departmental dumps	`fetched_attribute` with lineage	Attached to the Worker Master after identity matching, with source and as-of date.

7.12 Data quality rule catalogue

The rules below are the starting catalogue approved at gate G2. Each rule has a stable identifier, a hit count per run and an action. The Data Governance Committee can add rules and change thresholds, and every change is recorded in the decision log.

Rule	Dimension	Check	Action on failure
DQ-001	Completeness	Registration Number present	Quarantine the record
DQ-002	Completeness	Name present in at least one language	Quarantine
DQ-003	Completeness	Date of birth present	Load as Migrated-unverified and flag for review
DQ-004	Validity	Mobile has 10 digits starting 6 to 9 after removing +91 or a leading 0	Cleanse; otherwise null and flag
DQ-005	Validity	Aadhaar number has 12 digits and a valid Verhoeff check digit	Do not store; flag the record
DQ-006	Validity	Age on registration date between 18 and 60	Flag for review
DQ-007	Validity	Gender code in the vocabulary	Map legacy codes; otherwise flag
DQ-008	Validity	IFSC matches the 4 letters, `0`, 6 characters pattern	Null the field; bank marked to verify
DQ-009	Validity	Bank account number has 9 to 18 digits	Flag; bank marked to verify
DQ-010	Validity	PIN code has 6 digits and agrees with the LGD district	Flag
DQ-011	Validity	UAN has 12 digits	Null and flag
DQ-012	Validity	Days worked between 0 and 366	Null and flag
DQ-013	Validity	Disability percentage between 0 and 100	Null and flag
DQ-014	Uniqueness	Registration Number unique across the source	Earliest kept; later kept as alias
DQ-015	Uniqueness	Aadhaar reference key unique	Merge candidate for review
DQ-016	Uniqueness	One mobile number on no more than 5 records	Review flag on every record
DQ-017	Consistency	Registration date not later than renewal dates, and no date in the future	Flag
DQ-018	Consistency	A child's date of birth at least 12 years after the parent's	Flag the relationship
DQ-019	Consistency	Spouse or parent name differs from the worker's own name	Flag
DQ-020	Consistency	Nominee shares sum to 100	Flag; nominee marked to verify
DQ-021	Consistency	Date format resolved by source convention, never by guess	Quarantine ambiguous values
DQ-022	Referential	Address resolves to an LGD village or ward	Steward queue
DQ-023	Referential	Centre code exists in the centre master	Steward queue
DQ-024	Referential	Scheme code exists in the catalogue	Quarantine the claim
DQ-025	Referential	Claim amount is not negative and not above the scheme maximum	Flag
DQ-026	Referential	CESS entry links to a worker or payer	Load with `unlinked` flag
DQ-027	Referential	Evidence document reference resolves to a scanned file	Flag for rescan

7.13 Matching algorithm parameters

Stage	Specification
Normalisation	Unicode NFC; case folding for English; honorifics and initials removed; repeated spaces collapsed; Telugu text transliterated to a Latin key using a fixed scheme so Telugu and English spellings of the same name produce the same key.
Blocking keys	Records are compared only within blocks: Aadhaar reference; mobile plus date of birth; phonetic key of the first name plus year of birth plus LGD mandal; ration card number; UAN.
Deterministic rules	Aadhaar reference, Registration Number or e-Shram UAN equal: merge. Mobile, date of birth and normalised first name equal: merge unless the mobile exceeds the reuse threshold.
Probabilistic score	Weighted sum of name similarity (0.35), date of birth agreement (0.25), parent or spouse name similarity (0.20), village agreement (0.10) and gender agreement (0.10). Name similarity is the higher of Jaro-Winkler and trigram similarity on the normalised key.
Date of birth agreement	Exact match scores 1.0. A day and month swap or a one-year difference scores 0.6. Anything else scores 0.
Thresholds	0.92 or above: review queue, high priority. 0.80 to 0.91: review queue, normal priority. Below 0.80: not a candidate. No probabilistic match is merged without a human decision.
Reversibility	A merge writes a `merge_event` holding the pre-merge records and the decision. Un-merge restores both records and logs a second event.

7.14 Survivorship matrix

Field	Rule	Tie-breaker
Name, date of birth, gender, address	UIDAI-verified value, then operator-verified value with a document, then legacy declaration.	Latest verification date
BOCW Registration Number	Earliest issued number is kept. Later numbers are recorded as aliases.	None needed
Mobile number	Most recently OTP-verified number, otherwise the most recently used.	Latest date
Bank account	NPCI-validated, then passbook-evidenced, then legacy.	Latest validation
Days worked	Most recent 12-month declaration. Values from different sources are never summed.	Latest declaration date
Social category and income	GSWS certificate value, then evidence-backed declaration, then legacy.	Latest certificate date
Disability	SADAREM value, then certificate-backed declaration, then legacy.	Latest assessment date
Photograph	Latest live capture, otherwise the e-KYC photograph, otherwise the legacy image.	Latest capture time
Nominee	Latest signed declaration.	Latest signature date

Non-surviving values are retained in lineage with the rule identifier that excluded them. Nothing is deleted.

7.15 Production cut-over runbook timings

The window is planned for a low-traffic period, normally a weekend night, and is rehearsed in full at the dress rehearsal (gate G5). The durations below are planning figures that the dress rehearsal replaces with measured values.

Time	Activity	Owner	Checkpoint
T minus 7 days	Dress rehearsal accepted. Go/No-Go checklist reviewed by the Steering Committee.	Migration Lead	G5 signed
T minus 48 hours	Freeze date announced to centres through broadcast. Paper registers distributed.	Training and Change Lead	Acknowledged by all clusters
T minus 2 hours	Final backup of the target. Point-in-time recovery verified. Pre-checks on environment and credentials.	DevOps Lead	Backup verified
T	Legacy registration closed to new transactions. Running paper register starts.	Programme Manager	Freeze confirmed
T plus 0:15	Final extract of the delta since the last mock load, with checksums.	Migration Lead	Control totals taken
T plus 1:00	Delta load through the ten-stage pipeline. Reconciliation queries run.	Migration Lead	Checkpoint 1: balances
T plus 3:30	CESS ledger, scheme status, WLH and consent reconciliation signed by APBOCWWB accounts and PMU.	Migration Lead, PMU	Checkpoint 2: go or roll back
T plus 4:30	Release deployed to Production. Smoke suite run.	DevOps Lead, QA Lead	Smoke suite green
T plus 6:00	Canary centres opened, one per cluster, with cluster staff on site.	TSU Head	Canary healthy for 60 minutes
T plus 8:00	Statewide enablement and Go-Live broadcast. Paper-register entries keyed as assisted transactions.	Programme Manager	War room opened

7.16 Reconciliation queries

The migration control tables live in a temporary `mig` schema that is archived after cut-over certification. The queries below are written in portable SQL and are part of the gate evidence. Each is expected to return no rows.

```
-- 1. Control totals must balance for every source and run
SELECT run_id, source_id, extracted,
       loaded + rejected + merged_away + deferred AS accounted
FROM   mig_run_summary
WHERE  extracted <> loaded + rejected + merged_away + deferred;

-- 2. Every worker has exactly one migrated Registration event on the timeline
SELECT w.public_id
FROM   worker w
LEFT JOIN wlh_event e ON e.worker_id = w.id AND e.event_type = 'REGISTRATION_MIGRATED'
GROUP BY w.public_id
HAVING COUNT(e.id) <> 1;

-- 3. CESS ledger totals agree with the legacy ledger by financial year
SELECT l.fin_year, SUM(l.amount) AS loaded_total, m.legacy_total
FROM   cess_ledger_entry l
JOIN   mig_legacy_cess_total m ON m.fin_year = l.fin_year
GROUP  BY l.fin_year, m.legacy_total
HAVING SUM(l.amount) <> m.legacy_total;

-- 4. No migrated worker without a consent tag
SELECT w.public_id
FROM   worker w
WHERE  w.migrated = 1 AND w.consent_status NOT IN ('CONSENT_ON_FILE', 'CONSENT_REQUIRED');
```

8. Security, Privacy, Aadhaar/UIDAI, Audit, VAPT, Backup and Disaster Recovery Approach

Security is delivered once, as a platform property, through the shared services described in Section 2.2.1 — every one of the 22 modules inherits the same controls automatically.

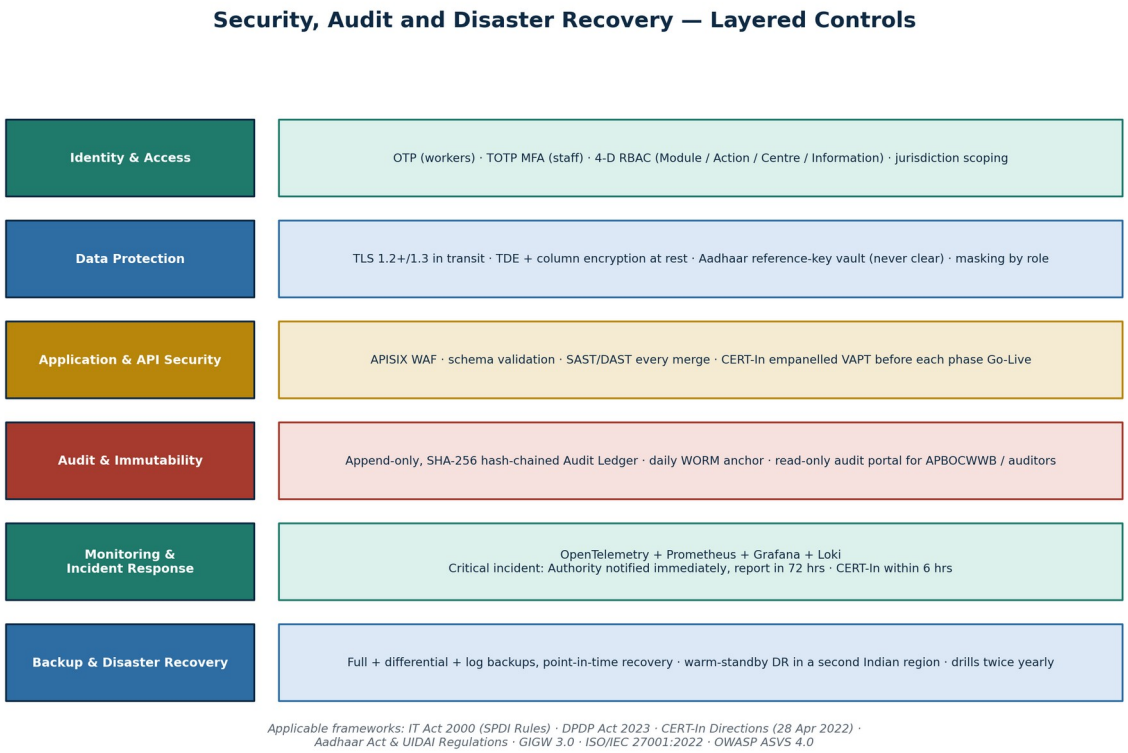


Figure 8.1 — Layered security, audit and disaster-recovery controls

8.1 Governing frameworks

The Platform is designed, built, hosted and operated against: the Information Technology Act, 2000 and the Sensitive Personal Data Rules, 2011; the Digital Personal Data Protection Act, 2023; the CERT-In Directions of 28 April 2022 (incident reporting within six hours, 180-day log retention within India, synchronised system clocks); the Aadhaar Act, 2016 and UIDAI's AUA/KUA/Aadhaar Data Vault regulations; the Guidelines for Indian Government Websites and Apps (GIGW 3.0); ISO/IEC 27001:2022 as the reference control set; and the OWASP Application Security Verification Standard 4.0.

8.2 Identity and access

Actor	Authentication
Worker (self-service, kiosk)	Mobile OTP, or QR card-token scan as an alternative identity fetch
Worker at assisted desk	OTP or card-token scan, confirmed in the presence of the Centre Data Entry Operator
Centre Data Entry Operator, counsellor, PMU staff, administrator	Username/password with policy, plus mandatory TOTP multi-factor authentication
Employer	Mobile OTP; job-posting access activated only after APBOCWWB staff

Actor	Authentication
	verification
Departmental user	Federated login where the department offers one, otherwise Platform-issued credentials with TOTP MFA
System-to-system (Integration Hub)	Mutual TLS or OAuth2 client credentials; credentials resolved only at call time from the secrets vault, never logged
Privileged administrator	Named individual account, hardware-backed or TOTP MFA, access only through a bastion host with session recording

Authorisation runs through a single policy decision point that every service consults — no module implements its own access-control logic. A Role carries four dimensions: Module access, Action access (View/Create/Edit/Delete/Approve/Escalate), Centre scope, and Information visibility. A centre-scoped user's effective access is mandatorily filtered at the data-access layer, applied identically to every screen, report, export and API — a Centre Data Entry Operator cannot see another centre's records through any interface. A Role change takes effect at the holder's next request; deactivating a User blocks login immediately while their historical actions remain correctly attributed to them.

8.3 Data protection and privacy

8.3.1 Data classification

Data is classified into seven tiers, each with its own controls — from Class 1 (Aadhaar, biometric-adjacent data: segregated Aadhaar Data Vault, tokenisation, never exported, never logged) through Class 2 (financial: field-level encryption, masked display), Class 3 (health/disability), Class 4 (social category/income), Class 5 (personal/contact), Class 6 (operational/case data), to Class 7 (public content, integrity-protected only).

8.3.2 Encryption

All traffic is encrypted in transit with TLS 1.2 or higher (TLS 1.3 preferred), HSTS enforced, mutual TLS toward departments that support it. At rest: Transparent Data Encryption plus column-level encryption on Class 1–2 fields for the system-of-record database, server-side encryption with object lock on the Document Vault, encrypted local storage on field devices keyed to the device keystore, and encrypted backups with keys distinct from production keys.

8.3.3 Aadhaar handling

As stated in Section 5.4: OTP-based authentication and e-KYC only, no biometric processing in this scope; the Aadhaar number is placed in a segregated, cryptographically isolated Aadhaar Data Vault on first authentication, and every other table holds only a reference key plus the last four digits for display. Repeat lookups resolve the reference key inside the vault boundary — the number is never returned to a calling module. The number is excluded from every export, report, notification and log.

8.3.4 Consent lifecycle

The Consent Manager is the single service through which consent is captured, checked and withdrawn. Every processing purpose is a configured, versioned entry naming the data classes and external systems it

may touch. Consent is an explicit affirmative action, shown in Telugu or English, read aloud by the operator at the assisted desk where relevant, and logged with notice version, channel and timestamp. Every Integration Hub call carries a purpose code and is refused with CONSENT_REQUIRED where no matching consent exists. Withdrawal is itself a logged consent event; processing that depended on it stops, and the worker is told what continues and why.

8.3.5 Retention and deletion

Retention periods are configured per data class and enforced by scheduled jobs, never by manual action, aligned to BOCW Act record-keeping obligations, the DPDP Act, UIDAI requirements and CERT-In's 180-day minimum for security logs. A worker's correction or erasure request is a Case with a service clock; corrections retain both the prior and corrected value with a reason; erasure of legally-required data is refused with a documented next step; other erasure is executed by tombstoning (personal-data fields overwritten, the record and the audit chain preserved) — physical deletion of ledger rows never occurs.

8.3.6 Data residency

All five environments, all backups, the disaster-recovery site, the Aadhaar Data Vault, logs and telemetry are hosted in India. No personal data leaves India. Web, kiosk and mobile clients carry no third-party analytics, advertising or crash-reporting SDK that transmits data outside the Platform.

8.3.7 Government infrastructure and VPN-protected connectivity

The Platform runs only on Government of India or Government of Andhra Pradesh infrastructure — government-owned servers at NIC MeghRaj or the AP State Data Centre, or a MeitY-empanelled government cloud service provider — never on general commercial hosting outside that empanelment (Section 16.5). Every network path into that infrastructure is closed by default and opened only over an encrypted channel:

- ILAC centres, District Field Support Cluster offices and the PMU reach the Platform over an encrypted site-to-site VPN or a NICNET leased line, never over the open internet in clear.
- Every departmental integration (Section 5) is reached over a VPN tunnel or a mutually authenticated private link where the department supports one, with fixed source and destination addresses so both sides can allow-list the connection.
- Primary-to-disaster-recovery replication between the two hosting regions runs over a dedicated, encrypted private link — database and object-store traffic never traverses the public internet even between the Platform's own environments.
- Administrative access (deployments, database administration, log access) is permitted only through a bastion host reachable over VPN, with every session recorded and multi-factor authentication enforced — there is no direct internet-facing administrative endpoint anywhere in the estate.

This is shown in the hosting topology diagram at Figure 16.1, and is a fixed design constraint, not a configurable option — the API gateway is the only internet-facing surface, and it in turn is reachable solely over TLS 1.2 or higher.

8.4 Application and API security

All inbound traffic passes through the API gateway, which enforces token/certificate authentication, per-client and per-route rate limiting (with stricter limits on OTP issuance), OpenAPI schema validation, a web application firewall rule set, and full access logging with sensitive fields redacted. Application controls include typed input validation at every trust boundary, parameterised queries only (no string-built SQL anywhere in the code base), context-aware output encoding, CSRF protection on state-changing requests, a strict content-security policy, and content-based (not extension-based) file-upload validation with antivirus scanning. The field application adds certificate pinning, rooted/jailbroken-device detection, an encrypted local database keyed to the device keystore, remote wipe on deactivation, and disabled screenshots on screens showing sensitive data. Kiosk sessions hold no data beyond the session — on timeout, sign-out or completion, tokens are revoked and local storage cleared.

8.5 Audit and immutability

The Audit Ledger is the single immutable record of what happened on the Platform, implemented as an append-only table with insert-and-select-only database permissions — enforced by the database engine, not left to application code. Each row carries a SHA-256 hash of its own content and the previous row's hash, so any alteration breaks the chain from that point forward; the head hash is anchored daily to write-once storage so an auditor can verify the ledger has not been rewritten, independent of the database itself. Every create, update, status change, configuration change, consent event, integration call and access to sensitive data is recorded, with actor, role, jurisdiction, correlation ID and timestamp. APBOCWWB and authorised auditors have a dedicated read-only audit portal with search by worker, Case ID, operator, centre, event type and date range.

8.6 Security testing and VAPT

Static analysis, dependency scanning, secrets scanning and container scanning run on every merge request and block release on any Critical or High finding; dynamic application security testing runs against every UAT deployment. An independent vulnerability assessment and penetration test by a CERT-In empanelled auditor is performed before each Phase Go-Live and at least annually during operations — covering the web portal, kiosk and assisted-desk clients, mobile applications, every API exposed through the gateway, the identity provider, infrastructure and network, the Integration Hub adapters, and the Audit Ledger controls. Every Critical and Major finding is remediated and retested before phase acceptance is requested; a Safe-to-Host certificate is obtained where the Platform is hosted on government infrastructure.

8.7 Security monitoring and incident response

Gateway, application, authentication, Audit Ledger and integration events are aggregated in the Platform's observability stack and evaluated against alert rules covering authentication anomalies, access anomalies (bulk reads, cross-centre attempts), integrity events (ledger chain mismatch), and infrastructure events. Incidents are classified Critical / Major / Moderate / Minor, each with a documented response playbook rehearsed at least annually.

Obligation	Timeline
APBOCWWB notification	Immediately on classification; a detailed written report within 72 hours, or sooner where a shorter legal requirement applies
CERT-In reporting	Within six hours of noticing, for incidents in CERT-In's listed categories
DPDP Act breach notification	As Data Processor, the Bidder notifies APBOCWWB (Data Fiduciary) without delay, who notifies the Data Protection Board and affected workers as prescribed
UIDAI reporting	Additional reporting for any incident affecting Aadhaar authentication or e-KYC data

8.8 Backup and recovery

Full, differential and continuous transaction-log backups enable point-in-time recovery of the system-of-record database within the retention window; the Document Vault replicates continuously to the DR region with object lock preserving evidence against deletion; configuration is versioned in the database and exported to the Authority repository on every change. Backups are encrypted, stored on infrastructure separate from production, and copied to the DR region. A restore test of the database and a sample of Document Vault objects runs monthly; a full-environment restore runs at least twice yearly as part of the DR drill.

8.9 Disaster recovery

Production runs in a primary Indian region; the Disaster Recovery environment, provisioned from the same infrastructure-as-code, runs warm-standby in a second Indian region with continuously replicated data stores. A documented runbook defines the declaration criteria, decision authority, failover sequence (freeze writes → promote the DR database → verify currency → switch gateway and DNS → verify integrations and notification channels → announce to centres), and a verification checklist. A DR drill — including at least one full failover and failback — runs twice yearly, witnessed by APBOCWWB, with the first drill completed before Phase 4 Certified Final Go-Live. During a DR event, the field application continues to capture offline and synchronises once the DR environment is reachable; departmental integrations resume from the Integration Hub's replay store so no request is lost.

8.10 Business continuity for centres

Where a centre loses connectivity independently of any regional disaster, the field application continues offline capture (Section 10); the assisted-desk client buffers locally for a short outage and, for a longer one, the operator uses a certified paper register mirroring the standard case record, with a provisional reference later keyed into the Platform and the clock back-dated to the paper intake time. Restoration priority is identity and case intake first, then scheme applications and grievances, then events and reporting.

8.11 Compliance mapping summary

RFP requirement	Where delivered
Encryption in transit and at rest	Section 8.3.2
Role-based, least-privilege access control	Section 8.2
Multi-factor authentication where applicable	Section 8.2
Consent, purpose limitation, masking, retention, deletion	Section 8.3.4–8.3.5
India-resident hosting	Section 8.3.6, Section 16
Government-only infrastructure, VPN-protected connectivity	Section 8.3.7, Section 16.1
VAPT and closure of Critical/Major findings before acceptance	Section 8.6
Backup, point-in-time recovery and disaster recovery	Section 8.8–8.9
Immediate incident reporting, 72-hour detailed report	Section 8.7
Audit access for APBOCWWB and authorised auditors	Section 8.5

8.12 Threat model by trust boundary

The threat model is kept in the repository and reviewed at each phase gate and after any change to a trust boundary. The table lists the boundaries, the principal threats using the STRIDE categories, the controls that answer them and the evidence that the controls work.

Boundary	Principal threats	Controls	Evidence
Internet to gateway	Spoofing through OTP abuse or SIM swap. Injection and tampering. Denial of service.	OTP throttling per mobile and per address. WAF rule set. Schema validation. TLS 1.2 or higher. Rate limits per route class.	DAST on each UAT build. VAPT. Load and abuse tests.
Staff client to gateway	Credential theft. Session hijack. Privilege escalation.	Mandatory TOTP. Short-lived tokens. Step-up MFA on administrator routes. Access enforced at the data-access layer. Revocation on deactivation.	VAPT. Quarterly access review.
Field device	Loss or theft. Rooted device. Tampering with offline data. Replay of change-sets.	Encrypted local database keyed to the device keystore. Root detection. Remote revoke. Media hashing. Idempotency keys on every change-set.	Device-lab test matrix (Section 10.8).
Gateway to services	Lateral movement. Spoofed caller.	Mutual TLS from the gateway to upstreams. Default-deny network policies. Service accounts with least privilege.	Network policy tests. VAPT.
Services to database	SQL injection. Over-privileged login. Bulk exfiltration.	Parameterised queries only. A separate login per family limited to its schema. Insert-only audit rights. Column encryption. Statement timeouts. Bulk-read alerts.	SAST. Code review. VAPT.

Boundary	Principal threats	Controls	Evidence
Hub to departments	Credential leakage. Server-side request forgery. Response tampering and contract drift. Over-collection.	Secrets vault with call-time resolution. Egress allow-list. Runtime response validation. A field allow-list per consent purpose.	Adapter conformance suite. Secrets scan.
Administrator access	Insider misuse.	Bastion with session recording. Step-up MFA. Dual approval for configuration publication and key operations. Read-only audit portal.	Session review. Audit sampling.
Document Vault	Malware upload. Evidence tampering.	Antivirus scan before an object is readable. Content-based type detection. Object lock. Hash verification on receipt and on read.	Upload abuse tests. Restore tests.
Supply chain	Malicious dependency. Compromised image.	Software bill of materials. Dependency and container scanning. Pinned image digests. Signed images. Reproducible build at each gate.	Pipeline gate. Reproducible-build witness.

8.13 Authentication flows and token policy

Actor	Flow	Access token	Refresh and session limits	Idle timeout
Worker	Mobile OTP through a Keycloak authenticator. Authorisation code with PKCE for apps.	15 min	Refresh 30 days with rotation and device binding on the app; 8 hours on the browser.	30 min on browser
Staff	Username, password and TOTP. Authorisation code with PKCE.	10 min	Refresh 8 hours, absolute session 12 hours.	15 min
Employer	Mobile OTP.	15 min	Refresh 24 hours.	30 min
Departmental user	Federated login, or Platform credentials with TOTP.	10 min	Refresh 8 hours.	15 min
System caller	Mutual TLS or OAuth2 client credentials.	5 min	None; a new token is requested.	Not applicable
Field device	Staff login on a registered device. Offline credential cache.	10 min online	Offline session lifetime 72 hours by default, set by configuration.	Screen lock 5 min

Policy	Setting
Token format	JWT signed with RS256 or ES256. Keys rotated every 90 days with a `kid` header so old and new keys overlap. Claims: `sub`, `roles`, `centreScope`, `amr` (including MFA), `sid`, `deviceId`, `jti`, audience per route class.
Revocation	The gateway checks `sid` against a Redis deny list on every request. Deactivation, logout and remote revoke add the session to the list.
OTP	Six digits, valid 5 minutes, 3 attempts, 60 second resend cooldown, at most 5 issues per mobile per 10 minutes. Stored only as a keyed hash with a time to live.
TOTP	RFC 6238, 30 second step, 6 digits, plus or minus one step of drift. Ten single-use recovery codes issued at enrolment.

Policy	Setting
Passwords	Minimum 12 characters, checked against a list of known breached passwords, history of 5, lockout for 15 minutes after 5 failures. Stored with Argon2id where the identity provider supports it, otherwise PBKDF2-HMAC-SHA256 with at least 600,000 iterations.
Step-up	Administrator routes and key operations require MFA authentication within the last 15 minutes.

8.14 Cryptographic standards and key hierarchy

Keys form a three-level hierarchy. A root key is held in a hardware security module or the key management service of the hosting facility. Each purpose has a key-encryption key under the root. Data-encryption keys for columns, buckets and backups are wrapped by the key-encryption key of their purpose. Root key operations need two authorised people.

Purpose	Algorithm	Custody	Rotation
Transport	TLS 1.2 or higher, TLS 1.3 preferred. ECDHE key exchange, AES-GCM or ChaCha20-Poly1305.	Certificates in the certificate manager	Automated renewal 30 days before expiry
Database at rest	Transparent Data Encryption, AES-256	Database master key wrapped by the root key	Annual
Column encryption (Class 1 and 2)	AES-256-GCM with a data key per column	Key-encryption key in the HSM or vault	Annual key-encryption key; data keys on rewrap
Aadhaar reference key	HMAC-SHA-256, 256-bit key	Vault-held, never exported	Versioned; lazy re-key plus a background job
Aadhaar number in the vault	AES-256-GCM, reference key as associated data	Separate key-encryption key	Annual
Secrets table	AES-256-GCM envelope	Master key held by the Authority from Production Go-Live	Annual and on any change of key personnel
Object storage	Server-side AES-256 with a key per bucket	Object-store key service	Annual
Backups	AES-256	Keys distinct from production keys	Annual
Local device database	AES-256 through SQLCipher	Per-installation key in the device keystore, wrapped by the session key	On logout, expiry or revoke
Configuration bundles	ECDSA P-256 signature	Signing key in the vault	Annual
Audit Ledger	SHA-256 hash chain	Not applicable	Not applicable

8.15 Aadhaar Data Vault operations

The vault is a separate service on a separate host group, in its own network zone, with its own database and its own key-encryption key. Only two callers can reach it: the vault service itself and the UIDAI adapter. The reference key is derived as follows.

```
refKey   = "v" + keyVersion + ":" + hex( HMAC_SHA256( vaultKey[keyVersion], digitsOnly(aadhaarNumber) ) )
stored   = AES256GCM( kek, aadhaarNumber, aad = refKey )          # inside the vault boundary only
display  = "XXXX XXXX " + last4(aadhaarNumber)                     # the only form shown or exported
```

Operation	Allowed caller	Output	Logging
Tokenise	Registration family through the vault API	Reference key and last four digits	Caller, purpose code, reference key
Duplicate check	Registration family	Match or no match against stored reference keys. The number is hashed inside the vault.	Caller, result
Detokenise	UIDAI adapter only, for OTP and e-KYC	Number held in the adapter's memory for the duration of the call	Caller, UIDAI transaction identifier
Re-key	Vault service	New reference key version for every record	Job identifier, counts
Export or bulk read	Nobody	Not offered	Any attempt raises a Critical alert

A detokenise request from any other caller is refused and raises a Critical alert. The reference key is the join key for every other table. A change of key version is invisible to modules because they only ever see the current reference key returned by the vault.

8.16 Network zones and flow matrix

The estate is divided into six zones. Every flow not listed in the matrix is denied by default at the network layer and again by Kubernetes network policy inside the cluster.

Flow	From	To	Port and protocol	Control
N1	Internet and VPN clients	Gateway (public subnet)	443 TLS 1.2 or higher	WAF, rate limits, token validation
N2	Gateway	Family API pods (private subnet)	8443 mutual TLS	Network policy: only the gateway can reach APIs
N3	Family pods	System-of-record database (data subnet)	1433 or 5432 TLS	Per-family login, schema-scoped rights
N4	Family pods	Redis, MongoDB, MinIO	6379, 27017, 9000 TLS	Authenticated, role-based
N5	Family and worker pods	Temporal frontend	7233 mutual TLS	Namespace-scoped credentials
N6	Registration family and Hub	Aadhaar Data Vault zone	8443 mutual TLS	Only two callers; allow-list by service identity
N7	Hub worker pool	Departments	443 or SFTP 22 over VPN or private link	Fixed egress addresses, egress allow-list
N8	Departments	Dump landing area	SFTP 22 over VPN	Key-based, per-source directory
N9	Primary region	DR region	Dedicated encrypted link	Replication only; no other traffic
N10	Bastion (VPN only)	Kubernetes API, hosts, databases	6443, 22, 1433 or 5432	MFA, session recording, just-in-time access
N11	All components	Observability collector	4317 TLS	Telemetry only, redacted fields
N12	Superset	Read replica	1433 or 5432 TLS	Read-only reporting login

8.17 Logging, retention and monitoring

All system clocks are synchronised to the NIC or NPL time servers, as the CERT-In Directions require, and every log carries a UTC timestamp and the correlation identifier. The retention periods below are proposed and confirmed against the Authority's retention schedule at inception. None is shorter than the 180 days the Directions require.

Log	Content	Retention	Store and access
Gateway access	Route, status, latency, client, correlation identifier. Sensitive fields redacted.	365 days	Loki, then archive. Security and TSU read-only.
Application	Structured events with correlation identifier. No personal data in messages.	180 days	Loki. Developers by request, TSU.
Authentication	Logins, failures, MFA events, token revocations, step-up.	365 days	Loki and Audit Ledger. Security Lead.
Audit Ledger	Every change, configuration change, consent event, integration call and access to sensitive data.	Life of the record plus the statutory period	Database, with daily anchors. Audit portal.
Database audit	Logins, privilege changes, schema changes, reads of Class 1 and 2 columns.	365 days	Write-once storage.
Integration calls	Outcome, latency, source, consent reference. Payload snapshots hashed and encrypted.	365 days; snapshots 90 days	Database and MongoDB. Integration Lead.
Administrator sessions	Recorded sessions through the bastion.	365 days	Write-once storage. Security Lead.
Infrastructure	Host, container and Kubernetes audit logs.	365 days	Loki and archive.

8.18 OWASP ASVS 4.0 control mapping

The Platform is built to ASVS Level 2 as its baseline. The Aadhaar Data Vault, the Audit Ledger and the identity provider configuration are verified to Level 3. The independent VAPT of Section 8.6 reports against these chapters.

ASVS chapter	Platform control	Section
V1 Architecture	Threat model per boundary. Layered services. Policy decision point. Shared services for audit, consent and case handling.	2.15, 8.12
V2 Authentication	OTP, TOTP, password policy, throttling, Keycloak configuration.	8.13
V3 Session management	Short-lived tokens, rotation, revocation, idle and absolute limits.	8.13
V4 Access control	Four-dimension role model enforced at the data-access layer.	2.5, 2.15
V5 Validation and encoding	Typed input validation, parameterised queries, output encoding, CSP.	8.4
V6 Stored cryptography	Key hierarchy and algorithms.	8.14
V7 Error handling and logging	Problem details without internals. Structured, redacted logs.	2.13, 8.17
V8 Data protection	Classification, masking, retention, tombstoning.	8.3
V9 Communication	TLS 1.2 or higher, HSTS, mutual TLS inside the estate.	8.3.2, 8.16

ASVS chapter	Platform control	Section
V10 Malicious code	Signed images, dependency scanning, reproducible build.	8.12, 14.8
V11 Business logic	Two-outcome closure, idempotency, immutable records, rate limits.	3.7
V12 Files and resources	Content-based validation, antivirus, object lock.	8.4
V13 API and web service	OpenAPI validation at the gateway, versioning, rate limits.	2.13, 2.14
V14 Configuration	Hardening baselines, secrets in the vault, infrastructure as code.	8.19

8.19 Hardening baselines

Layer	Baseline	Verification
Operating system	CIS benchmark Level 1 for the chosen Linux distribution, or the Authority's own standard where stricter. Unused services removed. Automatic security updates in the window of Section 2.21.	OpenSCAP scan at build and monthly
Containers	Non-root user. Read-only root filesystem. All Linux capabilities dropped. Seccomp runtime default profile. Minimal base image. Signed image with pinned digest.	Trivy scan and admission policy
Kubernetes	CIS Kubernetes benchmark. Pod Security Standard restricted. Default-deny network policy. Secrets encrypted in etcd. RBAC with least privilege. No privileged pods.	kube-bench at build and monthly
Database	Engine CIS benchmark. No application use of an administrative login. Per-family logins. Auditing on. Encryption on.	Quarterly review
Redis	Authentication on, TLS on, dangerous commands renamed or disabled, protected mode.	Configuration scan
MongoDB	Authentication and role-based access on, TLS on, network binding to the private subnet.	Configuration scan
MinIO	TLS, key service, object lock on evidence buckets, bucket policies denying public access.	Configuration scan
Gateway and identity provider	Admin interfaces reachable only from the management zone. Realm settings in version control.	Configuration review at each gate

8.20 Recovery objectives and backup schedule

The objectives below are proposed. The RFP service-level schedule replaces any figure it states, and the DR drill measures the achieved values against them.

Store	Backup method	Retention	RPO	RTO
System-of-record database	Full weekly, differential daily, log backup every 5 minutes. Synchronous in-region secondary.	35 days online; monthly full for 12 months	In region 0; cross-region 5 min	In region 5 min; cross-region 60 min
Document Vault	Continuous replication to DR with versioning and object lock.	Per data-class schedule	15 min	2 hours
MongoDB staging	Daily snapshot plus change log.	14 days	15 min	2 hours
Redis	None. Rebuilt on start.	Not applicable	Not applicable	15 min
Configuration	In the database and exported to the Authority repository on every change.	Life of the Platform	0	With the database

Store	Backup method	Retention	RPO	RTO
Secrets and keys	Encrypted export held by the Authority.	Per key policy	0	4 hours

8.21 Audit Ledger chain algorithm

The ledger is divided into 16 lanes selected by a hash of the subject key, so that the chain of one lane is never a bottleneck for the others. Within a lane the rows form a hash chain. The append happens in the same database transaction as the change it records, so the ledger cannot miss a committed change.

```
lane   = hash16(subjectKey)
BEGIN                                          -- the business transaction
  head = SELECT last_hash, last_seq FROM audit_chain_head WHERE lane = :lane FOR UPDATE
  seq  = head.last_seq + 1
  body = canonical_json({lane, seq, ts, actor, role, jurisdiction, action, subject,
                         correlationId, detail})
  hash = SHA256( head.last_hash || body )
  INSERT INTO audit_ledger(lane, seq, ts, actor, ..., prev_hash, hash)
         VALUES (:lane, :seq, :ts, :actor, ..., head.last_hash, :hash)
  UPDATE audit_chain_head SET last_hash = :hash, last_seq = :seq WHERE lane = :lane
COMMIT
```

- **Permissions.** The application login has `INSERT` and `SELECT` on `audit_ledger` and no other right. The small `audit_chain_head` table, which holds the current head of each lane, is the only related table it can update.
- **Contention.** At the peak rate of Section 2.20 the ledger takes about 12 appends per second per lane. A row lock held for under 2 milliseconds keeps lane utilisation near 2.5%.
- **Anchoring.** At 00:05 IST the AuditAnchor workflow reads the 16 lane heads, hashes them together and writes the result with the date to write-once storage. Nothing in the database can rewrite that record.
- **Verification.** A nightly job recomputes each lane from the previous anchor forward. A mismatch raises a Critical alert and starts the incident playbook of Section 8.7.
- **Reads.** Any read of a Class 1, 2 or 3 field is itself an audited event, with actor, purpose and the number of records returned.

9. Web, Mobile, Field, Kiosk and Assisted-Desk Approach

One set of module APIs, one design system, and one form/workflow definition per function are rendered on every surface. No surface carries its own logic; a function added to self-service is available at the assisted desk on the same release. This is what makes assisted-mode parity structural rather than a separate development effort repeated per surface.

9.1 Channel-to-locus coverage

Surface	Primary users	Primary locus	Authentication
Worker portal / PWA	Registered workers, dependants	L-5	Mobile OTP
Worker mobile app	Registered workers	L-5	Mobile OTP; card token for kiosk/desk interop
Kiosk mode	Workers at centres	L-1	OTP or card-token scan; no data persists after session
Assisted-desk console	Centre Data Entry Operators	L-1	Staff login with MFA; centre-scoped
Field application	Field officers, mobile-unit staff	L-2, L-3, L-4	Staff login; offline credential cache
Employer portal / app	Verified employers	L-5	Mobile OTP; posting access gated on verification
Departmental portal	Authorised departmental users	L-6	Federated login or Platform credentials with MFA
PMU / admin console	PMU, HR, MIS administrators, Board	L-6	Staff login with MFA
Helpline, IVR, WhatsApp, SMS	Any actor without app access	L-5	Identified by mobile number, OTP or Case/Ticket ID

9.2 Design system and accessibility

A single shared component library serves the web surfaces and is mirrored in the React Native field and worker apps — the same names, states, validation messages and Telugu/English strings appear on every surface. Web surfaces conform to WCAG 2.1 Level AA and the Guidelines for Indian Government Websites (GIGW 3.0); mobile surfaces follow the equivalent platform accessibility conventions. Automated accessibility checks run in the build pipeline, with a manual audit using assistive technology before each Go-Live.

Because many workers have limited reading ability and use entry-level Android phones on intermittent networks, the design applies icon-plus-label navigation, one question per screen on the registration wizard, a large-type read-back before commit, and an always-available "call an operator" action that hands a session to the assisted desk without losing entered data. Text-to-speech is not used — read-

aloud, where needed, is delivered by the operator, consistent with keeping AI-driven capabilities out of this Part 1 scope.

9.3 Worker portal and mobile app

A responsive web application, installable as a Progressive Web App, and a React Native (Expo) mobile app sharing the same functions: registration status and renewal, profile and household view, Eligible/Applied/Availed scheme views with Apply, job postings and one-tap enrolment, the opportunity and event calendar, a resumable learning library, the full Worker Longitudinal History timeline, notification history, support-ticket raising and tracking, and the QR card-token display. The PWA caches the worker's own read-only record for offline viewing but does not queue offline writes — write capability with conflict resolution is reserved for the staff field application, where an authorised operator can resolve conflicts.

9.4 Employer portal and app

Employer registration with GST/contractor-number validation, job posting and management, enrolled-worker review (contact numbers visible only after a worker has enrolled), engagement confirmation by Registration Number, daily muster, job-mela participation and support-ticket raising. Every employer action is attributed to a verified account; verification decisions are made by APBOCWWB staff in the PMU console, with the Platform recording, not making, the decision.

9.5 Kiosk mode

Any Authority-provided touch device at a centre or camp, running the worker web application in a locked, full-screen kiosk mode — no browser chrome, no file-system access. The Platform supplies no kiosk hardware, biometric device, scanner or camera; the kiosk client works with whatever the device has, degrading gracefully where a peripheral is absent (for example, an on-screen token with SMS acknowledgement where no printer exists). Sessions time out with a warning, mask sensitive fields, and leave no data behind.

9.6 Assisted desk

The Centre Data Entry Operator console is where the "no worker turned away" rule is delivered in person: identity fetch by OTP or card token, the worker's full 360-degree view, every worker-facing function in assisted mode with operator attribution, document verification against originals, manual fallback capture, the Work Request token queue, and printed or SMS-delivered acknowledgements. The console buffers form entries locally during a short connectivity loss and switches to the certified paper register for a longer outage, with the same Case ID discipline applied at keying.

9.7 Field application

A React Native (Expo) offline-first application for field officers and Centre Data Entry Operators on the move, covering the household baseline survey, worker pre-registration, field exposure trips with geo-tagged evidence, job-mela and camp operations, construction-site catalogue survey, functionary attendance, grievance intake, daily muster capture, and livelihood-group activity recording. Full detail — the offline architecture, data model, security and synchronisation protocol — is in Section 10.

9.8 Departmental portal and PMU console

Role-scoped web access for departmental officers to the Convergence Register filtered to their own department, committee agendas and decisions, department-scoped reports, and interface health for their own source. The PMU and administration console gives PMU, HR, MIS and Board users the unified case inbox, session/event creation, content and broadcast management, the configuration console, MIS, functionary monitoring and integration health — everything L-6 back-office work needs, in one authenticated surface.

9.9 Performance and device baseline

All worker-facing surfaces are designed for entry-level Android devices on 2G/3G connections: text-first pages, deferred images, a bounded page-weight budget, and adaptive-bitrate or downloadable video for low-bandwidth centres. Before Phase 1 Go-Live, the registration and scheme-results journeys are usability-tested with real workers and operators at pilot centres across multiple districts, in Telugu, on the kiosk, assisted-desk and mobile surfaces, with findings fed back into the design system before wider rollout.

9.10 Screen inventory by surface

The inventory lists the principal screens of each surface. All of them render the same module APIs, and a screen that exists on one surface exists on the others wherever the actor can reach it.

Surface	Principal screens and areas	Languages
Worker portal and app	Home. Profile. Household. Entitlements (Eligible, Applied, Availed). Apply. Jobs. Calendar. Learning library. Timeline. Notifications. Support tickets. QR card token.	Telugu, English
Employer portal and app	Registration. Verification status. Job postings. Applicants and enrolled workers. Engagement confirmation. Daily muster. Mela participation. Tickets.	Telugu, English
Kiosk	Language choice. Identify by OTP or QR token. Service menu. Status lookup. Work request token. Grievance. Sign-out.	Telugu, English
Assisted desk	Identify worker. 360-degree view. Registration wizard. Renewal. Document verification. Fallback capture. Work request queue. Applications. Tickets. Acknowledgement print or SMS.	Telugu, English
Field application	Today (assignments). Household survey. Pre-registration. Site visit. Mela desk. Site survey. Attendance. Grievance. Muster. Group meeting. Sync status and conflicts.	Telugu, English
Departmental portal	Convergence Register. Committees and decisions. Department reports. Interface health.	English, Telugu
PMU and administration console	Case inbox. Sessions and events. Content and broadcast. Configuration console. MIS and dashboards. Functionary monitoring. Integration health. Users and roles. Audit portal.	English, Telugu

9.11 Progressive web application and caching strategy

The service worker caches only what makes the application usable on a weak connection. It never queues a write, and it never holds another person's data.

Resource class	Strategy	Lifetime	Notes
Application shell	Precache, versioned	Until next release	Updated in the background. The user sees a prompt to reload when a new version is ready.
Static assets (scripts, styles, icons)	Immutable, content-hashed names	1 year	No revalidation needed.
Fonts	Precache of the Telugu and Latin subsets	Until next release	Self-hosted. No third-party font service.
Worker's own record (read-only)	Network first, cache fallback	24 hours	Only the signed-in worker's record. Cleared on logout. Masked fields stay masked.
Images and course thumbnails	Cache first with a size cap	30 days	Capped at 50 MB per origin.
API writes	Never cached or queued	Not applicable	The page shows an offline notice. Offline writes belong to the field application (Section 9.3).

9.12 Internationalisation and Telugu support

Aspect	Approach
Message catalogue	ICU MessageFormat strings in JSON, one namespace per module. Keys are never reused for a different meaning. A pipeline check fails the build when a key exists in English without a Telugu string, or the reverse.
Fonts	Noto Sans Telugu, subset to the characters in use, self-hosted, with a system font fallback. Line height is set for Telugu conjuncts so that stacked characters are not clipped.
Input	Unicode Telugu keyboards, including InScript, are supported. Names are captured in the language typed and a transliteration is stored alongside, so both forms are searchable.
Collation and search	Database collation for Telugu and an ICU collation on the client. Search normalises zero-width characters and common spelling variants.
Numbers and dates	Western digits throughout. Dates shown as dd/mm/yyyy in Indian Standard Time. Currency shown with the rupee symbol and Indian digit grouping.
Layout	Components tolerate 30% longer strings. No text is embedded in images.
Messaging templates	Telugu templates are Unicode. A Unicode SMS segment carries 70 characters, or 67 per part when concatenated, so templates carry a length budget checked when they are saved.
Translation workflow	Translator, then reviewer, then PMU sign-off. A glossary of programme terms is held in the repository so the same term is rendered the same way everywhere.

9.13 Performance budgets and accessibility checks

Measure	Budget on a mid-range Android phone with 2 GB of memory, 3G profile
First-view page weight (worker portal)	500 KB or less compressed
Initial JavaScript	170 KB or less compressed
Largest contentful paint	4 s or less
Time to interactive	5 s or less
Cumulative layout shift	0.1 or less

Measure	Budget on a mid-range Android phone with 2 GB of memory, 3G profile
Median API payload	50 KB or less
Registration wizard step transition	2 s or less end to end

Accessibility check	Method
Contrast	Text contrast of at least 4.5 to 1, checked automatically in the pipeline.
Keyboard and focus	Every control reachable and operable by keyboard with a visible focus indicator. Checked in the end-to-end suite.
Labels and errors	Every field has a programmatic label. An error names the field and says how to correct it, in the user's language.
Screen reader	Manual audit with NVDA on the web and TalkBack on Android before each Go-Live.
Zoom and reflow	Usable at 200% zoom with no horizontal scrolling.
Target size	At least 44 by 44 pixels on mobile and at least 64 by 64 pixels on the kiosk.
Timeouts	A warning is shown before a session expires, with a way to extend it.
Language	The `lang` attribute is set per page and per mixed-language fragment.

10. Offline Data Capture and Synchronisation Design

Field functions at worksites, households and camps must work with no assumption of connectivity. The Platform delivers four properties throughout: encrypted offline capture, local buffering that survives restarts, automatic synchronisation on reconnection, and conflict management that preserves both versions for authorised resolution — never a silent overwrite.

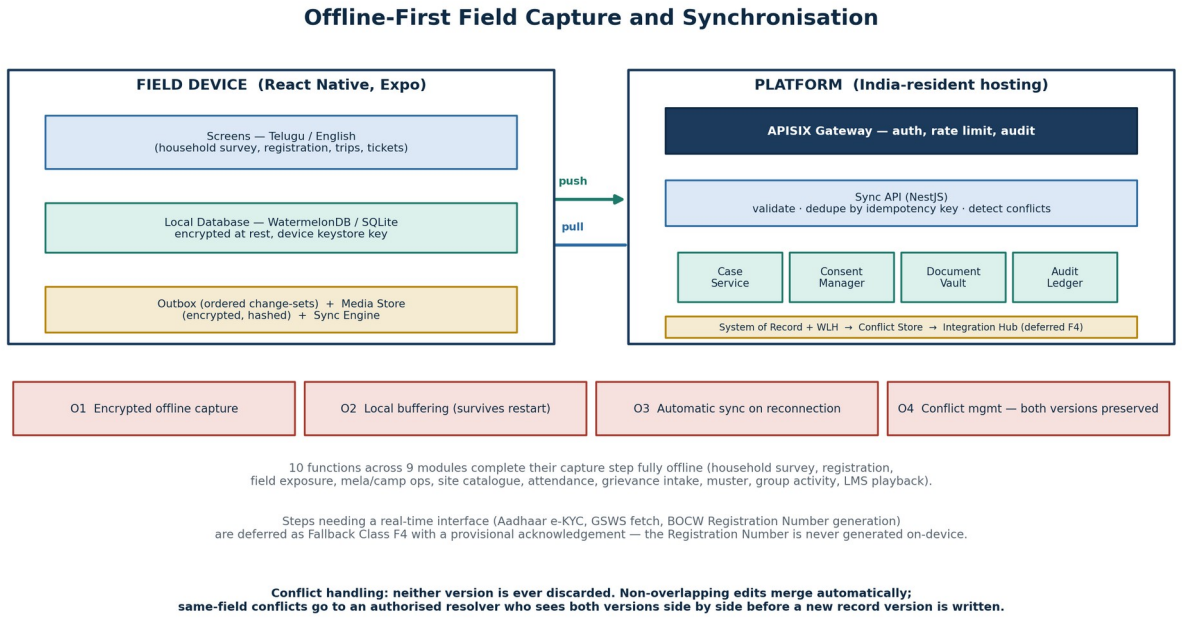


Figure 10.1 — Offline-first field capture and synchronisation architecture

10.1 What completes fully offline

Ten functions across nine modules complete their capture step entirely offline, each producing a case step with the same Case ID lineage, service clock, document linkage, audit trail and MIS visibility as any other channel: the household baseline survey, worker pre-registration, field exposure trips with geo-tagged media, job-mela and camp operations, construction-site catalogue survey, functionary attendance and turnout, grievance intake in the field, daily muster capture, livelihood-group activity recording, and offline learning-content playback.

10.2 What cannot complete offline, and how it is deferred

A small number of steps depend on a real-time external interface or a server-side generator and cannot complete on a device — Aadhaar e-KYC, GSWS fetch, EPFO/ESIC/e-Shram checks, NPCI validation, and BOCW Registration Number generation. Each is deferred, not blocked: the request is queued with a provisional identifier, the service clock runs from the moment of capture, and the step completes automatically once the device reconnects. The Registration Number is never generated on a device — this preserves the rule that it is a single, immutable, centrally issued sequence, and it means two devices can never issue the same number.

10.3 Device-side architecture

The field application writes every read and write to a local encrypted database (WatermelonDB over SQLite), with an ordered outbox of change-sets as the only path to the server — the user interface never calls a server API directly. A media store holds photographs, videos and document scans, each hashed and metadata-tagged with location and timestamp at capture. A background sync engine, triggered by connectivity change, foreground activation, timer or manual "Sync now", pushes the outbox and pulls server changes. Form definitions and validation rules are distributed as versioned configuration, so the device rejects the same invalid inputs the server would reject.

Server identifiers — Case IDs, Registration Numbers, Ticket IDs, Trip IDs — are always assigned or computed on the server; the device holds only a provisional client identifier until synchronisation confirms the server-issued one. Every change-set and media object carries an idempotency key, so a retried push never creates a duplicate record.

10.4 Security of offline data

The local database and media store are encrypted with a per-installation key held in the device keystore, wrapped under a key derived from the officer's authenticated session — logout, session expiry or remote revocation destroys the wrapping key, making the data unreadable without fresh authentication. The application refuses to run on a rooted or jailbroken device, disables screenshots on screens showing personal data, and excludes application data from operating-system cloud backups. Aadhaar numbers are held only in memory for the duration of capture and are persisted only as a reference key, never in full.

10.5 Synchronisation and conflict management

On push, each change-set carries a client identifier, the operation, the changed fields, a content hash, and the server version the device last saw for that record. A conflict is detected when that base version is stale, or when a new record matches an existing one by a deterministic key. Neither version is ever discarded: the server stores the incoming variant alongside the current one, both fully attributed to their author, device and capture time, and the conflict enters a resolution queue.

Conflict class	Handling
Non-overlapping field edits (e.g. two different fields updated offline and online)	Merged automatically; both changes applied and audited — no queue entry needed
Same field, different values	Both versions preserved; the field shows "under review" until an authorised resolver decides
Two officers capture the same worker independently	Deterministic-key matches merge as an update variant; probabilistic matches route to the identity-matching review queue
Record edited offline after it changed state online (e.g. a ticket already closed)	The offline edit is stored as a variant; the closed record is not silently reopened

An authorised resolver sees both versions side by side — author, device, capture time, location and any attached evidence — and chooses per field, accepts one version whole, or records a corrected value with

a reason. The resolution is written as a new record version; both original variants remain stored and linked, never overwritten, and both original authors are notified of the outcome.

10.6 Clock, location and integrity

Every captured record stores both the raw device-clock time and a server-corrected time, computed from the clock offset recorded at each sync; a suspicious clock jump between syncs flags the record for review without blocking capture, and a monotonic sequence number preserves ordering regardless. GPS location is recorded with an explicit status — ok, low-accuracy, unavailable, or permission-denied — so a null location is never written silently; media captured for field-exposure evidence requires a location fix by design. Every photograph and video is hashed at capture and the hash is re-verified on receipt at the Document Vault, providing tamper evidence without any bespoke per-module implementation.

10.7 Operational modes and recovery

Mode	Behaviour	Recovery
Fully offline	Full capture per Section 10.1; deferred steps queue as per Section 10.2	Automatic sync on return to coverage
Intermittent connectivity	Sync batches small, media deferred, no user impact	Automatic, with backoff to prevent thrashing
Platform degraded or unavailable	Field queues grow with a device-side alert after a configured wait; desks buffer locally; kiosks show a bilingual "temporarily unavailable" notice and the centre falls back to its certified paper register	Devices and desks sync in arrival order on restoration; paper entries are keyed within an agreed working-day window
Device lost or replaced	Remote revoke wipes local data after the offline session lifetime expires	A replacement device is enrolled and assigned lists pulled fresh

10.8 Testing

The offline application is tested against a defined matrix — airplane mode for a whole capture session, network flapping during sync, connection loss mid-media-upload, application kill mid-sync, device restart with pending records, low storage, clock skew, denied/unavailable location, and two devices editing or creating the same record — verifying no data loss, no duplication, correct ordering, and correct conflict presentation in every case. A field pilot at several District Field Support Clusters runs before Phase 1 Go-Live for the household-survey and pre-registration flows, and before Phase 2 for mela operations.

10.9 Local data model on the device

The device database is WatermelonDB over SQLite with SQLCipher encryption. The user interface reads and writes only these tables.

Local table	Contents	Direction
`draft_record`	Form data per entity type, with the client identifier and form definition version.	Device to server

Local table	Contents	Direction
`outbox_change`	Ordered change-sets: client identifier, operation, changed fields, content hash, base server version, monotonic sequence, idempotency key.	Device to server
`media_item`	Photographs, videos and scans with SHA-256, location fix and status, capture time, upload offset.	Device to server
`assignment`	The worklist assigned to the officer: households, sites, events and groups.	Server to device
`server_record_cache`	Records pulled for the assigned scope, with server version.	Server to device
`lookup_*`	LGD hierarchy, trades, document types, vocabularies, scheme names.	Server to device
`form_definition`	Versioned form and validation definitions.	Server to device
`conflict_variant`	Conflicts reported by the server, with both versions for display.	Server to device
`sync_state`	Cursors, last successful sync time, clock offset samples.	Local
`device_info`	Device identifier, registration state, offline session expiry.	Local

10.10 Synchronisation protocol

Synchronisation is a pair of calls, push then pull, plus a resumable media upload. The figure shows one complete cycle.

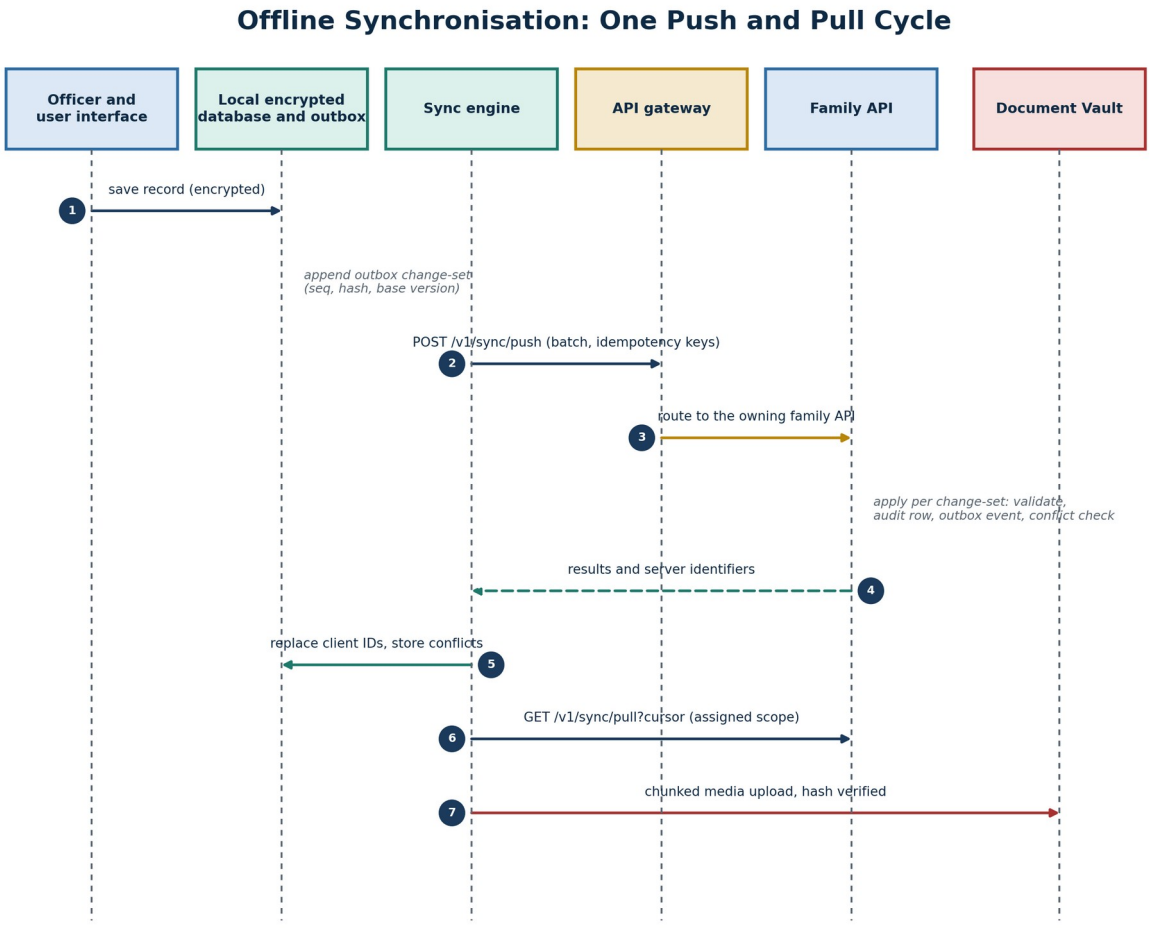


Figure 10.2 — One synchronisation cycle between the field application and the Platform

```
POST /v1/sync/push
{ "deviceId": "DEV-00417", "batchId": "01JB9C1X...", "clockOffsetMs": -4200,
  "changes": [
    { "clientId": "c-7f31", "seq": 412, "entity": "household_survey", "op": "CREATE",
      "idempotencyKey": "9d2f...", "baseVersion": null, "hash": "sha256:a1c4...",
      "capturedAt": "2026-11-12T09:14:55Z", "locationStatus": "ok",
      "fields": { "workerClientId": "c-7f30", "dependants": 3, "schoolGoing": 2 } } ] }

200 OK
{ "results": [
    { "clientId": "c-7f31", "status": "ACCEPTED", "serverId": "01JB9C2M4Q...", "version": 1 } ],
  "serverTime": "2026-11-12T09:21:03Z" }
```

Result status	Meaning and device action
ACCEPTED	Applied. The device replaces the client identifier with `serverId` and marks the change synced.
DUPLICATE	The idempotency key was seen. The original result is returned, and the device treats it as accepted.
CONFLICT	The base version was stale or a deterministic key matched. Both versions are stored by the server. The device keeps its copy marked under review and receives the conflict on the next pull.
REJECTED_VALIDATION	A rule failed on the server. The device shows the field errors to the officer and keeps the draft editable.
DEFERRED	The step needs a real-time interface (for example e-KYC) and has been queued with a provisional identifier. It completes automatically.

- **Order.** The outbox is pushed in ascending `seq`. Change-sets that reference another change-set's provisional identifier are applied after it, and a failure of the parent holds the children.
- **Pull.** `GET /v1/sync/pull?cursor=...` returns changes in the officer's assigned scope since the cursor, including tombstones, in pages of 500. The cursor is a per-scope server change sequence.
- **Retry.** Backoff of 5 s, 15 s, 45 s, 2 min and 5 min, then every 5 minutes, each with 20% jitter. The batch size falls from 100 to 20 change-sets on a poor connection and recovers gradually.
- **Power.** Automatic sync pauses below 15% battery. A manual Sync now always runs.
- **Time.** The device sends its clock offset on every push. The server stores the raw device time and the corrected time on each record (Section 10.6).

10.11 Resumable media upload

Element	Specification
Preparation on the device	Photographs scaled to a 1600 pixel long edge and encoded as JPEG at quality 80, about 300 KB each. Video limited to 720p, a bitrate cap of 1.5 Mbps and 60 seconds. Scans compressed to the document-type size cap.
Chunking	512 KB chunks, reduced to 256 KB on a 2G link. Each chunk carries its offset and SHA-256.
Resume	The server reports the last confirmed offset. After any interruption the device continues from that offset and never restarts the file.
Verification	When the last chunk arrives the server compares the SHA-256 of the whole object with the hash taken at capture. A mismatch discards the object and the device re-uploads.
Order and priority	Record data is pushed before media. Evidence media for a visit is uploaded before general photographs.
Network policy	Configurable per user group: any network, or unmetered networks only for video.
Device storage	Uploaded media is kept for 7 days and then purged. The application warns at 80% of its 2 GB quota and refuses new video above it.

11. Technical Support Unit, Helpdesk and Seven District Field Support Cluster Model

Support runs on three tiers, each with a defined scope, so that every issue reaches the right level of skill without unnecessary escalation.

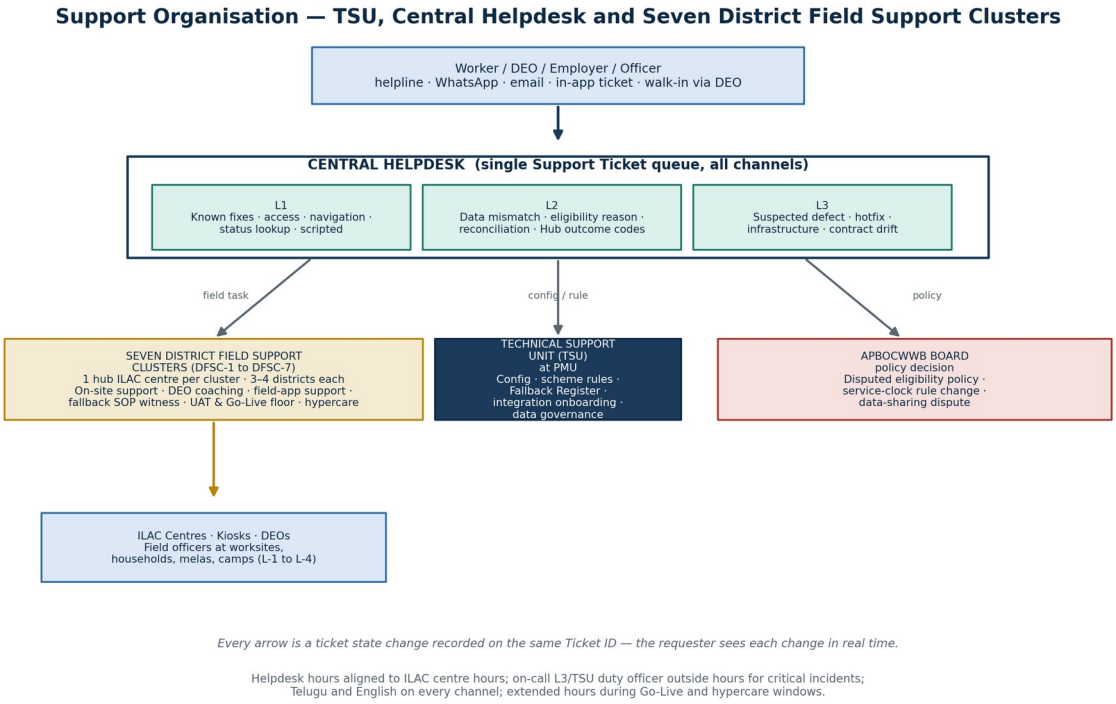


Figure 11.1 — Three-tier support organisation

11.1 The three tiers

Tier	Where	Who it serves	What it does
Technical Support Unit (TSU)	Embedded at APBOCWWB head office with the PMU	APBOCWWB, PMU, departmental nodal officers	Programme-level technical support: configuration, scheme-rule maintenance, the Fallback Register, integration onboarding, data governance, release management, MIS
Central Helpdesk (L1/L2/L3)	One central location, operating the PMU backoffice ticket queue	Every platform user — workers, employers, operators, PMU, departments	Ticket intake on every channel, triage, assignment, resolution, escalation, closure — using the Platform's own Support Ticket module as the ticketing system
Seven District Field Support Clusters (DFSC-1 to DFSC-7)	One hub ILAC centre per cluster, each covering 3–4 districts	ILAC centres, kiosks, Centre Data Entry Operators, field officers	On-site support, DEO coaching, field-app support, fallback-route witnessing and certification support, UAT/Go-Live floor presence, hypercare

11.2 Ticket levels

Level	Scope	Typical tickets
L1	Routine, well-understood issues with a known fix	Login/password, OTP not received, navigation, status lookup by Case/Ticket ID, printing
L2	Issues needing deeper platform or process knowledge	Data mismatch between modules, unclear eligibility results, manual-route reconciliation questions, offline sync conflicts referred from a cluster
L3	Complex technical issues needing specialist or development intervention	Suspected defects, hotfixes, infrastructure incidents, confirmed adapter contract drift, performance degradation
Board	Matters needing a policy or governance-level decision	Disputed eligibility policy, a request to change a service clock, a department data-sharing dispute

Every channel — toll-free helpline, WhatsApp, email, in-app ticket raising, walk-in via a Centre Data Entry Operator, or a cluster field visit — lands in the same queue with the same Ticket ID, so every role has equal access to raise a ticket, and every state change is visible to the requester in real time.

11.3 Seven District Field Support Clusters

Andhra Pradesh's districts are grouped into seven clusters of three to four districts each (exact composition confirmed with APBOCWWB at inception, accounting for tribal/Agency areas). Each cluster has a Cluster Lead, Field Support Engineers, and a Trainer/Change Agent, all with employee profiles scoped to the cluster's centres. Their duties span on-site software and configuration support, DEO coaching and refresher training, field-app support including sync-backlog and conflict-queue resolution, first-line explanation of Integration Hub outcome codes, fallback-route SOP support and certification witnessing, UAT and Go-Live floor presence, hypercare daily presence, and canary-district support during API cut-over. A cluster never closes a ticket it did not resolve — it escalates on the same Ticket ID to L2, L3 or the TSU as appropriate.

11.4 Interfaces between tiers

Every arrow in the flow diagram above is a ticket state change recorded on the same Ticket ID. A field task routes to the relevant District Field Support Cluster; a configuration or rule question routes to the TSU; a policy matter escalates to the APBOCWWB Board. Escalation carries the ticket's full history — nothing is re-explained from scratch at a higher level.

11.5 Service levels and measurement

Helpdesk hours are aligned to ILAC centre operating hours, with extended hours during Go-Live and hypercare windows, and an on-call L3/TSU duty officer outside hours for critical incidents. Every channel operates in Telugu and English — call scripts, ticket categories, the knowledge base and SMS/WhatsApp templates are all maintained bilingually. Severity is recorded at triage (Critical / High / Medium / Low) with response and resolution targets set from the RFP's service-level schedule at inception; a Critical issue

— a Phase-live module unavailable, a security event, data integrity at risk — triggers immediate L3 and TSU engagement with APBOCWWB informed.

11.6 Knowledge management

Call scripts, one knowledge-base article per L1 playbook and per Integration Hub outcome code, worker-facing FAQs published into the learning module, and release notes with configuration changes and known issues, are all maintained bilingually and owned by the TSU's Trainer and Knowledge Manager, with cluster Field Support Engineers contributing from the field.

11.7 Phase-wise ramp

The TSU is operational from Week 1 of the contract, well before any Go-Live, so that configuration, integration onboarding and data governance are already moving. The central helpdesk goes live with Phase 1 Go-Live, using the Support Ticket module as its ticketing system from day one. District Field Support Clusters are staffed for Phase 1 districts at first Go-Live and are fully operational statewide by Phase 2, with coverage extended to scheme desks in Phase 3 and the full operations-and-maintenance support model in place from Phase 4.

11.8 Tooling

No separate helpdesk product is introduced — the Support Ticket module is the ticketing system, so every ticket is auditable in the Platform's own Audit Ledger and visible in the Reporting Layer without any integration effort. Cluster staff carry a field kit: a laptop with Platform access, a test device with the field app, a bilingual SOP pack, and training materials.

11.9 Severity matrix and proposed service targets

Severity is set at triage from impact and reach. The targets below are proposals. The RFP service-level schedule replaces any figure it states, and the final values are fixed in Week 1 to 2. Response means a named person has acknowledged the ticket and started work. Resolution means a fix or an accepted workaround is in place.

Severity	Definition	Response	Resolution or workaround
Critical	A live module is unavailable, a security event is open, or data integrity is at risk. Affects more than one centre.	15 minutes, 24 by 7	Workaround in 4 hours, fix in 24 hours
High	A core function is degraded for one cluster, or a fallback route is failing.	30 minutes in service hours	Workaround in 1 working day, fix in 3 working days
Medium	A function fails for a small group and a workaround exists.	2 hours	5 working days
Low	A cosmetic fault, a question or a change request.	1 working day	Next scheduled release

11.10 Ticket categories and routing

Category	First owner	Typical next owner	Clock (proposed)
Login, OTP and password	L1 helpdesk	L2	4 working hours
Registration and renewal	L1 helpdesk	Cluster field engineer	1 working day
Scheme result or application	L2	TSU, scheme owner	3 working days
Integration outcome or delayed verification	L2	Integration Lead	2 working days
Offline sync backlog or conflict	Cluster field engineer	L2, then L3	1 working day
Notification not received	L1 helpdesk	L2	1 working day
Report or dashboard	TSU MIS analyst	L3	3 working days
Configuration or rule change request	TSU	PMU configuration owner	5 working days
Suspected defect	L2	L3	Per severity
Policy or data-sharing matter	TSU Head	Board	Per Steering Committee cadence

11.11 Helpdesk workload model and staffing basis

The staffing basis below is a planning model. It is replaced by measured volumes after Phase 1 hypercare, and the roster is adjusted at the end of each phase.

Assumption	Value	Source
Active operators and field officers at statewide scale	1,200	Planning assumption
Tickets per operator per month in steady state	0.8	Planning assumption, higher in hypercare
Tickets per month in steady state	About 960	1,200 x 0.8
Hypercare multiplier	x 3	First 4 weeks after each Go-Live
Share resolved at L1	65%	Target through knowledge articles
Average handling time at L1	12 minutes	Planning assumption
Peak tickets per working day in hypercare	About 140	960 x 3 / 21
L1 agent capacity per day	About 30 tickets	6 productive hours at 12 minutes

At 140 tickets a day with 65% handled at L1, hypercare needs about 3 L1 agents on duty plus cover, rising to a rostered team of 6. Steady state needs 2 on duty. Telephone traffic is counted on top of tickets and is sized with the helpline provider at inception.

11.12 Cluster field kit and visit pattern

- **Field kit.** A laptop with Platform access, a test device with the field application, a portable hotspot, the bilingual SOP pack, printed Case ID acknowledgement forms and the certified paper register template.
- **Visit pattern in hypercare.** Daily presence at centres with a fallback share above the agreed level. Weekly rounds elsewhere.

- **Visit pattern in steady state.** Each centre is visited at least monthly, and a centre whose scorecard falls below the agreed band is visited within 5 working days.
- **Evidence.** Each visit produces a visit record in SM-16 with the checklist, defects raised and training given.

11.13 Runbooks for common operational events

Each runbook is kept in the repository and tested in a tabletop exercise at least once a year. The table lists the trigger, the first response and the owner.

Event	Detection	First response	Owner
A department API is down	Breaker opens, source degraded alert	Confirm the fallback share is rising as designed. Tell clusters and the helpdesk. Contact the nodal officer. Set Fallback active, API suspended if the outage nears the clock threshold.	Integration Lead
Sync backlog grows at a cluster	Device queue age alert	Check gateway and sync API latency. Check the cluster connectivity. Ask the cluster to sync from a better location. Review conflicts queue.	Cluster Lead
SMS delivery falling	Delivery rate below threshold	Check provider status and DLT template state. Confirm in-app and WhatsApp legs continue. Switch to the secondary provider if configured.	TSU Head
Audit chain mismatch	Nightly verification	Declare a Critical incident. Preserve the database state. Compare against the last anchor. Notify APBOCWWB and the Security Lead at once.	Security Lead
Database failover	Availability group event	Confirm the new primary and the application reconnection. Check replication lag of the DR copy. Record the event.	DevOps Lead
Certificate near expiry	Certificate manager alert	Confirm automated renewal. If it failed, renew manually in the change window.	DevOps Lead
Suspected data exposure	Bulk-read alert or report	Contain, preserve evidence, classify, notify as in Section 8.7.	Security Lead

The runbook index, with the full text of each runbook, is in the Operations and Backup/DR Runbooks of Section 14.4.

12. Key Personnel and Governance Model

12.1 Team structure

The delivery team is organised around the programme's four core obligations: build and deploy the 22 modules, deliver the integrations and certified fallback routes, migrate and govern the data, and operate the TSU, helpdesk and District Field Support Clusters. Module-family leads (mirroring the eight service families of Section 2.2.2) report to the Solution Architect; the Integration Lead, Data Migration Lead, Security Lead, QA Lead, UX Lead, Mobile/Field App Lead and DevOps Lead report to the Programme Manager, who in turn reports to the Project Director — accountable to the APBOCWWB Steering Committee. The Technical Support Unit, headed from Phase 1 Go-Live, runs the central helpdesk and the seven District Field Support Cluster Leads in parallel to the build organisation.

12.2 Key personnel

Role	Responsibilities	Presence
Project Director	Single point of accountability to APBOCWWB; chairs the Bidder side of the Steering Committee; approves deviations and recovery levers; signs phase-gate submissions	All phases and O&M
Programme Manager	Day-to-day delivery; owns the integrated plan, RAID log, phase-gate readiness, weekly reporting, dependency management with departments	All phases; continues as O&M service manager
Solution Architect	Owns the architecture, the ten shared services, API-first contracts, non-functional requirements and design reviews	All phases; design authority through O&M
Integration Lead	Owns the Integration Hub, department onboarding, credential custody, LGD harmonisation, and the cut-over pipeline	All phases; leads integration change control in O&M
Data Migration Lead	Owns profiling, cleansing, matching, golden-record construction, dump ingestion, reconciliation, and the migration gates	Phase 1–3 full-time; part-time Phase 4/O&M
Security Lead	Owns security architecture, Aadhaar/DPDP controls, VAPT coordination and closure, backup/DR drills, incident response	All phases; annual audit and incident lead in O&M
QA / Test Lead	Owns the test strategy across all levels, UAT support and defect management; reports independently to the Project Director	All phases; regression owner in O&M
UX and Accessibility Lead	Owns worker-facing, kiosk, assisted-desk and field-app design; Telugu localisation; accessibility compliance	Phase 1–3 full-time; part-time thereafter
Mobile / Field App Lead	Owns the offline field application — capture, encryption, synchronisation, conflict management	Phase 1–3 full-time; part-time Phase 4/O&M
DevOps / Infrastructure Lead	Owns the five environments, infrastructure-as-code, pipelines, observability, backup/DR, source-deposit	All phases; O&M operations lead

Role	Responsibilities	Presence
	automation	
Training and Change Lead	Owns training-needs analysis, bilingual materials, Training-of-Trainers, DEO certification on fallback SOPs	All phases
TSU Head	Heads the Technical Support Unit; owns SLA compliance, release management, integration monitoring, reconciliation queues	From Phase 1 Go-Live through O&M
Helpdesk Manager	Runs the central L1/L2/L3 helpdesk; SLA reporting; escalation to development	From Phase 1 Go-Live through O&M
District Field Support Cluster Leads (7)	Each leads one cluster covering 3–4 districts: on-site support, DEO coaching, fallback-SOP adherence, field-app rollout	From Phase 1 Go-Live through O&M

Named individuals, CVs and any RFP-prescribed minimum qualifications for these roles are provided in the Key Personnel annexure accompanying this proposal; the roles and responsibilities above hold regardless of the specific individuals named.

12.3 Key personnel commitments

- Every key person is available from contract signing, and the Project Director, Programme Manager, Solution Architect, Integration Lead, Data Migration Lead and DevOps Lead join the Week 1 inception workshop.
- No key person is replaced before Phase 1 Go-Live except for reasons beyond the Bidder's control, and only with APBOCWWB's approval.
- A replacement holds equal or better qualification, is proposed with a CV in advance, and overlaps with the outgoing person for a defined handover period, evidenced in the decision log.
- The Programme Manager, Solution Architect, Integration Lead, Security Lead, TSU Head, Helpdesk Manager and Cluster Leads continue into O&M, so the people who built the Platform are the people who operate it.
- Every team member signs a confidentiality undertaking covering worker, departmental and Aadhaar-related data, and undergoes background verification before any access to production or migration staging data is granted.

12.4 Three-tier governance structure



Figure 12.1 — Three-tier governance and escalation matrix

Tier	Chair	Cadence	Decides
Steering Committee	APBOCWWB CEO	Monthly, and at every phase gate	Scope changes, phase-gate acceptance, deviations, recovery levers, O&M performance
Project Management Committee	PMU Head	Weekly	Plan adjustments within scope, risk responses, departmental escalations, priority of defects
Working Groups (6)	Family lead / relevant lead	Weekly or as scheduled	Detailed functional decisions within a module specification; day-to-day operational matters

The six Tier-3 working groups are: the module-family working groups, the Integration & Department Onboarding Forum, the Data Governance Committee (Section 7.2), the Security Working Group, the Training & Change Working Group, and the Operations Review (from Phase 1 Go-Live). A matter unresolved at one tier escalates to the next at its regular cadence, or sooner through the escalation matrix below.

12.5 Escalation matrix

Level	Trigger	Goes to
1	Working Group deadlock beyond an agreed number of working days	Programme Manager and PMU Head
2	Issue affects phase date, scope, a deviation, or a departmental agreement	Project Director and APBOCWWB Steering chair
3	Security incident of any severity, or DR invocation	Immediate — Project Director and APBOCWWB together, per Section 8.7
4	Dispute on acceptance, deviation or contract performance	Steering Committee, then the contractual dispute route

Escalation never pauses work — the lower tier continues on the agreed interim position while the higher tier decides, and every decision is recorded in a single, numbered decision log.

12.6 Department engagement governance

Because integration depends on departments outside the contract, the Integration & Department Onboarding Forum is the standing meeting with departmental nodal officers, with every source tracked as an onboarding case visible on the Integration Health Dashboard and reported weekly. A data-sharing agreement or credential issuance outstanding beyond an agreed number of working days is escalated by the Programme Manager, and further beyond that to the Steering Committee for APBOCWWB's direct intervention — while the affected source continues to serve workers on its certified fallback route throughout, so a departmental delay never stops service.

12.7 Quality governance

The QA Lead reports independently to the Project Director, not to the Programme Manager, and can withhold a phase-gate recommendation. Design review precedes build of each module family; code review runs on every merge; security review covers authentication, consent, Aadhaar handling and integration adapters; localisation and accessibility review runs before UAT. Quality metrics — defect density, test coverage, UAT first-pass rate, open Critical/High defects, performance-test results, SLA compliance, fallback share — are reported monthly to the Steering Committee.

12.8 O&M governance

From Certified Final Go-Live, the Platform is operated by the TSU under a defined cadence: a weekly-then-monthly Operations Review covering SLA compliance and integration health; a fortnightly Change Advisory Board reviewing release content, regression evidence and rollback plans; a quarterly Security and DR Review; an annual independent Security Audit; and a quarterly Contract Performance Review before each O&M payment instalment — since O&M payment is in equal annual instalments, subject to performance, SLA compliance and acceptance, the evidence produced through this governance is the basis of that acceptance.

12.9 Role-to-permission examples

The matrix gives the starting role set. Roles are configuration, so APBOCWWB can add roles and adjust permissions at inception and later without a release. Each permission is further limited by centre scope and information visibility.

Role	Module access	Actions	Scope and visibility
Centre Data Entry Operator	SM-01, 02, 03, 04, 05, 06, 11, 15, 18	View, Create, Edit	Own centre. Masked Aadhaar and bank number.
Centre Supervisor	As operator plus SM-16	View, Create, Edit, Approve, Escalate	Own centre. Can countersign F2 elements.
Field Officer	SM-01 (pre-registration), 04, 11, 15, 20, 21	View, Create, Edit	Assigned households and sites.
Counsellor	SM-08, 09, 10	View, Create, Edit	Own centre.
Para-legal Volunteer	SM-17	View, Create, Edit	Own cases and sessions.
PMU Case Manager	SM-04, 07, 17	View, Edit, Approve, Escalate	Statewide, no Class 1 data.
PMU Configuration Owner	SM-19	View, Create, Edit	Drafts configuration; cannot approve own drafts.
PMU Head	All, read; SM-19 approve	View, Approve	Statewide.
Department Officer	SM-07, 13	View	Own department rows only.
MIS Analyst	SM-13	View, Create (reports)	Statewide, aggregates and row exports with approval.
Administrator	SM-19	View, Create, Edit	No access to case data without a case-specific grant.
Auditor	Audit portal	View	Read-only, all events, no editing right anywhere.

12.10 Change control procedure

Step	Name	Description
1	Request	Any APBOCWWB or Bidder person raises a change request with the reason, the affected modules and the urgency.
2	Classification	Standard (pre-approved pattern such as a configuration publication), Normal (needs review) or Emergency (security or Critical defect).
3	Impact assessment	Solution Architect assesses scope, schedule, security, data, fallback and training impact. Output is a short impact note.
4	Decision	Change Advisory Board for operations (fortnightly). Steering Committee for a change in scope, schedule or a deviation.
5	Implementation	Built and tested in the lower environments and released through the release train of Section 4.12.
6	Verification and closure	The requester confirms the outcome. The decision log and the Fallback Register are updated where affected.

12.11 RAID log structure

The RAID log is kept as configuration data on the Platform itself, so entries, owners and actions are visible to APBOCWWB and appear in the weekly report. Every entry has the fields below.

Field	Content
Identifier and type	Risk, Assumption, Issue or Dependency, with a running identifier.
Description and cause	What may or did happen, and why.
Probability and impact	Scored 1 to 5 each, with the product used to rank. A score of 15 or more goes to the Steering Committee.
Owner and due date	One named owner and the date of the next action.
Response	Avoid, reduce, transfer or accept, with the specific action and any recovery lever of Section 4.8.
Status and history	Open, in progress, closed or realised, with every change dated.

13. UAT, Testing, Training, Go-Live and Hypercare Plan

13.1 Test strategy

Testing runs as a pyramid — automated at the lower levels, on every merge; structured and evidenced at the upper levels, per build candidate and per phase.

Level	Scope	Tooling	Frequency
Unit	Service logic, validation rules, Scheme Engine rules, status machines, sync-conflict logic	Vitest	Every commit
Service & API contract	Every API against its OpenAPI specification; RBAC and jurisdiction scoping at the boundary	Supertest, contract tests	Every merge
Integration	Module-to-shared-service interactions; Integration Hub adapters against mocks and sandboxes	Vitest, adapter conformance suite	Every build
End-to-end UI	Worker, employer, DEO, PMU and administrator journeys across every surface, in both languages	Playwright, Detox/Maestro (field app)	Every build
Performance & resilience	Load to 300 concurrent access nodes, soak, failover, backup restore, sync bursts	k6, chaos scripts	Per phase
Security	SAST/DAST on every merge; independent VAPT by an empanelled auditor before each acceptance	CI scanners; external VAPT	Every merge; per phase
Accessibility & localisation	WCAG 2.1 AA / GIGW conformance; Telugu/English parity	Automated axe checks; manual audit	Per phase
Offline & synchronisation	Field-app capture, buffering, reconnection, conflict preservation	Device-lab matrix (Section 10.8)	Per phase and on any sync change
UAT	Authority-executed acceptance against approved test cases	UAT scripts, defect board	Two cycles per phase

A Requirements Traceability Matrix links every functional and business rule to at least one test case and its latest result — a requirement with no passing test blocks the phase gate. Defect severity thresholds require zero open Critical and zero open Major defects at every Go-Live.

13.2 Functional testing

Every module has a test pack derived from its Module Specification (or, where none was supplied, from the proposed design in Section 3), covering every numbered functional requirement, a negative test per business rule, boundary tests per validation rule, and the specification's own acceptance criteria as end-to-end scenarios. Cross-cutting tests are applied to every module regardless of its own specification: the documented-next-step rule, assisted-mode parity, Telugu/English parity, role and jurisdiction scoping, consent gating, audit completeness, and every fallback route exercised with the relevant interface simulated as absent, unavailable, delayed, degraded and rate-limited.

13.3 Integration testing

Every adapter passes a fixed conformance suite before being enabled above Development: contract validation against its OpenAPI descriptor, forced coverage of every typed outcome code, retry/circuit-breaker behaviour under simulated timeouts, idempotency under duplicate submission, LGD harmonisation across a representative sample of codes, a static scan confirming no credential exists outside the secrets vault, and — where a department permits it — a live sandbox run. Where a department allows production calls before cut-over, the adapter runs in Shadow state during UAT with daily agreement-rate reporting feeding the Fallback Register.

13.4 Non-functional testing

Performance testing targets the registration wizard, Scheme Engine evaluation, the Work Request/token flow, notification bursts, MIS dashboards, sync bursts after connectivity restoration, and dump ingestion — all at or above the 300-concurrent-access-node baseline, with specific response-time targets set from the RFP's service-level schedule at inception. Resilience testing includes induced component failures, a full DR drill with measured recovery-point and recovery-time objectives, and backup-restore verification including Audit Ledger hash-chain integrity after restore. Security testing runs continuously in the pipeline and through independent VAPT (Section 8.6).

13.5 Data migration testing

Migration testing follows the gate structure of Section 7.9 without repetition here — profiling, rules acceptance, two mock migrations, a dress rehearsal, and cut-over certification. UAT verifies migrated data through the same read-back screen a worker sees at registration, on a stratified sample of records, confirming that Verified and Migrated-unverified statuses are correctly shown and actionable by operators.

13.6 User acceptance testing

UAT is chaired by an APBOCWWB or PMU-nominated UAT Lead, who approves scripts and issues the sign-off certificate; the Bidder's UAT Support Team prepares the environment and data, trains testers, and fixes defects within agreed turnaround times. Participants include PMU and backoffice staff, Centre Data Entry Operators and supervisors from pilot centres across multiple districts (including at least one KSK-only centre), departmental nodal officers, employer and worker representatives, and helpdesk staff. A defect triage board meets daily during UAT cycles. UAT for a phase exits only when every script has

passed (or has an accepted Minor defect with a fix date), zero Critical or Major defects remain open, and every fallback route for that phase's modules is certified.

13.7 Training plan

A Training-of-Trainers model draws a core group of trainers from the seven District Field Support Clusters and the PMU; each cluster then delivers role-based training to its own districts, with Bidder trainers present for the first delivery in each cluster and for any new-module release — giving every district a resident training capability for refreshers and new joiners throughout operations.

Curriculum	Covers	Audience
Centre operations — foundation	Registration and renewal (all steps), Household Master, document verification, work requests, grievance intake, card-token issue, next-step recording	DEOs, supervisors
Centre operations — services	Scheme results and applications, service requests, events and counselling attendance, IEC/LMS assistance, field exposure, mela desk, agency desk	DEOs, supervisors
Fallback SOPs	Every certified route, evidence lists, attestation and countersign, the Needs Review queue, reverse cut-over behaviour	DEOs, supervisors, PMU reviewers
Field operations	Household survey, worksite visit, camp registration, offline capture and sync, conflict resolution	Field officers, cluster staff
Backoffice	Case inbox, convergence register, session/event creation, broadcast, HR and attendance, para-legal intake	PMU staff
Administration	Roles, users, centres, Service Areas, configuration console, Fallback Register	Administrators
MIS and oversight	Dashboards, reports, exports, data-quality dashboard, integration health	Management, MIS staff, department users
Security and privacy	DPDP handling, consent, masking, incident-reporting duties	All staff

Attendance is recorded in the HR module and course completion in the learning module; a Centre Data Entry Operator is certified to operate a module only after passing the assessment and a supervised assisted transaction in the training environment — an uncertified operator holds a restricted role until certified. Training for each phase runs in the two weeks before that phase's Go-Live: trainers first, then centre and backoffice staff, then on-the-job coaching in the Go-Live week itself.

13.8 Go-Live plan

A fixed readiness checklist gates every phase Go-Live: the UAT sign-off certificate, VAPT Critical/Major closure, certified fallback routes for the phase's modules, the migration gate passed, DR drill and restore tests passed, performance targets met, DLT SMS templates approved, the helpdesk and clusters staffed and rostered, training complete with DEOs certified, monitoring live, source code deposited, and a rehearsed rollback plan. The Steering Committee holds a formal Go/No-Go meeting ahead of the planned

date, reviewing checklist evidence and recording the decision; a No-Go returns the phase to the fix window under the recovery mechanism of Section 4.8.

The cut-over runbook itself: freeze legacy transactions and start the paper register → take and verify the pre-load backup → execute the final delta migration and reconciliation → deploy to Production and run the automated smoke suite → enable canary centres (one per cluster) with cluster staff on site, monitoring service clocks, notification delivery and adapter health → enable statewide and announce Go-Live through broadcast → enter paper-register transactions as assisted transactions with full Case ID discipline → hand over to hypercare. A war room stays open from before the first centre opens until after the last closes on Day 1, with cluster staff at every canary centre and an hourly health summary to the PMU.

13.9 Hypercare

Hypercare runs for an agreed number of weeks after each phase Go-Live, with the Phase 4 hypercare extending through to Certified Final Go-Live. Three layers operate together: a central war room at PMU staffed by the delivery team and TSU; floor support at centres by the District Field Support Clusters, prioritised toward centres with higher fallback share or defect volume; and the helpdesk at full roster with an expedited path from L1 straight to the war room for Critical and Major issues. Critical defects receive a workaround or fix within hours, with the certified fallback route active in the meantime so no worker is ever turned away; hypercare exits only when defect, service-clock-breach, notification-delivery and sync-failure metrics hold steady for a defined number of consecutive working days and centres are confirmed to be operating without floor support.

13.10 Documentation and knowledge transfer during testing and training

The master test plan, module test packs, the traceability matrix, automated test suites and data generators, performance/security/accessibility/offline test reports, VAPT reports and closure evidence, UAT scripts and sign-off certificates, training curricula and materials in Telugu and English, attendance and certification registers, and Go-Live/hypercare reports are all deposited to the Authority repository at each phase gate and at least monthly — this feeds directly into the exit-management approach of Section 14.

13.11 Defect severity definitions

Severity	Definition	Example
Critical	No workaround. Data loss, a security breach, or a core journey blocked for all users.	Registration Number issue fails. A role can read another centre's workers.
Major	A core function fails or gives a wrong result and the workaround is costly.	A scheme result is wrong for a defined rule. A closure accepted without next-step fields.
Moderate	A function fails for a subset or in a rare condition and a simple workaround exists.	A report column is missing in one filter combination.
Minor	A cosmetic, wording or usability defect with no functional effect.	A label wraps badly on a small screen.

Exit rules at each Go-Live are zero open Critical and zero open Major defects. An accepted Minor or Moderate defect carries a fix date and an owner.

13.12 Performance test scenarios

Performance tests run in UAT with production-like data volumes. Each scenario has a pass criterion that comes from the response budgets of Section 2.20 or from the RFP schedule where the RFP states a number.

ID	Scenario	Load profile	Pass criterion
P1	Mixed interactive load at the 300-node floor	300 virtual users with the traffic mix of Section 2.20, 60 minutes	p95 within budget per interaction class. Error rate below 0.5%.
P2	Growth case at 500 nodes	Ramp to 500 users over 20 minutes, 30 minutes at plateau	p95 within 20% of P1. No autoscaling flapping.
P3	Registration wizard peak	150 concurrent registrations through all 15 steps	Step save p95 under 700 ms. No duplicate Registration Numbers.
P4	Scheme baseline batch	Evaluate all migrated households (Year-1 volume)	Completes within 24 hours without affecting P1 latency.
P5	Sync burst after an outage	200 devices push 100 change-sets each within 10 minutes	No loss, no duplicates, correct ordering, p95 push under 3 s per 100.
P6	Notification burst	Campaign of 500,000 messages over the provider limits	Provider limits respected, in-app and interactive traffic unaffected.
P7	Dashboard concurrency	60 concurrent dashboard users on the replica	Widget p95 under 3 s. Primary latency unchanged.
P8	Dump ingestion	Largest expected dump (about 2 million rows) while P1 runs	File completes inside its window. P1 latency unchanged.
P9	Soak	70% of the P1 load for 24 hours	No memory growth beyond 10%. No connection pool exhaustion.
P10	Failover under load	Fail a database primary and an API node during P1	Service restored within the RTO. No committed write lost.

13.13 Training schedule and certification

The durations below are proposed and are fixed in the Training Needs Analysis at inception. A DEO is certified after passing the written and practical assessment for the module.

Curriculum	Duration	Format	Pass mark	Refresher
Centre operations, foundation	3 days	Classroom and sandbox	80% and a supervised transaction	Annual, and on a new release
Centre operations, services	2 days	Classroom and sandbox	80% and a supervised transaction	Annual
Fallback SOPs	1 day	Scenario drills	100% on safety-critical steps	Six-monthly
Field operations and offline	2 days	Field exercise	80% and a field exercise	Annual
Backoffice	2 days	Classroom	75%	Annual
Administration	2 days	Hands-on	80%	On a new release

Curriculum	Duration	Format	Pass mark	Refresher
MIS and oversight	1 day	Hands-on	75%	Annual
Security and privacy	0.5 day	Classroom and quiz	85%	Annual

13.14 Hypercare exit criteria

The thresholds below are proposed and agreed with the PMU at inception. Each must hold for 10 consecutive working days before hypercare exits.

Metric	Threshold	Measured by
Open Critical and Major defects	Zero	Defect board
Service-clock breaches as a share of cases due	Below 2%	SM-13 ticket SLA mart
Notification delivery rate (in-app plus at least one channel)	97% or more	SM-14
Sync failures as a share of change-sets pushed	Below 0.5%	Sync metrics
Tickets per operator per day	Falling and below 0.1	Helpdesk
Centres operating without floor support	All centres	Cluster Leads
Fallback share outside agreed bands	No source above its band	SM-13 fallback mart

14. Source Code, Documentation and Exit-Management Plan

14.1 Principles

Principle	How it is delivered
APBOCWWB owns the bespoke Platform	A perpetual, irrevocable, transferable, royalty-free licence is embedded in every deposited repository; no bespoke component is withheld or delivered only as compiled/minified code
Everything needed to rebuild, deploy, operate and change the Platform is in the Authority's own repositories	A reproducible-build check runs at every phase gate, witnessed by the Authority — a competent third party with no contact with the Bidder can rebuild every environment from the repository alone
No bidder-held secret, licence, account or tool is needed to run the Platform	All third-party accounts (SMS/DLT registration, WhatsApp Business account, IVR numbers, TLS certificates, domain names, hosting subscriptions) are opened in APBOCWWB's name from day one; the secrets-vault master key is held by the Authority from Production Go-Live
Open-source components only, except where explicitly approved	A software bill of materials is deposited with every release; any addition passes through change control
No proprietary Bidder framework or runtime	The Platform is built entirely on public, documented, community-maintained frameworks
Documentation is code	Every document lives in the repository, versioned with the release it describes, and generated from source wherever possible
Knowledge transfer is continuous from Phase 2, not an end-of-contract event	Section 14.4
Exit is planned from Phase 1	The Exit Plan is a maintained deliverable, updated at every phase gate, not written only when exit approaches

14.2 Repository structure and deposit mechanism

All deposits go to Git repositories owned and administered by APBOCWWB. The Bidder holds contributor access only — repository ownership, protected-branch rules and audit logs remain with the Authority from the very first commit. A single monorepo holds the whole Platform: applications (web, field-mobile, admin console), the ten shared services, module packages, adapters (one per departmental source, with contract, mapping, mock and tests), shared TypeScript types, workflow definitions, database migrations, infrastructure-as-code per environment, CI/CD pipeline definitions, the full test suite, documentation, training materials and the generated software bill of materials.

Deposit is automated, not a manual end-of-month task: every merge to the main branch is already a deposit, because the repository is the Authority's own; a monthly tag generates a deposit manifest and report; each phase-gate deposit is a full manifest with a witnessed reproducible build. The deposit manifest is machine-readable and records the commit hash, build-artefact checksums, the software bill of materials reference, full test results, static/dependency scan results, VAPT status, and configuration-snapshot identifiers.

Reproducible-build verification. At each phase gate, and at exit, an Authority-nominated engineer rebuilds the tagged release from the Authority's own repository on a clean machine, using only the deposited pipeline definitions and public package registries, and deploys it to UAT. The rebuilt artefact checksums must match the deposit manifest; a failed reproducible build blocks gate acceptance until corrected. This is the strongest available proof that the Authority genuinely holds everything needed to operate the Platform independently.

14.3 What is deposited

- Application source — all apps, services, module packages, shared packages and workflow definitions.
- API contracts — an OpenAPI 3.1 document for every inbound API and every outbound adapter.
- Database schema, versioned migrations, seed data and the data dictionary.
- Configuration — every Master Data & Configuration export: forms, the scheme-rule catalogue, service clocks, escalation matrices, notification templates, the Fallback Register, jurisdiction and Service Area masters, role and permission definitions.
- Infrastructure-as-code for all five environments, and the build/test/scan/deploy/deposit pipeline definitions, written to be portable to another CI system if required.
- The full test suite — unit, contract, end-to-end, performance, security-scan configuration, fallback-route certification scripts, migration and reconciliation tests, and synthetic test-data generators.
- The software bill of materials and licence inventory, generated per release.
- The full documentation set (Section 14.4) and training materials.
- Runbooks — operations, incident, backup/DR, deployment, on-call, escalation, helpdesk.
- Security and audit evidence — VAPT reports and closure evidence, security policies, access reviews, incident reports, backup/DR test records.

Excluded from the repository by design: production secrets and credentials (held only in the secrets vault), personal data of any kind, and departmental production dumps. Each is handed over at exit through the mechanisms in Section 14.5.

14.4 Documentation set

A defined set of living documents — Solution Architecture Document, per-family Detailed Design, Data Dictionary, API Reference (generated from OpenAPI at build), an Integration Adapter Guide per source, the Configuration Guide, Fallback Route SOPs in Telugu and English, the Operations and Backup/DR Runbooks, security policies and VAPT evidence, the Deployment Guide, a Developer Onboarding Guide, per-role User Manuals in Telugu and English, the Kiosk and Field Application Guide, Training Manuals, the Test Plan and Traceability Matrix, Release Notes, a Decision Log, and the Exit Plan itself — all live in the repository, versioned with the release they describe, and are regenerated by the pipeline wherever possible so they cannot drift from the running code.

14.5 Knowledge transfer

Knowledge transfer begins in Phase 2 and runs through operations, so that by Certified Final Go-Live, APBOCWWB's own PMU and TSU-embedded staff can operate, configure and safely change the Platform — making exit a formality rather than an event. Authority-nominated engineers pair with Bidder engineers on adapters, configuration and operations tasks from Week 6; PMU administrators take over configuration changes in Staging/QA from Phase 2 and in Production from Phase 3; Authority staff shadow deployments, DR tests and incident handling from Phase 3, then lead them with Bidder backup from Phase 4; and Authority engineers are pull-request reviewers on the module families they will own, from Phase 2 onward.

A structured programme in operations covers architecture, code walkthroughs per module family, adapter development (a trainee builds a sandbox adapter end to end), configuration, operations (a trainee leads a DR drill), security (a trainee completes a key rotation and an incident tabletop), data stewardship, and helpdesk/cluster operations — every session recorded, every competency signed off per participant, and the completion register reviewed as an exit precondition.

14.6 Exit management plan

Exit management applies on contract expiry, on termination for any reason, or on transition to a successor vendor — the obligations are the same in every case; only the timeline compresses on termination. An Exit Management Committee, constituted under the Steering Committee, includes APBOCWWB, PMU, the TSU lead, the Bidder's exit manager and, when known, the successor vendor. The Exit Plan itself is first delivered at the Phase 1 gate and re-deposited at every subsequent gate and annual review.

Activity	What happens
Final deposit and reproducible build	A final tag with full manifest; a reproducible build witnessed by the Authority or successor engineers
Credentials and secrets	Every secret is rotated by the Authority with the Bidder present; all Bidder personal and service accounts are disabled and the disablement logged
Licences, accounts, domains, certificates	Ownership confirmed in APBOCWWB's name — a verification, not a transfer, because these were opened in the Authority's name from day one
Departmental integration credentials	Re-issuance to APBOCWWB requested through the department onboarding channel where a department originally issued credentials to the Bidder as an organisation
Data handover and verification	Full exports of the system-of-record database, the Audit Ledger with its hash-chain anchors, the Document Vault, and configuration snapshots, with control totals reconciled — a safeguard and verification, since the running environments are already Authority-owned
Environment handover	Administrator control over all five environments; optionally, the successor rebuilds Production from the repository into a fresh environment and cuts over, as the strongest possible verification
Support transition	A parallel-run period in which the successor handles tickets with Bidder shadow support; the open ticket backlog, knowledge base and vendor contact list are

Activity	What happens
	handed over
Field and cluster transition	District Field Support Cluster territory maps, centre contact lists, visit schedules and training registers handed over
Certificate of Exit Completion	Issued by the Exit Management Committee once every precondition is evidenced and signed by APBOCWWB

Service levels, SLAs, helpdesk and cluster coverage, and change delivery continue unchanged through the exit period — the Bidder does not reduce staffing before the Certificate of Exit Completion, and imposes no deployment freeze of its own. Confidentiality obligations survive exit for the period set in the contract, and any Platform data or credentials held on Bidder systems are securely destroyed within an agreed period, with a destruction certificate issued.

14.7 Vendor neutrality and reuse

The design is built so that APBOCWWB can reuse it without further dependence on the Bidder: a new scheme, district, form or service clock is a configuration change through the console; a new department is a descriptor, contract and mapping added by Authority engineers following the Integration Adapter Guide; a new module follows one documented template with clear extension points; and — because jurisdiction scoping is built into every role from the start — a second welfare board or department could in principle be onboarded as a separate tenant configuration, reusing the same registration, case, consent, audit, notification, integration and MIS services, without touching a line of core code.

14.8 Build pipeline, software bill of materials and reproducible build

The pipeline runs in the Authority's own repository and CI system, and its definitions are deposited with the code so that it can be moved to another CI product. Figure 14.1 shows the six stages. A failed stage stops the pipeline, and an artefact can be promoted only if the build stage produced it.

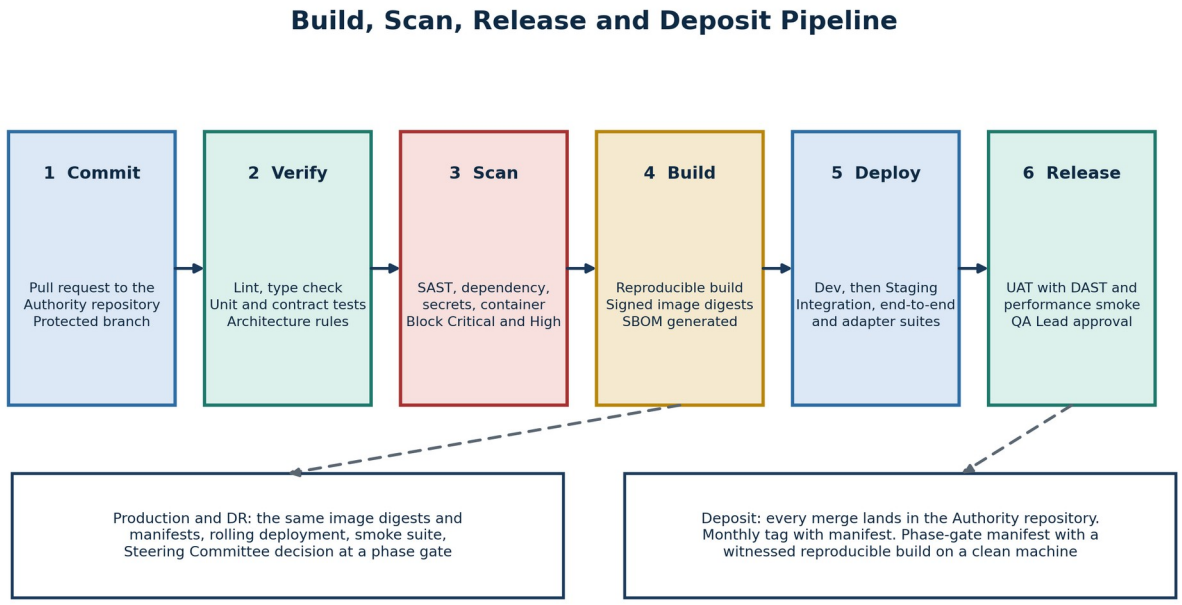


Figure 14.1 — Build, scan, release and deposit pipeline

Control	Specification
Branch protection	Main is protected. Merge needs two approvals, one of them from an Authority engineer on module families the Authority owns. Direct pushes are blocked.
Dependency control	A lock file per workspace. Packages are installed from a mirror controlled by the Authority. A new dependency needs a licence check against the approved list (MIT, Apache-2.0, BSD, MPL-2.0, ISC) and change control.
Software bill of materials	CycloneDX JSON generated for every image and every application at build, signed and stored with the release. Compared with the previous release to list added, removed and updated components.
Image build	Multi-stage builds from a pinned minimal base image. Build timestamps and file ordering normalised, so the same source gives the same digest.
Signing	Images are signed at build with a key held in the Authority's vault. The cluster admission policy admits only signed images from the Authority registry.
Scans	Static analysis, dependency scanning, secrets scanning and container scanning on every merge request. Any Critical or High finding blocks release. A waiver needs the Security Lead and is recorded with an expiry date.
Reproducible build test	At each phase gate a clean build runs on a fresh machine from the repository and public registries only. The resulting digests must equal those in the deposit manifest.
Deposit manifest	Commit hash, image digests, SBOM reference, test results, scan results, VAPT status, configuration bundle identifier and the infrastructure-as-code version.

14.9 Exit plan timeline

The timeline shows the planned order of exit activities when the contract expires. On termination the same activities run on a compressed schedule agreed by the Exit Management Committee. T is the contract end date.

When	Activity	Evidence
T minus 6 months	Exit Plan refreshed. Successor, if known, joins the Exit Management Committee. Knowledge-transfer completion register reviewed.	Updated Exit Plan
T minus 4 months	Successor engineers shadow deployments, DR drills and incident handling. Backlog of open tickets and changes reviewed.	Shadowing log
T minus 3 months	Departmental credentials re-issued to APBOCWWB. Third-party accounts verified in the Authority's name.	Verification record
T minus 2 months	Parallel run starts: successor handles tickets with Bidder shadow support. Reproducible build witnessed by successor engineers.	Parallel-run report
T minus 1 month	Final full export of the database, Audit Ledger, Document Vault and configuration with control totals. Secret rotation rehearsed.	Reconciled export report
T	Final deposit. Secrets rotated by the Authority. Bidder accounts disabled and the disablement logged.	Final manifest, access log
T plus 1 month	Support continues in a shadow role. Destruction of Bidder-held copies with a certificate.	Destruction certificate
After all conditions are met	Certificate of Exit Completion issued by the Exit Management Committee.	Signed certificate

14.10 Documentation set contents

Document	Principal contents	Generated or written
Solution Architecture Document	Principles, layers, shared services, decisions and their reasons	Written, versioned
Detailed Design per family	Entities, APIs, events, rules, workflows	Part generated from code
Data Dictionary	Every table and column with type, class, owner and retention	Generated from migrations
API Reference	Every endpoint, schema and example	Generated from OpenAPI
Integration Adapter Guide (per source)	Contract, mapping, credential procedure, fallback route, reconciliation	Written, tested by the conformance suite
Configuration Guide	Every configuration item type and how to change it	Part generated
Fallback Route SOPs	One per route in Telugu and English	Written
Operations and Backup/DR Runbooks	Procedures of Sections 8.8, 8.9 and 11.13	Written, drilled
Deployment Guide	Rebuild of every environment from the repository	Written, proven by the reproducible build
Developer Onboarding Guide	Repository, tooling, conventions, first change	Written
User Manuals by role	Telugu and English	Written
Test Plan and Traceability Matrix	Strategy, test packs, requirement links	Part generated
Decision Log, Release Notes, Exit Plan	Numbered decisions, per-release changes, exit procedure	Written

15. Compliance Matrix and Deviations

This section states plainly, clause area by clause area, how this proposal responds to the RFP, and declares every deviation and clarification in one place, so the evaluation committee does not need to search the rest of the document to find them.

15.1 Compliance summary by RFP area

RFP area	Response
Programme background & objectives (4E pathway, six service loci, household as unit of convergence, no-worker-turned-away mandate, dual-mode convergence)	Comply — Section 1, Section 3
Scope of work & operating principles (22 modules, notified integrations, fallback routes, web/mobile/field/kiosk/assisted-desk channels, API gateway, data governance, five environments, non-functional requirements, TSU/helpdesk/clusters, training, source deposit)	Comply — Sections 2–14
Functional modules & solution requirements (SM-01 to SM-22)	Comply — Section 3, with one qualifier (Deviation D2, below)
Architecture, security, privacy & data requirements (modular API-first architecture, database baseline, security & privacy controls, source-code and IPR terms)	Comply — Section 2, Section 8, with one qualifier (Deviation D1, below)
Delivery, testing & acceptance	Comply — Section 4, Section 13

15.2 Deviations

Exactly two deviations are declared. Every other requirement in this proposal is answered as written in the RFP.

Deviation D1 — system-of-record database platform

Position. The RFP names PostgreSQL 15 or higher, or an Authority-approved functionally equivalent open-source platform, as the system-of-record database, with PostGIS and pg_trgm for spatial and fuzzy-matching needs.

What is proposed. The recommended primary platform for the system-of-record database is stated in the commercial and technical annexures accompanying this proposal, reflecting the Bidder's current production experience and the fastest path to the 17-week schedule. Functional equivalence with PostGIS (point-in-area jurisdiction assignment, job-matching proximity) and pg_trgm (fuzzy name/address matching for deduplication) is demonstrated through native geography types and full-text/trigram matching capability, and the data-access layer is built to support both the recommended engine and PostgreSQL 15+ without any other change to the architecture.

Impact if not approved. If APBOCWWB does not approve the recommended platform, the Platform is delivered on PostgreSQL 15 or higher with PostGIS and pg_trgm, with no change to any module, any fallback route, any integration pattern, or any other section of this proposal. The decision is needed at

inception so that the first migration mock runs on the confirmed engine; a decision taken after Week 2 is treated as a schedule risk under the recovery mechanism in Section 4.8.

Deviation D2 — face-match at registration

Position. The Worker Registration module specification calls for an automated face-match between the live photograph captured at registration and the UIDAI e-KYC photograph, with routing to manual verification on a low confidence score. This is an AI face-verification capability that this scope (Part 1) excludes — such capability belongs to the advanced technology overlay procured separately.

What is proposed instead. The live photograph is captured and biometric consent is logged exactly as specified. The photograph is displayed beside the UIDAI e-KYC photograph, and the Centre Operator records a Verified or Not Verified decision with a reason; Not Verified routes to the existing manual-verification path. The verification step is implemented with a pluggable interface so that an automated comparison can be enabled later, when procured, without any redesign; the confidence threshold remains a configuration item that is simply unused in this scope.

Impact. Identity assurance at registration in this scope rests on UIDAI OTP-based e-KYC and an audited operator decision, rather than an automated score. No AI face-verification component is quoted, deployed or charged under this proposal.

15.3 Clarifications

The items below are not deviations — they are positions taken where a module specification and this scope needed reconciling, stated here for transparency and confirmed with APBOCWWB at inception without affecting any phase commitment.

Item	Position
Biometric smart card referenced in several module specifications	Delivered as a QR card token on existing centre hardware — the same tap-login and footfall-capture function without any new biometric device
Notification channel set (specification names only in-app and SMS)	In-app and SMS remain the mandatory baseline for every event; WhatsApp, IVR and helpline are additional channels on the same engine
Household as the unit of convergence (specification is worker-centric)	A Household Master is added under SM-01, with the household baseline survey captured in the field
Offline field operation (no specification describes it)	The field application is offline-first for worksite, household and camp functions (Section 10)
Service clocks and the documented next step (not defined per-module in the specifications)	Delivered once, as a shared Case Service property, applied identically to every module (Section 2.2.1, Section 3.2)
"Work request" terminology (used for both a worker's assistance ticket and an employer's job posting)	Both are delivered under SM-05 as distinct, clearly named object types
Renewal (excluded from the registration specification, but	An annual renewal wizard is added under SM-01, re-touching only fields that change year to year

Item	Position
implied by the lifelong-record requirement)	
Token printing (specification assumes kiosk hardware, excluded from this scope)	The token component uses the centre's existing display and printer, with on-screen and SMS fallback where none exists

15.4 Assumptions carried through this document

- The mandatory delivery schedule is planned as a 17-week programme from contract signing, to be aligned to the RFP's own schedule at contract signature without changing phase scope.
- Service-level targets — availability, recovery-point/recovery-time objectives, helpdesk response times, page-load and throughput targets — are set from the RFP's service-level schedule at inception; where a specific figure is not yet fixed in this document, the design and control are stated in full and the number is confirmed in Week 1–2.
- Named personnel, CVs and any RFP-prescribed minimum qualifications for key roles are provided in the accompanying Key Personnel annexure.
- Where two equivalent components are named in Section 2.8, the first is proposed and the second is an equivalent alternative acceptable to the Authority, to be fixed at inception.

None of the items in Sections 15.3 and 15.4 changes the Comply status recorded in Section 15.1 — they record how a position was reached, not a gap in compliance.

15.5 Cross-reference of RFP requirement areas to this proposal

The table links each requirement area stated in Section 15.1 to the places in this document where the design, the technical specification and the evidence are found. It is a navigation aid for the evaluation committee and does not change any compliance position.

Requirement area	Design	Technical detail and evidence
Programme background and objectives	Sections 1 and 3.3	1.9 parameters; 3.7 Case Service
Functional modules SM-01 to SM-22	Section 3.2	Technical specification per module; 3.6 to 3.9; Annex A
Modular, API-first architecture	Sections 2.1 to 2.7	2.11 to 2.19; Annex A
Technology stack and open-source terms	Section 2.8	2.21 baseline; 14.8 licence control
Integration with notified sources	Section 5	5.10 to 5.16; Annex C
Manual and assisted fallback	Section 6	6.12 to 6.16
Data migration and quality	Section 7	7.11 to 7.16
Security, privacy, Aadhaar and audit	Section 8	8.12 to 8.21
Channels and accessibility	Section 9	9.10 to 9.13
Offline field capture	Section 10	10.9 to 10.11
Support, helpdesk and clusters	Section 11	11.9 to 11.13

Requirement area	Design	Technical detail and evidence
Personnel and governance	Section 12	12.9 to 12.11
Testing, training, Go-Live and hypercare	Section 13	13.11 to 13.14
Source code and exit	Section 14	14.8 to 14.10
Hosting	Section 16	16.8 to 16.13

15.6 Deviation impact summary

Item	Position	Technical consequence	Decision needed by
D1	System-of-record database platform	The data-access layer supports both engines through the mapping of Section 3.6, so no module, adapter or fallback route differs between them.	Inception, Week 2 (M2)
D2	Operator decision in place of automated face match	The verification step is an interface. An automated comparison can be plugged in later and the confidence threshold is already a configuration item.	Inception

16. Hosting Landscape and Transition

This section sets out the recommended Year-1 hosting footprint for the Digital ILAC Platform — compute, storage and bandwidth — a CSP recommendation consistent with MeitY's empanelment framework, and the transition approach from an initial cloud or NIC deployment to the Andhra Pradesh State Data Centre (APSDC), should APBOCWWB direct that transition. Figures in this section are illustrative Year-1 sizing, built bottom-up from the scale parameters stated in the RFP and in this proposal — the 300-concurrent-access-node floor, five environments, and an assumed Year-1 register of up to 3 million worker records (existing APBOCWWB registrations migrated plus fresh statewide enrolment) across roughly 800,000 households and 25 ILAC centres scaling statewide. These figures are confirmed and, if necessary, revised against APBOCWWB's actual volumes and the RFP's service-level schedule during inception (Section 4.5), and every number below should be read as a recommended starting allocation, not a hard ceiling.

16.1 Hosting topology

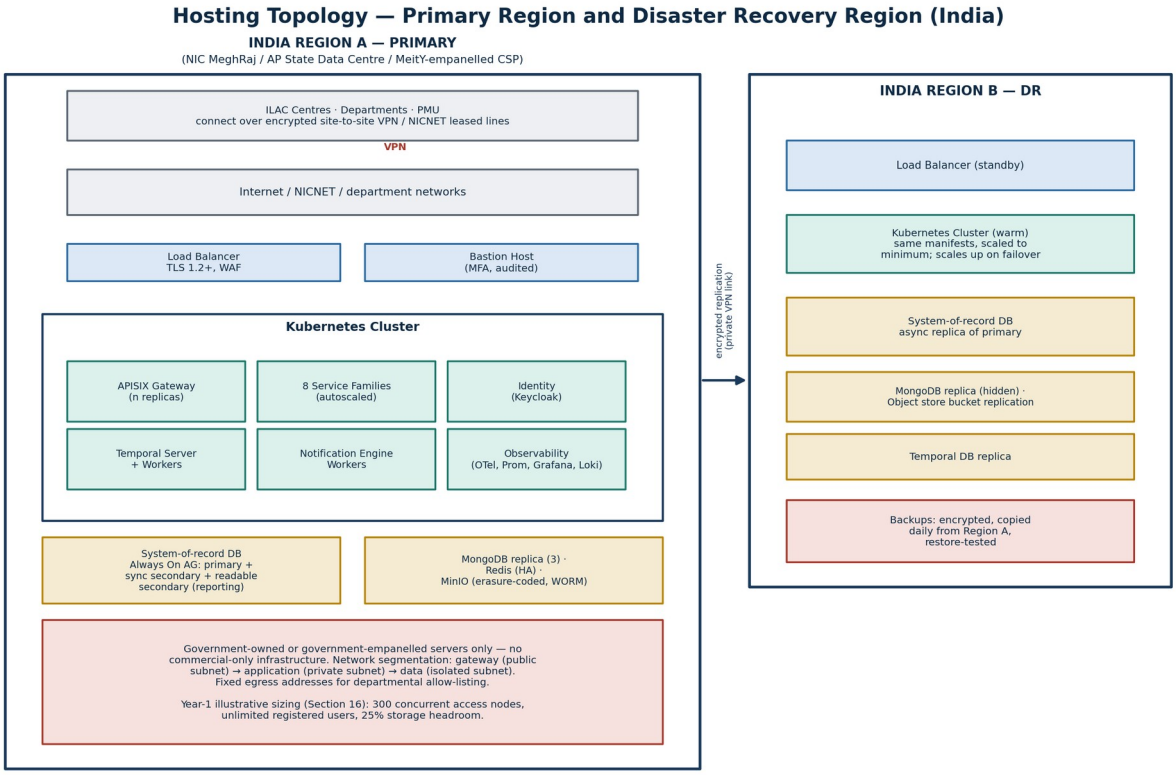


Figure 16.1 — Primary and disaster-recovery hosting topology

The Platform is hosted exclusively on Government of India or Andhra Pradesh government infrastructure — government-owned servers at NIC or the AP State Data Centre, or a MeitY-empanelled government cloud service provider in an Indian region — as APBOCWWB approves in writing, with the disaster-recovery environment in a second Indian region. No commercial hosting outside that empanelment is used at any point. Network segmentation places the API gateway in a public subnet, application services in a private subnet reachable only from the gateway, and data stores in an isolated subnet reachable only from application services.

All connectivity into and within this hosting footprint is VPN-protected, as set out in Section 8.3.7: ILAC centres, the PMU and the District Field Support Clusters reach the Platform over an encrypted site-to-site VPN or NICNET leased line; departmental integration traffic passes over a VPN tunnel or private link with fixed addresses for allow-listing; and primary-to-DR replication runs over a dedicated encrypted private link, never the open internet. This is shown as the VPN boundary in Figure 16.1.

16.2 Year-1 recommended compute

Compute is sized per environment. Production and Disaster Recovery run on Kubernetes for load balancing, health-based restart and rolling deployment; Development and Staging/QA are sized for build and test workloads and may run on lighter process management if preferred.

Environment	Indicative compute	Basis
Production	≈ 160 vCPU / 640 GB RAM across the tier (application/gateway nodes, database primary + two secondaries, cache and semi-structured-store nodes)	Sized to the 300-concurrent-access-node floor with headroom to grow toward 500 nodes within Year 1 without a re-architecture
Disaster Recovery	≈ 60–70 vCPU / 260 GB RAM warm standby, auto-scaling to the Production profile on declared failover	Warm-standby cost efficiency; full capacity available within the recovery-time objective agreed at inception
UAT	≈ 80 vCPU / 320 GB RAM	Full-volume masked migration data and pre-Go-Live performance testing require a production-like profile
Staging / QA	≈ 48 vCPU / 192 GB RAM	Continuous integration, regression and mid-scale load testing
Development	≈ 24 vCPU / 96 GB RAM	Developer workloads and sandbox adapters; synthetic data only

Application services scale horizontally within this compute envelope — each of the eight service families runs multiple replicas behind the gateway, autoscaled on CPU and request latency, so that a load spike in one family (for example, the Integration Hub during a dump-ingestion season, or Communication during a broadcast campaign) is absorbed by adding replicas of that family alone, without over-provisioning the rest of the Platform.

16.3 Year-1 recommended storage, with 25% headroom

Storage is sized bottom-up from the same Year-1 volume assumption, across the structured system-of-record database, the semi-structured staging store, the Document Vault (evidence, photographs and scans), and backups across all environments.

Store	Base Year-1 need	+25% headroom	Recommended provisioned
System-of-record database (Production)	≈ 650 GB	≈ 160 GB	≈ 800 GB (NVMe-backed, with room to a 1.5 TB ceiling before re-

Store	Base Year-1 need	+25% headroom	Recommended provisioned
			sizing)
Read replica (reporting)	Mirrors primary	—	≈ 800 GB
Semi-structured store (MongoDB — staging, dumps, snapshots)	≈ 400 GB	≈ 100 GB	≈ 500 GB
Document Vault (photographs, scans, certificates, evidence, WORM)	≈ 11 TB	≈ 2.8 TB	≈ 14 TB, erasure-coded
Backups (full + differential + transaction-log, all stores, rolling retention)	≈ 18 TB	≈ 4.5 TB	≈ 22.5 TB, in a second facility or region
Non-production environments (UAT, Staging, Dev — masked/synthetic data)	≈ 3.5 TB	≈ 0.9 TB	≈ 4.4 TB combined
Year-1 total (all stores, all environments)	≈ 34 TB	≈ 8.5 TB	≈ 42–43 TB provisioned

The 25% headroom is applied on top of the base bottom-up estimate for every store, not only in aggregate, so that no single store (most importantly the Document Vault, which grows fastest as registration volume rises) runs close to capacity before the Year-2 sizing review. Object storage is erasure-coded and horizontally expandable, so headroom can be added without downtime as actual registration volumes become clear during Phase 1 and Phase 2.

Sizing assumption stated for transparency: an average document footprint of roughly 4 MB per worker across the full document checklist (photograph, Aadhaar copy, work certificate, bank passbook and similar), applied to an assumed Year-1 population of up to 3 million workers, plus IEC/LMS video content. This assumption is confirmed against APBOCWWB's actual document policy and average file sizes at inception and may move the Document Vault figure in either direction.

16.4 Year-1 recommended bandwidth

Link	Recommendation	Basis
Primary data-centre internet/NICNET uplink	1 Gbps dedicated, burstable to 2 Gbps	300+ concurrent sessions, departmental API traffic, notification dispatch bursts, and dump-ingestion windows
Primary-to-DR replication circuit	500 Mbps dedicated, low-latency	Continuous database and object-store replication within the recovery-point objective
Centre-side connectivity (per ILAC centre)	2–4 Mbps broadband or equivalent (existing centre connectivity, not part of this scope)	Kiosk, assisted-desk and field-sync traffic is designed for 2G/3G-class links; centre connectivity itself is outside this RFP's scope, and the offline-first design (Section 10) absorbs outages gracefully
Departmental integration	Sized per department within the	31 API sources plus scheduled dump

Link	Recommendation	Basis
egress	primary uplink; fixed egress IPs for allow-listing	transfers from 16 more

Bandwidth, unlike compute and storage, is reviewed monthly rather than only at Year-1 — dump-ingestion seasons, broadcast campaigns, and job-mela check-in surges are bursty by nature, and the Integration Hub's rate-limiting and queueing design (Section 5.2) exists specifically so that a burst on one channel never starves interactive traffic on another.

16.5 MeitY-empanelled CSP recommendation

Three hosting options are available under RFP Clause 6.3, and this proposal recommends the following sequence, subject to APBOCWWB's written approval:

Option	Recommendation	Why
1 — NIC MeghRaj	Preferred starting point for Phase 1–2	Fastest provisioning path within the 17-week window; no fresh empanelment or procurement cycle; well-understood by APBOCWWB's existing IT function; India-resident by definition
2 — A MeitY-empanelled cloud service provider (GI Cloud / MeghRaj empanelment framework)	Acceptable alternative, or a bridge if NIC MeghRaj capacity or timelines do not fit Phase 1	Commercial-grade elasticity for the Year-1 compute and storage envelope above, with Safe-to-Host certification obtained before go-live; the current empanelment list should be confirmed with MeitY/NIC at inception, since empanelment status changes over time
3 — AP State Data Centre (APSDC)	Target steady-state for Year 2 onward	The state's own facility gives APBOCWWB the most direct long-term operational control and the lowest recurring hosting cost once the Platform is stable; Section 16.6 sets out the transition

Whichever option is selected, the disaster-recovery environment is provisioned in a second Indian region or facility from the same infrastructure-as-code used for Production, so that DR readiness (Section 8.9) does not depend on which primary option APBOCWWB ultimately chooses.

16.6 Cloud-to-APSDC migration and transition approach

Where APBOCWWB directs a transition from an initial NIC MeghRaj or CSP deployment to the AP State Data Centre — typically once APSDC capacity and connectivity are confirmed adequate for the live Platform — the transition follows the same disciplined, staged approach already used for departmental API cut-over (Section 6.9), applied to infrastructure instead of a single adapter:

Stage	What happens
1. Readiness assessment	APSDC capacity, network connectivity to ILAC centres and departmental systems, and compliance posture (ISO/IEC 27001, empanelment status) confirmed against the Year-1 sizing in this section

Stage	What happens
2. Parallel environment build	A full non-production environment (mirroring UAT) is stood up at APSDC from the same infrastructure-as-code already deposited with APBOCWWB (Section 14.3) — proving portability before any live data moves
3. Data replication (shadow)	The system-of-record database, Document Vault and configuration are continuously replicated from the current primary to APSDC, exactly as DR replication already works (Section 8.9) — no live traffic is served from APSDC yet
4. Cut-over rehearsal	A timed rehearsal — freeze, final delta sync, integrity verification, DNS and gateway switch, smoke test — executed in a maintenance window with a tested rollback, mirroring the migration cut-over runbook of Section 7.9
5. Canary cut-over	Traffic for one or two districts is switched to APSDC first, with the previous environment kept live and in sync as an immediate fallback
6. Full cut-over	Statewide traffic moves to APSDC once canary metrics (latency, error rate, integration health) hold steady for an agreed period
7. Decommission of the interim environment	The NIC MeghRaj / CSP environment is kept in a cold-standby or fully decommissioned state per APBOCWWB's direction, only after a defined stabilisation period at APSDC

Because the Platform's infrastructure is entirely defined as code (Section 2.8, Section 14.3) and every runtime component other than the system-of-record database is open-source and self-hosted, this transition is a hosting-location change, not a re-platforming exercise — the same containers, the same Kubernetes manifests and the same database engine run at APSDC as ran on the interim hosting. No module, no integration pattern and no fallback route changes as a result of this transition.

16.7 Ongoing capacity review

Capacity — compute, storage and bandwidth alike — is reviewed against actual usage at the end of each phase and formally at the Year-1 anniversary, using the same Reporting Layer dashboards that track integration health and fallback share (Sections 5.7, 6.11). Storage headroom is topped up ahead of need, in line with the 25% buffer principle applied throughout this section, rather than reactively once a store approaches capacity.

16.8 Kubernetes cluster design

Production and Disaster Recovery run one Kubernetes cluster each, spread across three availability zones inside the region where the hosting facility offers them. The figure shows the node pools and their zones. The vault zone and the data subnet are outside the shared application pools, which keeps the Aadhaar Data Vault and the stores on separate hosts from the application workloads.

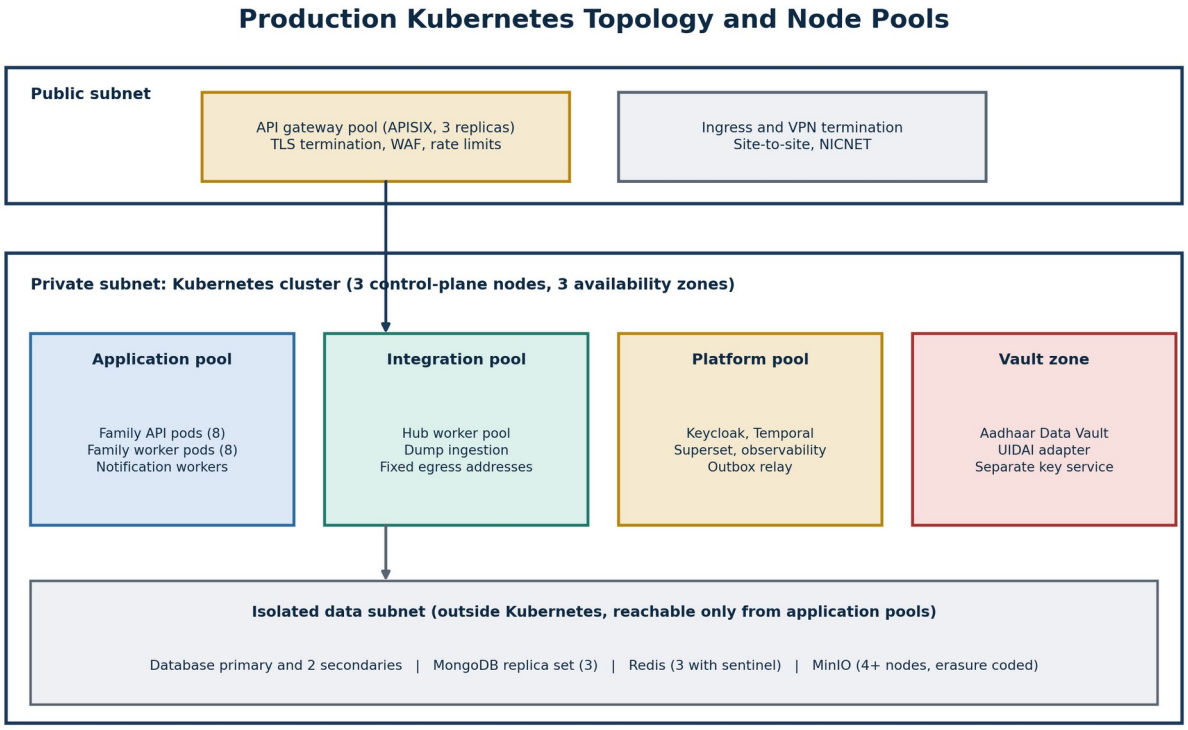


Figure 16.2 — Production Kubernetes topology and node pools

Node pool	Nodes	Node size	Workloads and scheduling
Application	3	12 vCPU, 48 GB	Family API and worker pods, notification workers. One node per zone, with pod anti-affinity spreading replicas across zones.
Integration	2	8 vCPU, 32 GB	Hub worker pool and dump ingestion. Tainted so only integration workloads schedule here. Fixed egress addresses.
Platform	2	6 vCPU, 24 GB	Keycloak, Temporal, Superset, observability collectors, outbox relay.
Vault zone	2	4 vCPU, 16 GB	Aadhaar Data Vault and the UIDAI adapter, in a separate network zone with its own key service.
Control plane	3	4 vCPU, 16 GB	One per zone. Counted outside the tier figure where the hosting facility supplies it as a managed service. etcd is encrypted and no application pods run here.

Autoscaling uses the horizontal pod autoscaler on CPU, memory and request latency for API pods, and on task-queue backlog for worker pods. A pod disruption budget keeps at least two thirds of the replicas of every family available during a node drain. Every pod sets resource requests and limits, and a namespace quota stops one family from consuming the cluster.

16.9 Capacity workings behind the Year-1 figures

The workings show how the compute and storage tables of Sections 16.2 and 16.3 are derived, so that APBOCWWB can replace an assumption and see the effect.

Tier	vCPU and RAM	Working
Application pool	36 vCPU, 144 GB	3 nodes of 12 vCPU, 48 GB
Integration pool	16 vCPU, 64 GB	2 nodes of 8 vCPU, 32 GB
Platform pool	12 vCPU, 48 GB	2 nodes of 6 vCPU, 24 GB
Vault zone	8 vCPU, 32 GB	2 nodes of 4 vCPU, 16 GB
Database tier (primary and two secondaries)	48 vCPU, 192 GB	3 nodes of 16 vCPU, 64 GB, with the primary taking transactional load and the secondaries serving failover and reporting
Redis and MongoDB nodes	24 vCPU, 96 GB	6 nodes of 4 vCPU, 16 GB
MinIO object store nodes	16 vCPU, 64 GB	4 nodes of 4 vCPU, 16 GB, erasure coded
Production total	160 vCPU, 640 GB	Matches the Production row of Section 16.2
Document Vault base need	About 11 TB	3,000,000 workers at 4 MB each, plus learning video and generated certificates
Database base need	About 650 GB	Rows of Section 3.6 with indexes at about 2.5 times the row size
Backups base need	About 18 TB	Retention of Section 8.20 across all stores

The pool sizes add up to the Production row of Section 16.2. Pods are small, with typical requests of 250 millicores, so each family can scale its replicas inside these pools. The Production figures are a starting allocation, and the capacity review of Section 16.7 revises them against measured usage.

16.10 Disaster Recovery environment

Aspect	Design
Posture	Warm standby. The Kubernetes cluster runs at reduced size and the stores run as replicas. Capacity scales to the Production profile on declaration.
Database	Asynchronous replication of the log stream over the dedicated link. Replica lag is a monitored metric with an alert at 2 minutes.
Document Vault	Bucket replication with object lock and versioning retained at the target.
Configuration and images	The same signed images, manifests and configuration bundle as Production.
Network	The gateway address is moved by DNS with a short TTL. Departments allow-list both regions' egress addresses in advance, so a failover does not need a new allow-listing request.
Drill	Twice a year with one full failover and failback, witnessed by APBOCWWB. Measured RPO and RTO are reported against Section 8.20.

16.11 Network and DNS design

- **Addressing.** A private address plan per environment with non-overlapping ranges, so that VPN peering between environments and with departments never collides.
- **DNS.** Public names for the gateway only. Internal names are resolved by the cluster and the private zone. TTL of 60 seconds on the gateway name.
- **Load balancing.** Layer 4 balancer in front of the gateway pool, health checked on `/readyz`. The gateway load balances to the family pods.

- **Egress.** All departmental traffic leaves through fixed addresses owned by the Authority. The list is published to departments at onboarding and any change goes through change control.
- **VPN.** Site-to-site tunnels for centres and clusters, with a standby tunnel over a second carrier where the centre has one.

16.12 Observability stack and retention

Signal	Tooling	Retention	Use
Metrics	OpenTelemetry to Prometheus, long-term store	15 days full resolution, 13 months downsampled	Service, adapter, service-clock and sync metrics. Capacity review.
Logs	Structured logs to Loki	180 to 365 days by log class (Section 8.17)	Incident analysis and audit support.
Traces	OpenTelemetry traces	7 days, with 100% of errors and slow requests sampled	Latency analysis across gateway, service and adapter.
Dashboards	Grafana	Not applicable	Operations, integration health, DR readiness.
Alerting	Alert rules to the Notification Engine and the on-call rota	Not applicable	Page the duty officer for Critical, queue the rest.

Indicator	Alert threshold	Owner
Gateway 5xx rate	Above 1% for 5 minutes	DevOps Lead
API p95 latency by class	Above budget for 10 minutes	Solution Architect
Database connections	Above 70% of the limit	DevOps Lead
Replica or DR lag	Above 2 minutes	DevOps Lead
Open circuit breakers	Any source open for more than 30 minutes	Integration Lead
Service clocks near breach	More than 5% of open cases above 75% elapsed	TSU Head
Notification failure rate	Above 5% in 15 minutes	TSU Head
Sync queue age on devices	Above 24 hours at a cluster	Cluster Lead
Audit chain verification	Any mismatch	Security Lead
Certificate expiry	Under 21 days	DevOps Lead

16.13 Infrastructure as code layout

```
infra/
  terraform/
    modules/        network, kubernetes, database, object-store, bastion, dns
    envs/           dev, staging, uat, prod, dr        (one state per environment)
  ansible/          host hardening, database and store provisioning, certificates
  helm/             one chart per family plus platform components
  policies/         network policies, pod security, admission rules
  runbooks/         automation used by DR, rotation and restore drills
```

Every environment is created from the same modules with different variables, which is why a late hosting decision costs days (Section 4.6.1) and why the Production environment can be rebuilt at the Authority's chosen facility from the repository alone.

Annex A. API Catalogue

The catalogue lists the principal endpoints of each service family. It is the index to the OpenAPI documents in `packages/contracts`, which are the authoritative contract and carry every request and response schema. The roles shown are the starting role set of Section 12.9. All paths are under `/v1`, except the health endpoints, and every endpoint follows the standards of Section 2.13. Scope means the data an actor can reach is further limited by centre and information visibility.

A.1 Registration & Records

Method	Path	Purpose	Roles
POST	`/registrations`	Start a registration draft	Operator, Field Officer
PUT	`/registrations/{id}/steps/{n}`	Save one wizard step with validation	Operator, Field Officer
POST	`/registrations/{id}/ekyc`	Run OTP e-KYC through the Hub	Operator
POST	`/registrations/{id}/photo`	Upload the live photograph and record the decision	Operator
POST	`/registrations/{id}/submit`	Submit and issue the Registration Number	Operator
GET	`/workers/{id}`	Worker record, masked by role	Operator, Supervisor, PMU
GET	`/workers/{id}/360`	Full 360-degree view	Operator, Supervisor
GET	`/workers?filter[...]`	Search within scope	Operator, Supervisor, PMU
POST	`/workers/{id}/renewals`	Annual renewal	Operator, Worker
GET	`/me`	Own profile	Worker
POST	`/households/{id}/survey`	Household baseline survey	Field Officer, Operator
GET	`/merge-candidates`	Matching review queue	Data Steward
POST	`/merge-candidates/{id}/decision`	Merge or reject a candidate	Data Steward

A.2 Entitlement & Convergence

Method	Path	Purpose	Roles
GET	`/schemes`	Catalogue with the active version	All staff
GET	`/workers/{id}/entitlements`	Eligible, Applied, Availed view	Operator, Worker
POST	`/scheme-evaluations`	Force re-evaluation of a household	Operator, TSU
POST	`/scheme-applications`	Initiate an application	Operator, Worker
POST	`/scheme-applications/{id}/status`	Record a status read from the department	Operator, Dept Officer, Integration
GET	`/scheme-applications/{id}/pack`	Manual desk pack for F3	Operator
GET	`/convergence-requests`	Register of departmental requests	PMU, Dept Officer (own rows)
POST	`/convergence-requests`	Log a manual departmental request	Operator
POST	`/committees/{id}/meetings`	Create a committee meeting	PMU Case Manager
POST	`/meetings/{id}/minutes`	Record minutes	PMU Case Manager
POST	`/meetings/{id}/decisions`	Record a decision and link cases	PMU Case Manager

A.3 Employment & Skilling

Method	Path	Purpose	Roles
POST	`/employers`	Employer registration	Employer
POST	`/employers/{id}/verification`	Record the verification decision	PMU Case Manager
POST	`/jobs`	Create a job posting	Employer
GET	`/jobs`	Search postings by district, trade and date	Worker, Operator
POST	`/jobs/{id}/enrolments`	Enrol a worker	Worker, Operator
POST	`/engagements`	Confirm engagement by Registration Number	Employer, Operator
POST	`/musters`	Daily muster entry	Employer, Field Officer
POST	`/work-requests`	Create an assistance token	Worker, Operator
POST	`/work-requests/{id}/call-next`	Call the next token	Operator
POST	`/site-visits`	Record a site visit	Field Officer
POST	`/skilling-events`	Create a training event	Operator, PMU
POST	`/rpl-batches`	Create an RPL batch	PMU
POST	`/mela-events`	Create a job mela	PMU
POST	`/mela-checkins`	Check in a worker or employer	Field Officer, Operator
POST	`/sites`	Create or update a site	Field Officer
POST	`/grouping-rules/{id}/run`	Run a grouping rule	PMU, TSU
POST	`/livelihood-groups`	Form a group	Field Officer
POST	`/federations`	Form a federation	PMU

A.4 Counselling & Learning

Method	Path	Purpose	Roles
POST	`/counselling-sessions`	Schedule a session	Counsellor
POST	`/counselling-sessions/{id}/attendance`	Record attendance	Counsellor
POST	`/counselling-sessions/{id}/outcome`	Record the outcome and referrals	Counsellor
POST	`/students`	Create a student profile	Counsellor
GET	`/courses`	Course library	Worker, Operator
PUT	`/course-enrolments/{id}/progress`	Save lesson progress	Worker
POST	`/assessments/{id}/attempts`	Submit an assessment attempt	Worker
GET	`/certificates/{id}`	Signed certificate	Worker
POST	`/iec-contents/{id}/publish`	Publish IEC content to an audience	PMU

A.5 Grievance & Service Requests

Method	Path	Purpose	Roles
POST	`/cases`	Open a case	Operator, Field Officer

Method	Path	Purpose	Roles
POST	`/cases/{id}/steps`	Add a step	Case owner
POST	`/cases/{id}/closures`	Close with the two permitted outcomes	Case owner, Supervisor
POST	`/cases/{id}/reopen`	Reopen within the window	Supervisor
POST	`/tickets`	Raise a ticket	All actors
GET	`/tickets/{id}/track`	Public status by Ticket ID and verified mobile	Worker, Employer
POST	`/tickets/{id}/escalations`	Escalate	Case owner, Supervisor
POST	`/tickets/{id}/resolution`	Resolve with a code	Case owner
POST	`/service-requests`	Raise a citizen-service request	Operator
POST	`/service-requests/{id}/output`	Attach the output document	Operator
POST	`/legal-aid-cases`	Open a legal-aid case	Operator, Para-legal
POST	`/legal-aid-cases/{id}/referrals`	Refer to DLSA or a clinic	Para-legal
POST	`/agency-requests`	Raise an agency request	Operator
POST	`/agency-requests/{id}/handoff`	Hand off to the agency	Operator

A.6 Communication, Administration, MIS and Platform

Method	Path	Purpose	Roles
POST	`/calendar-events`	Create a calendar event	PMU, Operator
POST	`/referrals`	Create a referral	Staff
POST	`/referrals/{id}/outcome`	Record a referral outcome	Receiving role
POST	`/campaigns`	Create a broadcast campaign	PMU
POST	`/campaigns/{id}/approve`	Approve a campaign (second user)	PMU Head
PUT	`/me/channel-preferences`	Set channel and language	Worker, Employer
POST	`/inbound/delivery-reports`	Provider callback	System
POST	`/users`	Create a user	Administrator
PUT	`/users/{id}/roles`	Assign roles	Administrator
POST	`/config-items/{key}/versions`	Draft a new configuration version	PMU Configuration Owner
POST	`/config-versions/{id}/publish`	Approve and publish	PMU Head
POST	`/config-bundles/export`	Export a signed bundle	PMU Head
POST	`/attendance`	Record attendance	Staff, Supervisor
GET	`/scorecards`	Centre and functionary scorecards	Supervisor, PMU
POST	`/reports/{id}/runs`	Run a report	MIS Analyst, PMU
POST	`/exports`	Request an export	MIS Analyst (approval for row data)
GET	`/admin/integration-health`	Hub health	TSU, Integration Lead
PUT	`/admin/adapters/{id}/cutover-state`	Change a cut-over state	Integration Lead
POST	`/sync/push`	Push change-sets	Field device
GET	`/sync/pull`	Pull changes in scope	Field device
GET	`/healthz`, `/readyz`	Liveness and readiness	Internal

Annex B. Data Dictionary Extract and Conventions

The full Data Dictionary is generated from the migrations and deposited with each release (Section 14.10). This extract lists the shared and principal tables with their data class (Section 8.3.1), the way they change, and their retention. The retention periods are proposals that the Authority confirms at inception against the BOCW Act, the DPDP Act and its own schedule.

B.1 Shared platform tables (schema core)

Table	Principal columns	Class	Mutability	Retention
`case`	public_id, worker_id, module_code, channel, mode, status, clock_started_at, clock_due_at	6	Status only	Life of the worker record
`case_step`	case_id, seq, type, mode, fallback_class, reason_code, operator_id, evidence_refs, is_provisional, received_at	6	Insert only	Life of the worker record
`case_closure`	case_id, type, outcome_code, evidence_doc_id, next_step fields	6	Insert only	Life of the worker record
`service_clock`	case_id, calendar, target, warn_at, due_at, paused_total	6	Updated by workflow only	With the case
`consent_record`	worker_id, purpose, notice_version, channel, granted_at, withdrawn_at	5	Insert only	Life of the record plus statutory period
`audit_ledger`	lane, seq, ts, actor, role, action, subject, correlation_id, prev_hash, hash	6	Insert only	Per Section 8.17
`audit_chain_head`	lane, last_hash, last_seq	6	Updated in the append transaction	Current only
`document`	public_id, case_id, type, sha256, size, bucket_key, retention_class	3 to 5 by type	Insert only, write-once object	By document type
`wlh_event`	worker_id, event_type, module_code, occurred_at, summary_ref	6	Insert only	Life of the worker record
`outbox_event`, `inbox_event`	event_id, type, version, aggregate, payload, state	6	State only	30 days then archive
`integration_call`	source_id, purpose, outcome, latency_ms, consent_ref, correlation_id	6	Insert only	365 days
`fetched_attribute`	worker_id, attribute, value, source_id, as_of, fetched_at	By attribute	Superseded by new row	Life of the worker record
`adapter_descriptor`	source_id, version, descriptor, state	6	Versioned	Life of the Platform
`config_item`, `config_version`	type, key, version, value, effective_from, approved_by	6	Versioned, immutable	Life of the Platform

B.2 Registration & Records tables (schema reg)

Table	Principal columns	Class	Mutability	Retention
`worker`	public_id, registration_no, status, registered_at, centre_id, service_area_id, household_id	5	Versioned	Life of the worker record
`worker_identity_ref`	worker_id, aadhaar_ref_key, last4, verified_at, verification_status	1	Versioned	Life of the worker record
`worker_address`	worker_id, lgd_code, line1, pin, source, as_of	5	Versioned	Life of the worker record

Table	Principal columns	Class	Mutability	Retention
`worker_bank_account`	worker_id, account_no (encrypted), ifsc, validated_at	2	Versioned	Life of the worker record
`worker_employment_profile`	worker_id, trade, days_worked_12m, employer_ref	5	Versioned	Life of the worker record
`worker_nominee`	worker_id, name, relationship, share_pct	5	Versioned	Life of the worker record
`household`, `household_member`	household_id, member_id, relationship, dob, education_stage	5	Versioned	Life of the worker record
`household_survey`	household_id, answers snapshot, surveyed_by, surveyed_at	4	Insert only	Life of the worker record
`registration_application`	draft_id, step_state, started_by, updated_at	5	Updated until submit	90 days after submit
`merge_event`	surviving_id, merged_id, rule, decided_by, decided_at	6	Insert only	Life of the worker record

B.3 Standard columns and conventions

Convention	Value	Detail
Primary key	`id bigint`	Never exposed outside the database tier.
Public identifier	`public_id char(26)`	Time-ordered Crockford base32. Used in URLs, events and exports.
Created and updated	`created_at`, `created_by`, `updated_at`, `updated_by`	UTC timestamp and `usr_` identifier. Absent on insert-only tables except the created pair.
Concurrency	`row_version`	Incremented on every update and returned as `ETag`.
Tombstone	`is_tombstoned`, `tombstoned_at`	Set on erasure. Personal fields are overwritten with a marker.
Provenance on fetched values	`source_id`, `as_of`, `fetched_at`, `consent_ref`	Present on every attribute from a department.
Case ID	`ILC-yy-xxxxxxxx-c`	Two-digit year, 8-character Crockford sequence and a check character computed with the Luhn mod-32 algorithm.
Registration Number	Issued once from one sequence	Never reused. Format fixed by APBOCWWB at inception.
Ticket ID	`TKT-yy-xxxxxxxx-c`	Same construction as the Case ID.

Annex C. Event Catalogue and Error Codes

C.1 Event catalogue

Events are published with the outbox pattern of Section 2.16. Each type is versioned and carries identifiers and non-personal attributes only. The producer is the family that owns the change.

Event type	Producer	Principal consumers and effect
`worker.registered`	Registration	Entitlement (baseline evaluation), Communication (welcome notice), Employment (matching), WLH
`worker.updated`	Registration	Entitlement (re-evaluation), Communication, Employment
`worker.renewed`	Registration	WLH, MIS
`worker.merged`	Registration	All families update references. WLH records the merge.
`household.updated`	Registration	Entitlement (re-evaluation), Counselling
`consent.granted`, `consent.withdrawn`	Core	Integration Hub (gate), all families
`case.opened`	Core	Starts the CaseLifecycle workflow
`case.step.added`	Core	WLH, MIS
`case.closed.completed`	Core	WLH, Notification, MIS
`case.closed.next_step`	Core	WLH, Notification to the named owner, MIS
`clock.warning`, `clock.breached`	Core workflow	Escalation workflow, Notification, MIS
`integration.call.completed`	Hub	Convergence Register, MIS
`integration.attribute.fetched`	Hub	Registration, Entitlement
`reconciliation.mismatch_raised`	Hub	Case Service opens a Needs Review case
`source.degraded`, `source.recovered`	Hub	Notification to TSU, MIS fallback mart
`scheme.evaluated`	Entitlement	Communication, Calendar matching, WLH
`scheme.application.submitted`	Entitlement	Convergence Register, WLH
`scheme.application.status_changed`	Entitlement	Notification, WLH, MIS
`ticket.opened`, `ticket.assigned`, `ticket.escalated`, `ticket.resolved`	Grievance	Notification, MIS, Legal aid for tagged tickets
`service_request.status_changed`, `service_request.output_ready`	Grievance	Notification, WLH
`employer.verified`, `job.posted`, `job.enrolled`	Employment	Notification, Calendar, MIS
`engagement.confirmed`, `muster.recorded`	Employment	WLH, MIS, Site and group updates
`visit.recorded`, `skilling.attended`, `rpl.nominated`, `rpl.certified`	Employment	WLH, MIS, Notification
`mela.checked_in`, `offer.made`, `placement.confirmed`	Employment	Notification, WLH, Employment tracking

Event type	Producer	Principal consumers and effect
`group.formed`, `group.meeting_held`, `federation.decision_recorded`	Employment	WLH, MIS
`counselling.session_held`, `counselling.outcome_recorded`	Counselling	Referral creation, WLH, MIS
`course.completed`, `certificate.issued`, `iec.published`	Counselling	Notification, WLH
`referral.created`, `referral.outcome_recorded`	Communication	Receiving module, Notification
`notification.requested`	Any family	Notification dispatch workflow
`notification.delivered`, `notification.failed`	Communication	Calling module, MIS
`config.published`	Administration	All families evict cached configuration
`user.role_changed`, `user.deactivated`	Administration	Policy cache, session end
`sync.conflict.raised`	Core	SyncConflict workflow, Notification to resolvers

C.2 Platform error codes

Every error response carries a stable `code`. A code never changes meaning. The prefix names the family and the first digit pair the HTTP status class. The messages are translated and are not part of the contract.

Code	HTTP	Meaning	Typical cause and client action
GEN-4001	400	Malformed request	Invalid JSON or unknown filter. Correct the request.
GEN-4011	401	Authentication required or token expired	Sign in again or refresh the token.
GEN-4031	403	Action not permitted for this role	Role lacks the action. Ask an administrator.
GEN-4032	403	Outside centre scope	The record belongs to another centre.
GEN-4033	403	Step-up authentication required	Repeat MFA to continue.
GEN-4041	404	Resource not found in scope	Wrong identifier or no access. The response does not say which.
GEN-4091	409	Duplicate request	The idempotency key was seen with a different body.
GEN-4121	412	Version conflict	The record changed. Reload and retry.
GEN-4221	422	Validation failed	See `fieldErrors`.
GEN-4291	429	Rate limit reached	Wait for `Retry-After`.
GEN-5001	500	Unexpected error	Report the correlation identifier to the helpdesk.
GEN-5031	503	Service temporarily unavailable	Retry with backoff. The fallback route applies where offered.
CASE-4221	422	Documented next step incomplete	One or more of the five fields is missing.
CASE-4222	422	Closure evidence missing	A completed closure needs an evidence document.
CASE-4091	409	Case already closed	Reopen with a reason if permitted.
CASE-4223	422	Target date in the past	Choose a future date.
REG-4091	409	Aadhaar reference already registered	A merge review has been opened.
REG-4222	422	Eligibility rule not met	The detail names the rule and the configured threshold.
REG-4223	422	Biometric consent missing for photograph	Capture consent first.
REG-4224	422	Step out of sequence	Complete earlier steps first.

Code	HTTP	Meaning	Typical cause and client action
INT-2001	200	Hub outcome NO_RECORD	Business result. The department has no record.
INT-4031	403	Hub outcome CONSENT_REQUIRED	Capture consent for the purpose.
INT-5031	503	Hub outcome SOURCE_UNAVAILABLE	The step continues on the fallback class.
INT-4292	429	Hub outcome RATE_LIMITED	Re-attempt scheduled after the source's retry-after.
INT-4093	409	Hub outcome MISMATCH	Needs Review queue entry created.
INT-5022	502	Hub outcome CONTRACT_DRIFT	Response quarantined. TSU alerted.
INT-2022	202	Hub outcome PENDING	Case stays open with the clock running.
SCH-4091	409	Application already open for this scheme	See the existing application.
SCH-4222	422	Scheme not active on the evaluation date	Inactive or not yet effective.
DOC-4151	415	Unsupported file type	Content-based detection did not match the allowed types.
DOC-4131	413	File too large	Above the size cap for the document type.
DOC-4222	422	Antivirus check failed	The file was rejected.
SYN-4091	409	Sync conflict	Both versions stored. Await resolution.
SYN-4221	422	Change-set rejected by validation	Fix the draft and push again.
SYN-4031	403	Device not registered or revoked	Enrol the device again.

Annex D. Configuration Reference, Reporting Marts and Glossary

D.1 Reporting mart refresh cadence

Every dashboard and report shows the refresh time of the mart behind it. The cadences below are proposals that the PMU confirms at inception.

Mart	Content	Refresh	Principal users
`mart_registration_daily`	Registrations, renewals and verification status by centre and day	15 minutes	Centres, PMU
`mart_scheme_funnel`	Eligible, Applied, Sanctioned, Rejected, Availed by scheme and district	Hourly	PMU, Departments
`mart_ticket_sla`	Tickets by category, level, age and clock status	15 minutes	Helpdesk, TSU
`mart_employment`	Postings, enrolments, engagements, placements	Hourly	PMU, Board
`mart_skilling`	Visits, training, RPL batches and certification	Hourly	PMU
`mart_fallback_share`	Fallback share, deferred share, reconciliation ageing by source and district	15 minutes	TSU, Clusters
`mart_integration_health`	Success rate, latency, breaker state and last sync by source	5 minutes	TSU, Clusters
`mart_4e_funnel`	Enrollment, Entitlement, Empowerment, Employment counts	Nightly	Board, PMU
`mart_data_quality`	Completeness, verification, duplicates, LGD coverage, consent coverage	Nightly	Data Governance Committee
`mart_functionary`	Attendance, activity counts and scorecards	Nightly	Supervisors, PMU

D.2 Configuration item keys

Key pattern	Type	Example value
`form.registration.step.{n}`	Form definition	Fields, validation expressions, labels in te and en
`scheme.{code}`	Scheme rule	Rule document of Section 3.8
`overlap.{code}`	Scheme rule	Rule type and scheme set
`clock.{module}.{category}`	Service clock	Target, warning percentage, pause reasons, escalation
`escalation.{module}.{district}`	Escalation matrix	Owner per level
`template.{event}.{channel}.{lang}`	Notification template	Text, DLT identifier, length budget
`fallback.{source}.{element}`	Fallback Register entry	Permitted classes, evidence list, countersign rule, cut-over state
`master.document_type.{code}`	Master data	Name, size cap, retention class
`master.lgd`	Master data	LGD hierarchy version
`threshold.mobile_reuse`	Business threshold	5
`threshold.kiosk_idle_seconds`	Business threshold	90
`threshold.offline_session_hours`	Business threshold	72
`sync.batch.max_changes`	Platform setting	100

Key pattern	Type	Example value
`campaign.approval.required`	Platform setting	true

D.3 Glossary

Term	Meaning
4E	Enrollment, Entitlement, Empowerment, Employment: the stages of a worker's pathway.
AUA, KUA	Authentication User Agency and e-KYC User Agency under UIDAI.
BOCW	Building and Other Construction Workers.
Case ID	The globally unique identifier of a request, issued by the Case Service.
CESS	Welfare cess levied on construction cost and credited to the Board.
Circuit breaker	A control that stops calls to a failing system for a period so that failures do not spread.
DEO	Centre Data Entry Operator.
DFSC	District Field Support Cluster.
DLT	Distributed Ledger Technology registration required for SMS templates in India.
DPDP	Digital Personal Data Protection Act, 2023.
F1 to F6	The six fallback classes of Section 6.2.
Golden record	The single surviving record for a worker after matching and survivorship.
Hub	The Integration Hub (SM-12).
ILAC	Integrated Labour Amenities Complex.
Idempotency key	A client-supplied value that makes a repeated request return the original result.
LGD	Local Government Directory, the source of location codes.
Outbox	A table written in the business transaction from which events are published.
PMU	Programme Management Unit at APBOCWWB.
RPL	Recognition of Prior Learning.
RPO, RTO	Recovery point objective and recovery time objective.
SBOM	Software bill of materials.
TSU	Technical Support Unit.
VAPT	Vulnerability assessment and penetration testing.
WLH	Worker Longitudinal History.